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OF MATHEMATICS, INFORMATICS
AND NATURAL SCIENCES

DISSERTATION

# Spin Ruijsenaars–Schneider models from supersymmetric gauge theories

Lukas Hardi

Department of Mathematics
Faculty of Mathematics, Informatics and Natural Sciences

Universität Hamburg
Hamburg, Germany

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Supervisors:
Prof. Dr. Gleb Arutyunov, Universität Hamburg
Prof. Dr. Jörg Teschner, Universität Hamburg
Prof. Dr. Paul Wedrich, Universität Hamburg

Committee:
1st Examiner: tba, Universität Hamburg
2nd Examiner: tba, Universität Hamburg

Universität Hamburg, Hamburg, Germany
Faculty of Mathematics, Informatics and Natural Sciences
Department of Mathematics

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I hereby declare upon oath that I have written the present thesis independently and have not used further resources and aids than those stated in the dissertation.

| | |
|---|---|
| Date | Signature<br>(Lukas Hardi) |

To my family

بۆ خێزانەکەم

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# Abstract

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions. (Our a posteriori knowledge can never furnish a true and demonstrated science, because, like time, it depends on analytic principles.) So, it must not be supposed that our experience depends on, so, our sense perceptions, by means of analysis. Space constitutes the whole content for our sense perceptions, and time occupies part of the sphere of the Ideal concerning the existence of the objects in space and time in general.

# Zusammenfassung

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# List of Publications

## Dissertation Related Publications


[AH26a] Gleb Arutyunov and Lukas Hardi. Elliptic spin Ruijsenaars-Schneider integrable models from 5d $\mathcal{N}=1$ gauge theories. arXiv:2608.17837 [hep-th]. Aug. 2026. DOI: 10.48550/arXiv.2608.17837. URL: http://arxiv.org/abs/2608.17837 (visited on 11/08/2026).

[AH26b] Gleb Arutyunov and Lukas Hardi. Quantized Quiver Varieties and the Quantum Spin Ruijsenaars-Schneider Model. In: *Communications in Mathematical Physics* 407.6 (June 2026). arXiv:2508.07862 [hep-th], p. 117. ISSN: 0010-3616, 1432-0916. DOI: 10.1007/s00220-026-05595-4. URL: http://arxiv.org/abs/2508.07862 (visited on 11/09/2026).

[AH26c] Gleb Arutyunov and Lukas Hardi. Spin Ruijsenaars-Schneider models are Coulomb branches. arXiv:2603.03048 [hep-th]. July 2026. DOI: 10.48550/arXiv.2603.03048. URL: http://arxiv.org/abs/2603.03048 (visited on 11/08/2026).

[AHK26] Gleb Arutyunov, Lukas Hardi, and Rob Klabbers. Quantum Trigonometric Spin Ruijsenaars-Schneider Models from $K$-theoretic Coulomb Branches. arXiv:2607.28043 [hep-th]. July 2026. DOI: 10.48550/arXiv.2607.28043. URL: http://arxiv.org/abs/2607.28043 (visited on 11/10/2026).


## Other Publications


[Ber+21] Sebastian Berndt, Kilian Grage, Klaus Jansen, Lukas Johannsen, and Maria Kosche. Robust Online Algorithms for Dynamic Choosing Problems. arXiv:2104.09803 [cs.DS]. Apr. 2021. DOI: 10.48550/arXiv.2104.09803. URL: http://arxiv.org/abs/2104.09803 (visited on 11/10/2026).

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# List of Abbreviations

**BFN** . . . . . Braverman–Finkelberg–Nakajima (and others)

**CM** . . . . . Calogero–Moser (and Sutherland in the trigonometric case)

**QFT** . . . . . Quantum field theory

**RS** . . . . . Ruijsenaars–Schneider

# 1 Introduction

**This chapter introduces the main ideas that went into this thesis. The first section gives a historically inspired motivation, beginning with the inception of quantum field theory, specifically quantum electrodynamics and quantum chromodynamics. We then explain how the understanding of strong-coupling phenomena in quantum field theories motivates the introduction of supersymmetry, which allowed Seiberg and Witten to draw a connection to integrable models. This thesis is introduced as a natural extension of these developments. The second section recounts the temporal progression of ideas that led to the results of this thesis. Since the author comes from a mathematical background, our exposition will be targeted towards mathematicians.**

## 1.1 Motivation

Two great revolutions in physics—quantum mechanics and relativity—occurred at the beginning of the 20th century. On the one hand, quantum mechanics describes observables in the world by hermitian linear operators and symmetries of the world by unitary linear operators, both acting on a Hilbert space of physical states. On the other hand, relativity describes observables in the world by sections of bundles over a four-dimensional spacetime manifold $M$ equipped with a metric with Lorentzian signature $(+,-,-,-)$, and the symmetries of the world are described by the Lorentz group $\mathrm{SO}(1,3)$ (or the simply connected double cover $\mathrm{Spin}(1,3)$ of its identity component) of linear transformations preserving the metric and its extension by spacetime translations $P_\mu$ for coordinate indices $\mu = 0,1,2,3$, the Poincaré group $\mathrm{SO}(1,3) \ltimes \mathbb{R}^4$.

Restricting to Minkowski spacetime $M = \mathbb{R}^{1,3}$ with the standard Minkowski metric $\eta = \mathrm{diag}(+1,-1,-1,-1)$, meaning

$$(ds)^2 = \eta_{\mu\nu} dx^\mu dx^\nu = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2, \tag{1.1}$$

where repeated indices are understood to be summed over, the approaches of quantum mechanics and relativity are unified in the framework of quantum field theory [Pes18], which posits that the Hilbert space of physical states and the algebra of observables

acting on it come equipped with compatible unitary actions of the Poincaré group. Observables $\Phi$ acting on the Hilbert space then become *fields*, *i.e.* dependent on a position $x = (x^0, x^1, x^2, x^3)$ in spacetime, via

$$\Phi(x) := e^{\mathrm{i}x^\mu P_\mu} \Phi e^{-\mathrm{i}x^\mu P_\mu}. \tag{1.2}$$

Rigorously speaking, fields should be interpreted as operator-valued distributions, which may be integrated over regions in spacetime. It then becomes relevant to ask how the fields transform under the action of the Lorentz group. Due to the isomorphism

$$\mathfrak{spin}(1,3)_{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}) \tag{1.3}$$

of complex Lie algebras, representations can be described by two half-integers $(j_L, j_R)$ which are the spin labels or weights of finite-dimensional irreducible representations of the left- and right-handed $\mathfrak{sl}(2,\mathbb{C})$. Physically relevant fields usually transform in the scalar representation $(0,0)$, the left- or right-handed spinor representations $(\frac{1}{2},0)$ or $(0,\frac{1}{2})$, or the vector representation $(\frac{1}{2},\frac{1}{2})$.

A simple example of a quantum field theory is the free massive scalar quantum field theory, which describes a real valued field $\Phi(x)$ (think: hermitian operator-valued distribution) that transforms in the scalar representation $(0,0)$ of the Lorentz group and whose dynamics is described by the action functional

$$S_{\text{free}}[\Phi] := \int_{\mathbb{R}^{1,3}} d^4x \left[\frac{1}{2}\partial^\mu\Phi\partial_\mu\Phi - \frac{m^2}{2}\Phi^2\right] \tag{1.4}$$

for some mass $m > 0$, where the spacetime index is raised via the inverse metric $\eta^{\mu\nu} = \eta_{\mu\nu}$, *i.e.* $\partial^\mu = \eta^{\mu\nu}\partial_\nu$. This means that, classically, the field $\Phi$ must extremize the action functional,

$$\frac{\delta S_{\text{free}}}{\delta\Phi} = 0, \tag{1.5}$$

and that, quantum mechanically, observables $\mathcal{O}$ (diagonal in the field basis) have expectation values

$$\langle\mathcal{O}\rangle := Z^{-1}\int [d\Phi] e^{\frac{\mathrm{i}}{\hbar}S_{\text{free}}[\Phi]}\mathcal{O}[\Phi], \qquad Z := \int [d\Phi] e^{\frac{\mathrm{i}}{\hbar}S_{\text{free}}[\Phi]}, \tag{1.6}$$

where $Z$ is the partition function and $[d\Phi]$ is the notoriously elusive path integral measure. In the particular case of the free massive scalar, however, expectation values can be calculated explicitly, making the free massive scalar quantum field theory an *exactly solvable* quantum field theory.

A more involved example of a quantum field theory that is also relevant to the real world is *quantum electrodynamics*, describing the interactions of electrons, positrons, and photons, which is also our first example of a *gauge theory*. The theory includes two fields: The *gauge field* $A(x) = A_\mu(x)dx^\mu$ transforming in the vector representation $(\frac{1}{2},\frac{1}{2})$—the photon field; and the *Dirac spinor field* $\psi(x)$ transforming in the Dirac spinor representation $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$—the electron-positron field. What makes this quantum field theory a gauge theory is that the fields transform under an additional symmetry, called *gauge symmetry*. Specifically, smooth maps $h : \mathbb{R}^{1,3} \to U(1)$ generate the transformations

$$A_\mu(x) \mapsto h(x)A_\mu(x)h(x)^{-1} + e^{-1}h(x)\partial_\mu h(x)^{-1}, \qquad \psi(x) \mapsto h(x)\psi(x), \tag{1.7}$$

where $e$ is the coupling constant controlling the strength of electromagnetic interactions. This exhibits the gauge field $A(x)$ as a principal $U(1)$-connection 1-form for the trivial principal $U(1)$-bundle over $\mathbb{R}^{1,3}$. The dynamics of quantum electrodynamics are then described by the essentially unique gauge invariant action functional

$$S_{\mathrm{QED}}[A,\psi] \coloneqq \int_{\mathbb{R}^{1,3}} d^4x \left[-\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \bar{\psi}(\mathrm{i}\gamma^\mu D_\mu - m)\psi\right], \tag{1.8}$$

with the electron mass $m > 0$, the curvature 2-form $F \coloneqq dA$, the covariant derivative

$$D_\mu \coloneqq \partial_\mu + eA_\mu, \tag{1.9}$$

and the Dirac matrices $\gamma^\mu$, which map the tensor product of the Dirac spinor representation with its dual to the vector representation.

Despite the mathematically beautiful differential-geometric formulation of quantum electrodynamics, the interactions induced by the presence of the term $e\bar{\psi}\gamma^\mu A_\mu\psi$ in the action place closed form expectation values out of reach of analytical methods. Even perturbative methods, where $e$ is assumed to be small, are famously plagued by divergent integrals. Perturbative expectation values can nonetheless be extracted by *renormalization*, which regularizes the divergent integrals by cutting off the integration range at an energy scale $k$ and introducing $k$-dependent counter terms that modify the values of the coupling constant $e$ and the electron mass $m$ in a way that cancels the divergent terms. For this reason, one says that both the coupling and the mass must *run* with the energy scale $k$. Importantly, the coupling constant $e$ must become small at low energy scales, which makes the perturbative predictions of quantum electrodynamics (coming in the form of asymptotic series in $e$) valid at ordinary energy scales.

However, not every physically relevant gauge theory has a valid perturbative description at low energy scales. In fact, *quantum chromodynamics*, describing the interactions of quarks and gluons relevant in nuclear physics, has the opposite behavior: Its coupling constant becomes small at high energies—a property known as *asymptotic freedom*—and large at low energies [GW73]. This invalidates perturbative methods for calculations at ordinary energy scales, requiring theoreticians to develop non-perturbative methods valid in the strong coupling regime. Everyday experience and numerical lattice simulations suggest that strongly coupled quantum chromodynamics exhibits a phenomenon known as *confinement*, binding charged quarks into neutral composite configurations known as hadrons.

To further explore strongly coupled phenomena in quantum field theories, a simplified testing ground is instructive. The most basic example of an asymptotically free quantum field theory is Yang–Mills theory [YM54] with gauge group $G_c$, describing the dynamics of a gauge field $A(x)$ valued in $\mathfrak{g} \coloneqq \mathrm{Lie}\, G_c$ (think: principal connection 1-form) transforming in the vector representation $(\frac{1}{2},\frac{1}{2})$ of the Lorentz group with the action functional

$$S_{\mathrm{YM}}[A] \coloneqq \int_{\mathbb{R}^{1,3}} d^4x \left[-\frac{1}{4}\operatorname{Tr} F^{\mu\nu}F_{\mu\nu}\right], \qquad F \coloneqq dA + gA\wedge A, \tag{1.10}$$

where $g$ is the coupling constant, determining the strength of non-abelian self-interaction of the gauge field. Indeed, Yang–Mills theory for $G_c = \mathrm{SU}(N)$ faces the same problems as quantum chromodynamics: The weak coupling approximation becomes invalid at low energies.

Surprisingly, confinement can be seen semiclassically after placing the theory on the spacetime $\mathbb{R}^{1,2} \times S^1_R$ with small circle radius $R$ and adding additional fields to Yang–Mills theory [PSU12]. Specifically, adding massless $\mathfrak{g}$-valued fields $\lambda(x), \bar{\lambda}(x)$ transforming in the *Majorana* spinor representation of the Poincaré algebra leads to the action functional

$$S_{\text{SYM}}[A, \lambda] \coloneqq \int_{\mathbb{R}^{1,3}} d^4x \, \text{Tr} \left[ -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} - \lambda \text{i} \sigma^\mu D_\mu \bar{\lambda} \right], \tag{1.11}$$

where $\sigma^\mu$ are the Pauli matrices, which map the tensor product of the Majorana spinor representation with its dual to the vector representation. This quantum field theory is known as $\mathcal{N} = 1$ supersymmetric Yang–Mills theory, since it possesses an extended symmetry which extends the Poincaré algebra to a Lie superalgebra using additional generators of odd degree, called *supercharges*, $Q$ and $\bar{Q}$, which transform in the $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ representations under the Lorentz group, and importantly satisfy

$$[Q, \bar{Q}]_+ \coloneqq Q\bar{Q} + \bar{Q}Q = 2\sigma^\mu P_\mu. \tag{1.12}$$

This extended Lie superalgebra is known as the $\mathcal{N} = 1$ super Poincaré algebra, capturing the spacetime symmetries of super Minkowski spacetime $\mathbb{R}^{1,3|4}$. Using the formalism of supermanifolds, the fields $A(x)$ and $\lambda(x), \bar{\lambda}(x)$ combine to transform in the *vector multiplet representation* of the $\mathcal{N} = 1$ super Poincaré algebra.

Continuing in this direction, a revolutionary breakthrough was achieved by Seiberg, Witten, and others after further extending supersymmetry to $\mathcal{N} = 2$ supersymmetric Yang–Mills theory with gauge group $\text{SU}(N)$ [SW94a; SW94b; Gor+95; MW96; DW96; DP98]. This quantum field theory extends $\mathcal{N} = 1$ supersymmetric Yang–Mills theory by another set of massless $\mathfrak{su}(N)$-valued Majorana spinor fields $\psi(x)$, $\bar{\psi}(x)$ as well as a massless $\mathfrak{su}(N)_{\mathbb{C}}$-valued scalar field $\varphi(x) = \phi_1(x) + \text{i}\phi_2(x)$. All of these fields may be combined into a field $\Psi(x, \theta, \bar{\theta})$ on super Minkowski spacetime $\mathbb{R}^{1,3|8}$, which transforms in the *vector multiplet representation* of the $\mathcal{N} = 2$ super Poincaré algebra, capturing the spacetime symmetries of $\mathbb{R}^{1,3|8}$. The action functional may be conveniently packaged as an integral over one chiral half of super Minkowski spacetime

$$S_{\text{SYM2}}[\Psi] \coloneqq \text{Im} \, \frac{1}{4\pi} \int d^4x \, d^4\theta \left[ \frac{1}{4} \, \text{Tr} \, \mathcal{F}(\Psi) \right], \tag{1.13}$$

where the *prepotential* $\mathcal{F}$ is some holomorphic function. At high energies, the theory has the prepotential $\mathcal{F}(\Psi) \coloneqq \frac{\tau}{2}\Psi^2$ with $\tau = \frac{\theta}{2\pi} + \frac{4\pi \text{i}}{g^2}$, where $g$ is the coupling constant and $\theta$ is an angular variable called the *$\theta$-angle*.

The important new ingredient present in $\mathcal{N} = 2$ super Yang–Mills theory is the adjoint-valued complex scalar field $\varphi(x) = \phi_1(x) + \text{i}\phi_2(x)$. At minimum energy, the potential forces

$$[\phi_1(x), \phi_2(x)] = 0, \tag{1.14}$$

so that the expectation values $\langle \varphi \rangle$ can be taken to lie in the diagonal Cartan subalgebra of $\mathfrak{su}(N)_{\mathbb{C}}$, whose components $a^i$ (modulo the Weyl group $S_N$) coordinatize the moduli space of Coulomb vacua, the *Coulomb branch* $\mathfrak{B}$, as a complex variety. Expanding the field $\varphi$ in the action functional (1.13) around the point $\langle \varphi \rangle$, one can see that all off-diagonal components of the $\mathfrak{su}(N)$-valued fields acquire a non-zero mass proportional to $a^i - a^j$. At low energy scales, we may integrate over all massive modes so that

only the massless fields survive in the effective theory. It follows that all fields are diagonal in the resulting *effective theory*, which is characterized by the values of the gauge invariant expectation values $u = (u_2, \dots, u_N)$ with $u_k = \langle \operatorname{Tr} \varphi^k \rangle = \sum_{i=1}^{N} (a^i)^k$. This simplifies the effective description enormously.

Crucially, supersymmetry tells us that the actional functional of the effective theory must still be of the form (1.13), albeit with a possibly different prepotential $\mathcal{F}$. Expanding the prepotential to quadratic order, the effective coupling matrix has components

$$\tau_{ij} = \frac{\partial^2 \mathcal{F}}{\partial a^i \partial a^j} \tag{1.15}$$

and $\operatorname{Im} \tau$ must be positive definite to ensure that the effective theory is unitary. This equips the Coulomb branch $\mathfrak{B}$ with a special Kähler structure where the Kähler metric is given by

$$(ds)^2 = (\operatorname{Im} \tau)_{ij} da^i d\bar{a}^j. \tag{1.16}$$

The effective theory can thus be written as a four-dimensional non-linear sigma model on $\mathbb{R}^{1,3}$ whose target space is the Coulomb branch $\mathfrak{B}$ with this Kähler metric.

Arguably, the main motivation of Seiberg and Witten to look at this construction was that it allows to show the existence of a confining phase in a mass deformation of the theory. Indeed, one can use the algebraic relations of the $\mathcal{N} = 2$ super Poincaré algebra to show that the mass of particles of electric charge $n$ and magnetic charge $m$ must satisfy the BPS bound [WO78]

$$M \geq \sqrt{2} |n \cdot a + m \cdot a_D|, \qquad a_D \coloneqq \frac{\partial \mathcal{F}}{\partial a}. \tag{1.17}$$

Particles that saturate this bound are called *BPS particles*. Usually, BPS particles of non-zero charge are heavy and thus do not contribute to the dynamics of the effective theory at low energy scales. However, it can happen at particular points of the Coulomb branch $\mathfrak{B}$ that $a \to 0$ or $a_D \to 0$. In this case, certain BPS particles become light and cannot be integrated out of the effective theory. Turning on a non-zero mass for the coordinate $u_2$ (breaking supersymmetry to $\mathcal{N} = 1$) to drive the theory towards the points $a \to 0$ or $a_D \to 0$ also drives the light BPS particle to a non-zero expectation value—it *condenses*. If this BPS particle is a magnetic monopole (carrying no electric charge), the monopole condensate increases the energy of electric flux times area. The electric flux therefore gets squeezed into tubes with a small cross section which have a *string tension*, energetically favoring shorter flux tubes. It follows that sources of (color-)electric charge connected by electric flux tubes are strongly attracted and bind into (color-)neutral configurations, *i.e.* become *confined*.

In order for the description of the light BPS monopoles to be valid perturbatively, we need the magnetic interactions to be weak. It thus becomes necessary to study the behavior of the coupling matrix $\tau$ more closely. Importantly, $\tau$ defines a local system of lattices $\Gamma$, where the lattice over a point $u$ is given by

$$\Gamma_u \coloneqq \mathbb{Z}^{N-1} + \tau(u) \mathbb{Z}^{N-1}. \tag{1.18}$$

This, in turn, defines a torus fibration $\mathfrak{M} \coloneqq T^*\mathfrak{B}/\Gamma \to \mathfrak{B}$ over the Coulomb branch $\mathfrak{B}$, which comes with the canonical holomorphic symplectic form of the cotangent bundle.

*The punchline is that the coordinates u form a Poisson-commutative family turning the torus fibration $\mathfrak{M}$ into the phase space of an algebraic integrable system in the sense of the Liouville theorem.* Specifically, it turns out that $a^i$ and their canonical conjugates are the action and angle variables of the periodic Toda chain (in the center-of-mass frame), whose Hamiltonian may be written as

$$u_2 = H_{\text{Toda}} = \frac{1}{2}\sum_{i=1}^{N} p_i^2 + \Lambda^2 \sum_{i=1}^{N} e^{x_i - x_{i+1}}, \tag{1.19}$$

where $x_i$ are particle coordinates, $p_i$ the canonical momenta, and $i$ is taken modulo $N$. This Hamiltonian has the stationary points

$$p_i = 0, \qquad x_i - x_{i+1} = 2\pi \mathrm{i}\, k/N, \tag{1.20}$$

at which $u_2 = N\Lambda^2 e^{2\pi \mathrm{i}\, k/N}$. Since these points are stationary, the Liouville torus at these energy levels (the fiber sitting over a point $u$) must degenerate in the direction of the flow. The modulus $\tau$ of the torus therefore blows up in this direction, making the effective theory of the corresponding light BPS particle weakly coupled.

A very general family of 4d $\mathcal{N} = 2$ gauge theories (referred to as class $\mathcal{S}$ theories) can be constructed by compactifying the 6d $\mathcal{N} = (2,0)$ superconformal field theory of type $A_{N-1}$ on a punctured Riemann surface $C$. In this case, [GMN11] showed that the analogous torus fibration $\mathfrak{M} \to \mathfrak{B}$ should be identified with the *Hitchin fibration*, whose total space $\mathfrak{M}$ is the moduli space of Higgs bundles and whose base $\mathfrak{B}$ (the Coulomb branch of the theory on $\mathbb{R}^{1,3}$) is given by the Hitchin base

$$\mathfrak{B} = \bigoplus_{k=2}^{N} H^0(C, K_C^{\otimes k}), \tag{1.21}$$

where $K_C$ is the canonical bundle of $C$. The corresponding integrable model is the *Hitchin system* associated to the punctured Riemann surface $C$.

To further explore the integrable structure of the Coulomb branch, we can again compactify the spacetime to $\mathbb{R}^{1,2} \times S^1_R$ [SW96; GMN10; GMN11; GK13]. According to the Kaluza–Klein construction, only the Fourier zero modes of the fields along the circle are massless. All other modes are massive and do not contribute to the dynamics of the effective theory on $\mathbb{R}^{1,2}$. However, the compactification introduces new scalars to the effective theory: The zero mode $\phi_3$ of the gauge field component $A_3$ along the circle and the dual photon $\sigma$ parametrizing the three-dimensional Hodge dual of the zero mode $F_3$ of the curvature via

$$\star_3 F_3 = d\sigma. \tag{1.22}$$

Large gauge transformations send $\phi_3$ and $\sigma$ to

$$\phi_3 \mapsto \phi_3 + \mathbb{Z}^{N-1}, \qquad \sigma \mapsto \sigma + \tau\mathbb{Z}^{N-1} \tag{1.23}$$

Therefore, the gauge invariant combination of $\phi_3$ and $\sigma$ is the monopole operator

$$v \coloneqq e^{\frac{2\pi \mathrm{i}}{\tau}(\tau\phi_3 + \sigma)}. \tag{1.24}$$

The transformation law (1.23) says that the monopole operator $v$ is a complex coordinate on the torus fibers $T^*_u\mathfrak{B}/\Gamma_u$. The special Kähler structure on $\mathfrak{B}$ may then be extended to a hyperkähler structure on $\mathfrak{M}$ with the *semiflat metric*

$$(ds)^2 = R(\operatorname{Im}\tau)_{ij}da^i d\bar{a}^j + R^{-1}(\operatorname{Im}\tau)^{-1}_{ij}\frac{dv^i}{v^i}\frac{d\bar{v}^j}{\bar{v}^j}. \tag{1.25}$$

The sigma model on $\mathbb{R}^{1,2}$ with target space given by $\mathfrak{M}$ equipped with the semiflat metric approximates the theory on $\mathbb{R}^{1,2}\times S^1_R$, but crucially misses the dynamics of BPS particles with worldlines wrapping around the circle. Taking these dynamics into account, the complex symplectic variety obtained from the small circle limit $R\to 0$ (which is independent of the choice of complex structure) was closely considered in [BDG15] and rigorously defined by Braverman, Finkelberg, Nakajima, and others (BFN) in a seminal series of papers [Nak16; BFN18b; BFN18a]. This construction generalizes to the case of general gauge groups $G_c$ and additional "matter" fields transforming in a cotangent-type representation $T^*V$ under the gauge group and in the hypermultiplet representation under the $\mathcal{N}=2$ super Poincaré algebra.

Looking instead at the holomorphic symplectic form of the complex structure $J$ at finite radius $R$ (as opposed to the complex structure $I$ coming from the coordinates $u$ and their canonical conjugates), the Darboux charts form cluster charts in the sense of cluster varieties, where the Poisson bracket of the Darboux coordinates $X_\gamma$ is

$$\{X_\gamma, X_{\gamma'}\} = \langle\gamma,\gamma'\rangle X_{\gamma+\gamma'}. \tag{1.26}$$

Following Braverman–Finkelberg–Nakajima, the resulting complex symplectic variety also has a mathematically rigorous construction for general gauge groups $G_c$ and hypermultiplet fields in cotangent-type representations $T^*V$, known as *$K$-theoretic Coulomb branches*. Their cluster structure was conjectured in [GMN10; GK13] and found by Schrader and Shapiro [SS26]. Generally, $K$-theoretic Coulomb branches also furnish algebraic integrable systems. Note, however, that the complex structure $J$ is distinct from the complex structure $I$, and the case $G_c=\mathrm{SU}(N)$ and $V=0$ does not yield the periodic Toda chain as in the complex structure $I$, but the open relativistic Toda chain. For class $\mathcal{S}$ theories, the complex structure $J$ yields the character variety of the Riemann surface $C$, instead of the moduli space of Higgs bundles in complex structure $I$.

Following [Nek98], the idea of Seiberg and Witten can be repeated for five-dimensional gauge theories with $\mathcal{N}=1$ supersymmetry placed on $\mathbb{R}^{1,2}\times T^2$: The effective theory can be rewritten as a sigma model on $\mathbb{R}^{1,2}$ with values in the Coulomb branch $\mathfrak{M}$ of the theory, which is again a hyperkähler manifold whose metric now depends on the radii $R_3$ and $R_4$ of $T^2$. The holomorphic symplectic forms of the complex structures $I$ and $J$ again define complex symplectic varieties. For example, in the case $G_c=\mathrm{SU}(N)$, $V=0$, the complex structure $I$ gives rise to the periodic *relativistic* Toda chain. The complex symplectic variety coming from the complex structure $J$ again fits (conjecturally) into the BFN framework for any $G_c$ and $V$ using equivariant elliptic cohomology.

To summarize, the correspondence between supersymmetric gauge theories with gauge group $\mathrm{SU}(N)$ and integrable models living on the Coulomb branch is given in table 1.1 without matter, and in table 1.2 with adjoint matter, which is the class $\mathcal{S}$ theory for the once-punctured torus. The bottom right entries are extrapolations based on the rational–trigonometric–elliptic hierarchy of the momentum dependence and the entries of the first row may be obtained from the entries of the second row by degeneration limits.

| $G_c = \mathrm{SU}(N), V = 0$ | complex structure $I$ | complex structure $J$ |
|---|---|---|
| $\mathcal{N} = 4, \mathbb{R}^{1,2}$ | open Toda | open Toda |
| $\mathcal{N} = 2, \mathbb{R}^{1,2} \times S^1$ | periodic Toda [Gor+95; MW96] | open relativistic Toda [BFM26] |
| $\mathcal{N} = 1, \mathbb{R}^{1,2} \times T^2$ | periodic relativistic Toda [Nek98] | dual periodic Toda? |

**Table 1.1:** Integrable models living on the Coulomb branches of gauge theories with gauge group $G_c = \mathrm{SU}(N)$ without matter.

| $G_c = \mathrm{SU}(N), V = \mathfrak{su}(N)$ | complex structure $I$ | complex structure $J$ |
|---|---|---|
| $\mathcal{N} = 4, \mathbb{R}^{1,2}$ | trigonometric CM | trigonometric CM |
| $\mathcal{N} = 2, \mathbb{R}^{1,2} \times S^1$ | elliptic CM [DP98] | trigonometric RS [BEF20] |
| $\mathcal{N} = 1, \mathbb{R}^{1,2} \times T^2$ | elliptic RS [Nek98] | dual elliptic RS? |

**Table 1.2:** Integrable models living on the Coulomb branches of gauge theories with gauge group $G_c = \mathrm{SU}(N)$ with adjoint matter.

The main results of this thesis can be understood as a natural continuation of the ideas above to the case of the *necklace quiver gauge theory*. The gauge group and matter content of the necklace quiver gauge theory can be read off of the necklace quiver shown in figure 1.1. For every node $\alpha \in \mathbb{Z}_\ell$ in the quiver (drawn as circles filled with the number $N$), we get a factor $\mathrm{U}(N)_\alpha$ in the gauge group (each associated to a $\mathfrak{u}(N)$-valued vector multiplet), and for every arrow, we get a hypermultiplet valued in the cotangent representation of the bifundamental representation $V_\alpha \otimes V_{\alpha+1}^\vee$, where $V_\alpha$ is the defining representation of $\mathrm{U}(N)_\alpha$ and $V_\alpha^\vee$ is its dual. This means

$$G_c = \prod_{\alpha \in \mathbb{Z}_\ell} \mathrm{U}(N)_\alpha, \qquad V = \bigoplus_{\alpha \in \mathbb{Z}_\ell} V_\alpha \otimes V_{\alpha+1}^\vee. \tag{1.27}$$

The results of this thesis show that the Coulomb branches of necklace quiver gauge

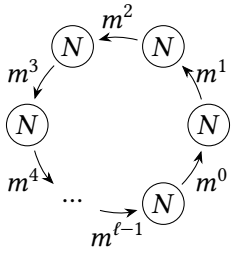



**Figure 1.1:** The necklace quiver with $\ell$ nodes of rank $N$. The bifundamental hypermultiplets corresponding to the arrows carry masses $m^\alpha$.

theories with $\ell$ nodes of rank $N$ are the phase spaces of the spin Ruijsenaars–Schneider models of $N$ particles with $\ell$ spin states. However, care should be taken, as the Hamiltonians defining the spin Ruijsenaars–Schneider models are actually *spectrally dual* to the Hamiltonians Seiberg–Witten theory would dictate. The precise meaning of this statement will be explained in the main text.

Table 1.3 exhibits the dictionary that the remainder of this thesis makes precise. Reading down the column, the compactification data of the spacetime fixes the degeneration type of the integrable system: for $\mathbb{R}^{1,2}$ the model is rational, for $\mathbb{R}^{1,2} \times S^1$ the model is trigonometric, and for $\mathbb{R}^{1,2} \times T^2$ the model is elliptic, meaning that the

| Necklace quiver | complex structure $J$ |
|---|---|
| $\mathcal{N} = 4, \mathbb{R}^{1,2}$ | dual rational spin RS |
| $\mathcal{N} = 2, \mathbb{R}^{1,2} \times S^1$ | dual trigonometric spin RS |
| $\mathcal{N} = 1, \mathbb{R}^{1,2} \times T^2$ | dual elliptic spin RS |

**Table 1.3:** Integrable models living on the Coulomb branches of necklace quiver gauge theories.

particles move on $\mathbb{C}$, $\mathbb{C}^\times$, or an elliptic curve, respectively. This is the same hierarchy that relates ordinary cohomology, $K$-theory and elliptic cohomology, which is precisely the input datum distinguishing the three versions of the Braverman–Finkelberg–Nakajima construction. Read off from the quiver of figure 1.1, the gauge-theoretic data fixes the kinematics of the spin Ruijsenaars–Schneider model: the rank $N$ of the gauge group at each node is the number of particles, and the number $\ell$ of nodes is the number of spin states each particle carries. The momenta, by contrast, are multiplicative in all three rows, since in the complex structure $J$ the fiber coordinates are the monopole operators $v$ of (1.24). This is why every entry of table 1.3 is of Ruijsenaars–Schneider rather than Calogero–Moser type.

What this identification buys is uniformity. Poisson structures for spin Ruijsenaars–Schneider models have so far been produced one degeneration at a time and by different means: by Hamiltonian reduction in the rational case [AF98], by quasi-Hamiltonian reduction in the trigonometric case, and in the elliptic case only on the spectral data of the Lax matrix [Kri98], with no expression in the natural variables $x_i$, $a_i^\alpha$, $c_i^\alpha$ beyond $N = 2$ [Sol08]. The Coulomb branch construction is indifferent to this distinction: one and the same quiver, evaluated in the appropriate cohomology theory, yields the Poisson algebra together with a distinguished commuting family of Hamiltonians, and its quantization is obtained by replacing the Coulomb branch with its quantized counterpart.

## 1.2 Progression of the doctorate

As often happens in research, the starting point for this thesis was of a very different nature than what was laid out in the previous section. The research question this thesis set out to solve was to write down a Poisson algebra, Hamiltonian, and quantization for the equations of motion of the elliptic spin Ruijsenaars–Schneider integrable model found by Krichever and Zabrodin [KZ95]. We recount them here for the readers convenience:

$$\begin{aligned} \dot{x}_i &= -f_{ii}, \\ \dot{a}_i^\alpha &= \sum_{j=1}^{N} (V(x_i - x_j) f_{ij} a_i^\alpha - V(x_i - x_j) f_{ij} a_j^\alpha), \\ \dot{c}_i^\alpha &= \sum_{j=1}^{N} (V(x_j - x_i) f_{ji} c_j^\alpha - V(x_i - x_j) f_{ij} c_i^\alpha). \end{aligned} \tag{1.28}$$

These equations of motion describe the dynamics of $N$ particles with $\ell$ spin states. The variables $x_i$ with particle index $i = 1, \dots, N$ denote particle positions, and the variables $a_i^\alpha$ and $c_i^\alpha$ denote the components of the spin covector and spin vector, respectively,

where $\alpha \in \mathbb{Z}_\ell$ is the spin state index. The inner product between the spin covectors and spin vectors is abbreviated by

$$f_{ij} \coloneqq \sum_{\alpha \in \mathbb{Z}_\ell} a_i^\alpha c_j^\alpha \tag{1.29}$$

and the potential is

$$V(z) \coloneqq E_1(z) - E_1(z - \eta), \tag{1.30}$$

where $E_1(z)$ is the logarithmic derivative of the odd Jacobi theta function $\theta(z)$ depending on the modulus $\tau$ of the elliptic curve

$$E_\tau \coloneqq \frac{\mathbb{C}}{\mathbb{Z} + \tau\mathbb{Z}} \tag{1.31}$$

on which the particles move. The equations of motion possess a symmetry which allow a simultaneous rescaling of the spin covector and spin vector:

$$a_i^\alpha \mapsto k_i a_i^\alpha, \qquad c_i^\alpha \mapsto k_i^{-1} c_i^\alpha. \tag{1.32}$$

This symmetry can be fixed by imposing the constraint

$$a_i^0 = 1. \tag{1.33}$$

Furthermore, the equations of motion are symmetric under the action of the spin rotation group $\mathrm{GL}_\ell$ by acting on the spin covectors and spin vectors as covectors and vectors, respectively. The constraint $a_i^0 = 1$ is preserved by the mirabolic subgroup $\mathrm{M}_\ell$ of $\mathrm{GL}_\ell$. The invariant dynamical variables of the elliptic spin Ruijsenaars–Schneider model therefore respect the constraint $a_i^0 = 1$ and must be $\mathrm{M}_\ell$-invariant combinations of the variables $x_i, a_i^\alpha, c_i^\alpha$. Before taking $\mathrm{M}_\ell$-invariants, the phase space has dimension $2N\ell$, while after taking invariants, the resulting phase space has dimension $2N\ell - \ell(\ell-1)$. Note that this is in fact an even number, allowing for the existence of a symplectic form. In particular, in the case $\ell = 1$, the only dynamical variables are $x_i$ and $c_i^0$ for $i = 1, \dots, N$, whose equations of motion reduce to the equations of motion of the elliptic (spinless) Ruijsenaars–Schneider model [Rui87], also known as the relativistic elliptic Calogero–Moser model.

Krichever and Zabrodin showed that the spin Ruijsenaars–Schneider model is an integrable model by exhibiting a Lax pair. This means that there are two matrices $L(z)$ and $M(z)$ satisfying the Lax equation

$$\dot{L}(z) = [M(z), L(z)]. \tag{1.34}$$

Specifically, the Lax pair of Krichever and Zabrodin is given by

$$\begin{aligned} L(z) &\coloneqq \sum_{i,j=1}^{N} \Phi(z, x_j - x_i + \eta) f_{ij} e_{ij}, \\ M(z) &\coloneqq \sum_{i=1}^{N} E_1(z) f_{ii} e_{ii} + \sum_{i,j=1}^{N} V(x_i - x_j) f_{ij} e_{ii} + \sum_{i \neq j}^{N} \Phi(z, x_j - x_i) f_{ij} e_{ij}. \end{aligned} \tag{1.35}$$

Subsequently, Krichever constructed a symplectic form for the $M_\ell$-invariant variables from the spectral data of the Lax matrix [Kri98]. However, it was not immediately obvious how to reexpress this symplectic form in terms of the natural variables $x_i$, $a_i^\alpha$, $c_i^\alpha$, except in the spinless case $\ell = 1$. In the case $N = 2$, [Sol08] wrote down a symplectic form for the variables

$$x_1, x_2, f_{11}, f_{22}, \tag{1.36}$$

where the dynamics of the variables $f_{12}, f_{21}$ are integrated out and $\eta$ is replaced by an effective coupling $z_0$ defined by the equation $\det L(z_0) = 0$.

Constructing Poisson structures in more general cases required degenerating the underlying elliptic curve, yielding the trigonometric and rational spin Ruijsenaars–Schneider models. In the rational case, where the elliptic curve degenerates to a plane, a Poisson structure was found by Hamiltonian reduction [AF98]. There, it was shown that the rational spin Ruijsenaars–Schneider model of $N$ particles and $\ell$ spin states can be obtained as a Hamiltonian reduction

$$(T^*\mathrm{GL}_N \times (T^*\mathbb{C}^N)^{\times \ell}) /\!\!/ \mathrm{GL}_N, \tag{1.37}$$

where $\mathrm{GL}_N$ acts on $\mathrm{GL}_N$ by the adjoint action and on $\mathbb{C}^N$ by the defining representation. Via the multiplication map on $\mathrm{GL}_N$, we can trivialize the cotangent bundle, giving an isomorphism

$$T^*\mathrm{GL}_N \cong \mathrm{GL}_N \times \mathfrak{gl}_N^* \cong \mathrm{GL}_N \times \mathfrak{gl}_N, \tag{1.38}$$

where we used the trace inner product for the last isomorphism. We thus obtain a parametrization of $T^*\mathrm{GL}_N$ by $g \in \mathrm{GL}_N$ and $x \in \mathfrak{gl}_N$, and of $(T^*\mathbb{C}^N)^{\times \ell}$ by $a \in \mathbb{C}^{N\times \ell}$ and $c \in \mathbb{C}^{\ell \times N}$. The action of $\mathrm{GL}_N$ can then be expressed as

$$g \mapsto hgh^{-1}, \qquad x \mapsto hxh^{-1}, \qquad a \mapsto ha, \qquad c \mapsto ch^{-1}. \tag{1.39}$$

Again identifying $\mathfrak{gl}_N^* \cong \mathfrak{gl}_N$, the momentum map $\mu : T^*\mathrm{GL}_N \times (T^*\mathbb{C}^N)^{\times \ell} \to \mathfrak{gl}_N^*$ for the action (1.39) turns out to be

$$\mu = x - gxg^{-1} - ac. \tag{1.40}$$

The phase space (1.37) of the (spinless) rational Ruijsenaars–Schneider model is obtained by imposing the momentum map equation $\mu = \eta \cdot \mathrm{id}$, where $\eta$ is the coupling constant, and quotienting by the action of $\mathrm{GL}_N$. The Poisson-commuting Hamiltonians of the Ruijsenaars–Schneider model are given by $H[n] = \mathrm{Tr}\, g^n$. In particular, the Hamiltonian $H[1]$ generates the equations of motion (1.28).

Given this construction of the phase space of the rational spin Ruijsenaars–Schneider model by Hamiltonian reduction, chapter 3 quantizes the model using the method of quantum Hamiltonian reduction. The main motivation for carefully quantizing the rational case was the observation that it already contained an unexplored structure that develops its full traction in the quantization. Namely, besides finding a Poisson algebra that reproduces the equations of motion (1.28), the paper [AF98] also identified two important symmetry algebras inside the Poisson algebra generated by the combinations

$$T(z) \coloneqq 1 + c(z - x)^{-1}a, \qquad J[n] \coloneqq cg^n a, \tag{1.41}$$

whose Poisson brackets are of the form

$$\{T_1(z), T_2(w)\} = \frac{1}{z-w}[P, T_1(z)T_2(w)], \tag{1.42}$$

$$\{J_1[n], J_2[m]\} = \frac{1}{2}[P, J_1[n+m] - J_2[n+m]], \tag{1.43}$$

where $P = \sum_{\alpha,\beta\in\mathbb{Z}_\ell} e_{\alpha\beta} \otimes e_{\beta\alpha}$ is the permutation operator on $\mathbb{C}^\ell \otimes \mathbb{C}^\ell$. Now, the important insight is that these Poisson brackets are the defining Poisson brackets of the classical limit of the Yangian $Y(\mathfrak{gl}_\ell)$ and the loop algebra $L(\mathfrak{gl}_\ell)$. More precisely, $T(z)$ and $J[n]$ generate the $N$-truncated quotients thereof, which comes from the fact that $T(z)$ and $J[n]$ are defined in terms of $N \times N$ matrices. Moreover, we find that

$$\operatorname{Tr} J[n] = \operatorname{Tr} cg^n a = \operatorname{Tr} acg^n = -\eta H[n], \tag{1.44}$$

where we used the momentum map equation for the last equality. The Hamiltonians of the rational spin Ruijsenaars–Schneider model therefore come from the *center* of the loop algebra $L(\mathfrak{gl}_\ell)$.

At this point, it was natural to conjecture that the $N$-truncated Yangian and loop algebra combine into a larger algebra which is the $N$-truncated loop Yangian $LY(\mathfrak{gl}_\ell)$. This conjecture was also motivated by the construction of other spin generalizations of Ruijsenaars–Schneider Hamiltonians from the quantum toroidal algebra [LPS22], whose rational degeneration is the loop Yangian. Through many conversations, it became known to the author that truncated shifted Yangians are examples of Coulomb branches of three-dimensional quantum field theories with $\mathcal{N} = 4$ supersymmetry. The natural continuation of the search for a Poisson structure of the elliptic spin Ruijsenaars–Schneider model thus was to have a closer look at Coulomb branches.

The paper [BDG15] offered very concrete insight into the Poisson structures defined by Coulomb branches. In particular, section 6 included a description of the momentum map of the enhanced topological symmetry $\mathfrak{sl}_\ell$ acting on the Coulomb branch of linear quiver gauge theories. Importantly, the $(\alpha, \beta)$-coefficients of the momentum map were expressed as sums of the form

$$\sum_{i^\alpha, i^{\alpha+1}, \dots, i^\beta} \frac{u^\alpha_{i^\alpha} \cdots u^\beta_{i^\beta}}{(\varphi^{\alpha+1}_{i^{\alpha+1}} - \varphi^\alpha_{i^\alpha}) \cdots (\varphi^\beta_{i^\beta} - \varphi^{\beta-1}_{i^{\beta-1}})} \tag{1.45}$$

that could be written as matrix products of $L$-operators of the form

$$L^\alpha = \sum_{i,j=1}^N \frac{u^{\alpha+1}_j}{\varphi^{\alpha+1}_j - \varphi^\alpha_i} e_{ij}, \tag{1.46}$$

resembling the Lax matrix of the (spinless) Ruijsenaars–Schneider model. Comparing these formulas to the ideas of [NT25] made it clear that such a construction should also be possible in the case of the necklace quiver. Constructed correctly, the momentum map for the necklace quiver then takes values in the loop algebra $L(\mathfrak{gl}_\ell)$, and the trace of the first mode should give the defining Hamiltonian $H[1]$ of the rational spin Ruijsenaars–Schneider model as in equation (1.44). From the knowledge of the Hamiltonian $H[1]$ and the identification $x_i = \varphi^0_i$, we may compute $f_{ii}$ from the equations of motion (1.28), which inspires an ansatz for the spin (co)vector variables $a^\alpha_i$ and $c^\alpha_i$. Again using the

equations of motion as a check, this ansatz finally yielded the correct identification of variables between the Coulomb branch variables and the variables of the rational spin Ruijsenaars–Schneider model.

After finding the correct identification of variables in the rational case, the next step was the extension to the trigonometric and elliptic cases, corresponding to the Coulomb branches constructed from equivariant $K$-theory and equivariant elliptic Borel–Moore homology instead of equivariant Borel–Moore homology. The extension to the trigonometric case followed swiftly, but the elliptic case remained resistant due to the necessity of a spectral parameter, which in the rational and trigonometric cases may simply be taken to infinity using the non-compactness of $\mathbb{C}$ and $\mathbb{C}^\times$. Only after a lot of careful matching to the references [ACF97; MOY21] could the identification of variables be found in the elliptic case. This identification contained a big surprise: The Poisson structure underlying the elliptic model includes dynamical inhomogeneities without which the Jacobi identity fails. Arguably, the apparent necessity of dynamical inhomogeneities is the crux of why the Poisson structure of the elliptic model has eluded us for over three decades. We hope that the requirement for dynamical inhomogeneities can be lifted in the future and a rigorous comparison with Krichever's symplectic structure can be made.

## 1.3 Structure of the thesis

# 2

# Mathematical background

**This chapter is dedicated to the introduction of the necessary mathematical background used in the body of this thesis. Section 2.1 is a mini primer on supersymmetry for mathematicians. Section 2.2 discusses various methods of reducing algebras by symmetries relevant to chapter 3. Specifically, we discuss Poisson reduction of a Poisson algebra by a Lie bialgebra, which in the special case of the trivial bialgebra structure reduces to ordinary Hamiltonian reduction, as well as quantum Hamiltonian reduction, which applies to Hopf algebras acting on non-commutative algebras. Section 2.3 discusses the construction of Coulomb branches following Braverman–Finkelberg–Nakajima relevant to chapter 4.**

## 2.1 Supersymmetric gauge theory

### 2.1.1 Vector multiplets and hypermultiplets

What was missing from the discussion of supersymmetric field theories in the introduction was a detailed description of the field content of supersymmetric field theories with eight supercharges, specifically the representation theory of the super Poincaré algebras in 3d, 4d, and 5d, which are the symmetries of the super Minkowski spacetimes $\mathbb{R}^{1,2|8}$, $\mathbb{R}^{1,3|8}$, and $\mathbb{R}^{1,4|8}$.

The general pattern that these super Lie algebras follow is explained in [Str87]. Firstly, it contains generators $J$ of the Lorentz algebra of the underlying Minkowski spacetime, as well as translations $P$. Secondly, it contains supercharges $Q$, which take values in the spinor representation of the Lorentz algebra and in the defining representation of an internal symmetry algebra with generators $R$, called the *R-symmetry*. Finally, there

are "central charges" $Z$, valued in a representation of the $R$-symmetry and commuting with everything else. The Lie superbracket follows the schematic form

$$[J,J] = J, \qquad [P,J] = P, \qquad [P,P] = 0, \tag{2.1}$$
$$[Q,J] = Q, \qquad [Q,P] = 0, \qquad [Q,Q]_+ = P + Z, \tag{2.2}$$
$$[R,R] = R, \qquad [J,R] = 0, \qquad [Q,R] = Q, \tag{2.3}$$
$$[Z,R] = Z, \qquad [J,Z] = [P,Z] = [Z,Z] = 0. \tag{2.4}$$

The $R$-symmetry group in 3d is $\mathrm{SU}(2)_H \times \mathrm{SU}(2)_C$, in 4d is $\mathrm{SU}(2)_R \times \mathrm{U}(1)_r$, and in 5d is $\mathrm{SU}(2)_R$.

In any such dimension, there are two particularly important representations that we want to focus on: *Vector multiplets* and *hypermultiplets*. In 5d, the *vector multiplet* comprises a gauge field $A$, a real-valued scalar field $\phi$, as well as fermionic fields[1]. On the other hand, the *hypermultiplet* consists of a scalar field $q$ valued in the defining representation of $\mathrm{SU}(2)_R$, as well as fermionic fields.

The field content of vector multiplets and hypermultiplets in 4d can be obtained by compactifying five dimensions on a circle $S^1$. This means that all fields are decomposed into Fourier modes along the compactification circle $S^1$. In particular, the zero mode of the gauge field $A = (A_0, \dots, A_4)$ of the 5d vector multiplet splits into a 4d gauge field $A_{4\mathrm{d}} = (A_0, \dots, A_3)$ and a 4d scalar field $A_4$, which combines with the real-valued scalar field $\phi$ to yield a complex scalar field

$$\varphi \coloneqq \phi + \mathrm{i}A_4. \tag{2.5}$$

In the presence of a compactification circle of non-zero radius, the large gauge group acts on $A_4$ as follows:

$$A_4 \mapsto A_4 + n/R, \qquad n \in \mathbb{Z}. \tag{2.6}$$

Gauge invariant combinations of the complex scalar field $\varphi$ thus become functions of $\exp(2\pi R\varphi)$. This mechanism will also be relevant later for the description of $K$-theoretic Coulomb branches. The 4d hypermultiplet remains a scalar field valued in the defining representation of $\mathrm{SU}(2)_R$.

We can instead also compactify the 5d field theory on a torus $T^2$ of non-zero volume with modular parameter $\tau$ (an elliptic curve). In this case, the zero mode of the 5d gauge field $A$ of the 5d vector multiplet splits into a 3d gauge field $A_{3\mathrm{d}} = (A_0, A_1, A_2)$ and a 3d complex scalar

$$\varphi \coloneqq A_3 + \tau A_4, \tag{2.7}$$

which in the presence of the compactification torus gets acted upon as

$$\varphi \mapsto \varphi + n + \tau m, \qquad n, m \in \mathbb{Z} \tag{2.8}$$

by the large gauge group. Gauge covariant combinations must thus be expressed in terms of theta functions $\theta(\varphi;\tau)$. The scalar field $\phi$ in the vector multiplet remains a scalar field, and the scalar in the 3d hypermultiplet are valued in the defining representation of $\mathrm{SU}(2)_H$.

[1] We will not need to discuss the fermionic (odd-graded) field content in detail for our purposes.

Therefore, we have the following recipe for constructing 5d $\mathcal{N} = 1$, 4d $\mathcal{N} = 2$, and 3d $\mathcal{N} = 4$ supersymmetric gauge theories. The gauge group—a choice of a compact Lie group $G_c$—gives rise to a massless vector multiplet valued in the adjoint representation of $G_c$. Given a complex representation $V$ of the complexified gauge group $G$, called the *matter representation*, we can then add possibly massive hypermultiplets—called *matter content*—valued in the cotangent representation $T^*V$, which is a *quaternionic* representation on which $\mathrm{SU}(2)_R$ or $\mathrm{SU}(2)_H$ act by unit quaternions. Given this field content, there is then an essentially unique action functional which is invariant under the action of the super-Poincaré and gauge group actions.

Of course, there are many more supersymmetric quantum field theories which do not fall under this construction, which however still retain some of the flavor of gauge theories. Many of these do not even have a description in terms of an action functional, including, but not restricted to, Argyres–Douglas theories [AD95], which do not possess natural weak-coupling descriptions. Computations in these theories are thus restricted to their non-perturbative content, such as the Seiberg–Witten torus fibration and Coulomb branches. The class of supersymmetric quantum field theories which will be discussed in this thesis and generically does not fall under the construction described above are the theories of class $\mathcal{S}$ and $\mathcal{S}_k$, which we briefly mentioned in the introduction, which are constructed by compactifying the 6d $\mathcal{N} = (2,0)$ superconformal field theory of type $A_{N-1}$ or the 6d $\mathcal{N} = (1,0)$ superconformal field theory of type $(A_{N-1}, A_{k-1})$, respectively, on a punctured Riemann surface. As discussed in the introduction, the Coulomb branches of class $\mathcal{S}$ theories have rigorous mathematical definitions in terms of Hitchin moduli spaces and character varieties.

### 2.1.2 Quiver gauge theories

Quiver gauge theories in 5d $\mathcal{N} = 1$, 4d $\mathcal{N} = 2$, or 3d $\mathcal{N} = 4$ are supersymmetric gauge theories whose gauge group and matter content are determined by the data of a quiver with a dimension vector as follows.

**Definition 2.1.** (i) An *(unframed) quiver* $Q = (Q_0, Q_1)$ is a set of nodes $Q_0$ and a set of arrows $Q_1$ together with *source* and *target maps* $s, t : Q_1 \to Q_0$. Diagrammatically, we draw nodes $\alpha \in Q_0$ as circles and arrows $a \in Q_1$ with $s(a) = \alpha$ and $t(a) = \beta$ as arrows from node $\alpha$ to node $\beta$.

(ii) A *dimension vector* for a quiver $Q$ is a tuple of positive integers $\boldsymbol{N} = (N_\alpha)_{\alpha \in Q_0}$.

(iii) The *compact* and *complexified gauge group* associated to a quiver $Q$ with dimension vector $\boldsymbol{N}$ are

$$G_c := \prod_{\alpha \in Q_0} \mathrm{U}(N_\alpha)_\alpha, \qquad G := \prod_{\alpha \in Q_0} \mathrm{GL}(N_\alpha)_\alpha, \tag{2.9}$$

where the subscripts are just labels to distinguish the gauge group factors.

(iv) The *matter representation* $V$ associated to a quiver $Q$ with dimension vector $\boldsymbol{N}$ is

$$V := \bigoplus_{a \in Q_1} V_{s(a)} \otimes V_{t(a)}^\vee, \tag{2.10}$$

where $V_\alpha$ is the defining representation of $\mathrm{GL}(N_\alpha)_\alpha$ and $V_\alpha^\vee$ is its dual.

In this way, the data of a quiver $Q$ and a dimension vector $\boldsymbol{N}$ define a supersymmetric gauge theory with gauge group $G_c$ (determining the vector multiplets, which have

to be valued in the adjoint representation of $G_c$) and hypermultiplets valued in the cotangent representation $T^*V$.

Quiver gauge theories generally provide a large class of tractably computable theories, which furnish a useful testing ground for establishing and testing conjectures. A particularly nice example of an invariant that can be computed simply from the data of the quiver is the moduli space of Higgs vacua of the quiver gauge theory, or *Higgs branch* for short. Concretely, it is the hyperkähler variety defined as the hyperkähler quotient which is the Hamiltonian reduction of (a certain stable locus of) $T^*V$ by the complexified gauge group $G$. The precise construction of *Hamiltonian reductions* is the topic of the next section.

## 2.2 Hamiltonian reduction

### 2.2.1 Poisson reduction

Suppose we are handed a Poisson manifold $(M, \{-,-\}_M)$, which we interpret as the phase space of a classical system, on which some symmetries act. Following [Sem85; AM94], the action of the symmetry may be formalized by the notion of Poisson–Lie groups:

**Definition 2.2.** (i) A *Poisson–Lie group* $(G, \{-,-\}_G)$ is a Lie group $G$ which is also a Poisson manifold with Poisson bracket $\{-,-\}_G$, such that the multiplication map $G \times G \to G$ is a map of Poisson manifolds.

(ii) Any Poisson–Lie group $(G, \{-,-\}_G)$ defines a *Lie bialgebra* $(\mathfrak{g}, \delta)$, where $\mathfrak{g}$ is the Lie algebra of $G$ and $\delta : \mathfrak{g} \to \mathfrak{g} \wedge \mathfrak{g}$ is a 1-cocycle called the *cobracket* defined by the differential of the bivector $\{-,-\}_G$ at the identity, equipping $\mathfrak{g}^*$ with the structure of a Lie algebra.

(iii) The connected, simply connected Lie group $G^*$ whose Lie algebra is $\mathfrak{g}^*$ is called the *dual Poisson–Lie group*.

*Example* 2.3. (i) Every Lie group $G$ becomes a Poisson–Lie group by endowing it with the trivial Poisson bracket $\{-,-\}_G = 0$. Its dual Poisson–Lie group $G^*$ is the abelian Lie group that comes from exponentiating the abelian group structure on $\mathfrak{g}^*$, meaning $\exp : (\mathfrak{g}^*, +) \to (G^*, \cdot)$ is an isomorphism of Lie groups.

(ii) Let $G$ be a Lie group with Lie algebra $\mathfrak{g}$. Given an element $r \in \mathfrak{g} \wedge \mathfrak{g}$ satisfying the modified classical Yang–Baxter equation

$$[r_{12}, r_{13}] + [r_{13}, r_{23}] + [r_{12}, r_{23}] + [s_{12}, s_{13}] + [s_{13}, s_{23}] + [s_{12}, s_{23}] = 0, \tag{2.11}$$

for some $s \in S^2(\mathfrak{g})^{\mathfrak{g}}$, we can endow $G$ with the *Sklyanin bracket*

$$\{g, h\}_G = [r, g \otimes h], \tag{2.12}$$

where the commutator should be understood as $r$ acting via the infinitesimal action coming from the adjoint action. The Sklyanin bracket turns $G$ into a Poisson–Lie group. The cobracket $\delta$ of the corresponding Lie bialgebra is the coboundary of $r$, defined by

$$\delta : \mathfrak{g} \to \mathfrak{g} \wedge \mathfrak{g}, \qquad \xi \mapsto [r, \xi \otimes 1 + 1 \otimes \xi]. \tag{2.13}$$

Importantly, the combinations $r_\pm := \frac{1}{2}(r \pm s)$ induce natural maps $r_\pm^\# : \mathfrak{g}^* \to \mathfrak{g}$ and

$$\bar{\iota} : \mathfrak{g}^* \to \mathfrak{g} \oplus \mathfrak{g}, \qquad x \mapsto (r_+^\#(x), r_-^\#(x)), \tag{2.14}$$

$$\bar{\sigma} : \mathfrak{g} \oplus \mathfrak{g} \to \mathfrak{g}, \qquad (x_+, x_-) \mapsto x_+ - x_-. \tag{2.15}$$

which integrate to maps of Lie groups

$$\iota : G^* \to G \times G, \qquad g \mapsto (g_+, g_-) \tag{2.16}$$

$$\sigma : G \times G \to G, \qquad (g_+, g_-) \mapsto g_+ g_-^{-1}. \tag{2.17}$$

The Lie bialgebra $\mathfrak{g}$ is *factorizable* if $s^\# = \bar{\sigma} \circ \bar{\iota}$ is an isomorphism, making $\sigma \circ \iota$ a local isomorphism (giving the factorization $g = g_+ g_-^{-1}$). In the connected, simply connected case, this exhibits $G$ as a model for the dual Poisson–Lie group $G^*$ with the modified multiplication law

$$g \star h := g_+ h g_-^{-1}. \tag{2.18}$$

In this model of $G^*$, the Poisson bracket $\{-,-\}_{G^*}$ is given by

$$\{g_\pm, h_\pm\}_{G^*} = [r, g_\pm \otimes h_\pm] \tag{2.19}$$

$$\{g_\pm, h_\mp\}_{G^*} = [r_\pm, g_\pm \otimes h_\mp] \tag{2.20}$$

$$\{g, h\}_{G^*} = r(g \otimes h) + (g \otimes h)r + (g \otimes 1)r_-(1 \otimes h) + (1 \otimes h)r_+(g \otimes 1), \tag{2.21}$$

where $r$ standing to the left or the right indicates that it acts by the infinitesimal action coming from left or right translation on the group.

(iii) Take $G = \mathrm{GL}_N$ and $\mathfrak{g} = \mathfrak{gl}_N$ and let

$$r = \sum_{i,j=1}^{N} \mathrm{sign}(i-j) e_{ij} \otimes e_{ji}, \qquad s = \sum_{i,j=1}^{N} e_{ij} \otimes e_{ji}. \tag{2.22}$$

Then $r$ and $s$ together satisfy the modified classical Yang–Baxter equation. Under the identification $\mathfrak{gl}_N^* \cong \mathfrak{gl}_N$ coming from the trace pairing, we see that $\bar{\iota}$ becomes the projection onto the upper and lower triangular Borel subalgebras $\mathfrak{b}_\pm$ (sharing the diagonal equally), which integrate to the upper and lower triangular Borel subgroups $B_\pm$ of $\mathrm{GL}_N$. The factorization property becomes the Gauss decomposition of the form

$$L = L_+ L_-^{-1}, \qquad \mathrm{diag}(L_+)\,\mathrm{diag}(L_-) = 1, \qquad L_\pm \in B_\pm \tag{2.23}$$

and the Poisson bracket defined by (2.19)–(2.21) is the classical limit of the RLL presentation of the quantum group $U_q(\mathfrak{gl}_N)$ [FRT89; RS88].

**Definition 2.4.** (i) A Lie group action $G \times M \to M$ of a Poisson–Lie group $(G, \{-,-\}_G)$ on a Poisson manifold $(M, \{-,-\}_M)$ is a *Poisson–Lie action* if it is a map of Poisson manifolds.

(ii) A Poisson–Lie action with infinitesimal action $\rho : \mathfrak{g} \to \Gamma(TM)$ is called *Hamiltonian* if there exists a *momentum map* $\mathcal{M} : M \to G^*$ satisfying

$$\langle \{\mathcal{M}, f\}_M \star \mathcal{M}^{-1}, \xi \rangle = \rho(\xi), \qquad f \in C^\infty(M),\ \xi \in \mathfrak{g}. \tag{2.24}$$

Importantly, this definition makes use of the modified product $- \star -$ in the dual Poisson–Lie group $G^*$.

(iii) Given any symplectic leaf $C$ of the dual Poisson–Lie group $G^*$, the Poisson reduction of a Hamiltonian Poisson–Lie action with momentum map $\mathcal{M}: M \to G^*$ is defined to be

$$M /\!\!/ G \coloneqq \mathcal{M}^{-1}(C)/G. \tag{2.25}$$

*Example* 2.5. (i) A Hamiltonian Poisson–Lie action $G \times M \to M$ on a Poisson manifold $(M, \{-,-\}_M)$ where $G$ is endowed with the trivial Poisson bracket $\{-,-\}_G = 0$ has a momentum map which can be written as $\mathcal{M} = \exp(\mu)$ for a map $\mu: M \to \mathfrak{g}^*$. The momentum map condition (2.24) then becomes

$$\langle \{\mu, f\}_M, \xi \rangle = \rho(\xi). \tag{2.26}$$

In this case, symplectic leaves of $G^*$ correspond to coadjoint orbits $\mathcal{O}$ of $\mathfrak{g}^*$ and the Poisson reduction reduces to ordinary Hamiltonian reduction

$$M /\!\!/ G \coloneqq \mu^{-1}(\mathcal{O})/G. \tag{2.27}$$

(ii) To make (i) more concrete, consider a particle moving freely in the plane $\mathbb{R}^2$. Its phase space is the symplectic manifold $T^*\mathbb{R}^2$ with coordinates $x, y$ and momenta $p_x, p_y$, with the non-zero Poisson brackets of $\mathfrak{A} \coloneqq C^\infty(T^*\mathbb{R}^2)$ coming from

$$\{x, p_x\} = 1, \qquad \{y, p_y\} = 1. \tag{2.28}$$

There is a Hamiltonian action of $\mathrm{SO}(2)$ on $\mathbb{R}^2$ by rotations

$$(x, y) \mapsto (\cos(\theta)x + \sin(\theta)y, \cos(\theta)y - \sin(\theta)x) \tag{2.29}$$

of angle $\theta$ whose momentum map is

$$\mu: S(\mathfrak{so}(2)) \to \mathfrak{A}, \qquad t \mapsto xp_y - yp_x, \tag{2.30}$$

where $t$ is defined by the isomorphism $S(\mathfrak{so}(2)) \cong \mathbb{R}[t]$. The combination $xp_y - yp_x$ is also known as *angular momentum*, which is where the name "momentum map" stems from. Moving to polar coordinates $(r, \theta)$, the Hamiltonian becomes $H_{\text{free}} = p_x^2 + p_y^2 = p_r^2 + \frac{1}{r^2}p_\theta^2$ and the momentum map becomes $\mu(t) = p_\theta$, so $\mathfrak{A}^{\mathfrak{g}}$ is the algebra of functions in $p_\theta, p_r$ and $r$. Plugging the Poisson ideal $I$ generated by $t - \eta$ into the Hamiltonian reduction gives $\mathfrak{A} /\!\!/ \mathrm{SO}(2) = C^\infty(T^*(0,\infty))$ with Hamiltonian

$$H_{\text{red}} = p_r^2 + \frac{\eta^2}{r^2}. \tag{2.31}$$

This phase space and Hamiltonian define the rational Calogero–Moser model of two particles on a line in the center-of-mass frame. In this interpretation, $r$ is the relative separation between the particles, which is always positive, since the first particle cannot overtake the other.

(iii) Here, we give a table of integrable $N$-body systems and how they arise from Poisson (or Hamiltonian) reduction by an action of $\mathrm{GL}_N$ either with the trivial Poisson bracket or the one from example 2.3 (iii):

| Integrable model | obtained from | reference |
|---|---|---|
| Rational CM | $T^*\mathfrak{gl}_N$ | [KKS78] |
| Trigonometric CM | $T^*\mathfrak{gl}_N(t)$ | [GN94a] |
| Trigonometric CM | $T^*\mathrm{GL}_N$ | [KKS78] |
| Elliptic CM | $T^*\mathfrak{gl}_N(z,\bar{z})$ | [GN94b] |
| Rational RS | $T^*\mathrm{GL}_N$ | [Foc+00] |
| Trigonometric RS | Heisenberg double of $\mathrm{GL}_N$ | [FK09] |
| Trigonometric RS | $T^*\mathrm{GL}_N(t)$ | [GN95] |
| Elliptic RS | $T^*\mathrm{GL}_N(z,\bar{z})$ | [AFM97a] |
| Elliptic RS | Affine Heisenberg double of $\mathrm{GL}_N$ | [AFM97b] |

**Table 2.1:** Integrable models obtained from Poisson reduction of various spaces.

### 2.2.2 Quantum Hamiltonian reduction

Quantum Hamiltonian reduction seeks to generalize the idea of Poisson reduction of a Poisson manifold to deformation quantizations thereof.

**Definition 2.6.** An *(algebraic) deformation quantization* $\mathfrak{A}_\hbar$ of a Poisson algebra $\mathfrak{A}_0$ is a possibly non-commutative algebra $\mathfrak{A}_\hbar$ over $\mathbb{C}[\hbar]$ such that

(i) $\mathfrak{A}_\hbar \cong \mathfrak{A}_0[\hbar]$ as vector spaces,
(ii) $[\mathfrak{A}_\hbar, \mathfrak{A}_\hbar] \subseteq \hbar\mathfrak{A}_\hbar$,
(iii) $\mathfrak{A}_\hbar/\hbar\mathfrak{A}_\hbar \cong \mathfrak{A}_0$ as Poisson algebras with the induced Poisson bracket

$$\{f, g\} \coloneqq \frac{1}{\hbar}[f, g] + \hbar\mathfrak{A}_\hbar. \tag{2.32}$$

*Remark* 2.7. Similarly, we define (algebraic) deformation quantizations of Poisson algebras $\mathfrak{A}_1$ which are algebras $\mathfrak{A}_q$ over the ring $\mathbb{C}[q^{\pm 1}]$.

The notion of a Poisson–Lie group then has to be replaced by the notion of a *Hopf algebra* over $\mathbb{C}[\hbar]$:

**Definition 2.8.** A *Hopf algebra* $H_\hbar$ over $\mathbb{C}[\hbar]$ is an algebra over $\mathbb{C}[\hbar]$ together with $\mathbb{C}[\hbar]$-linear maps $\Delta : H_\hbar \to H_\hbar \otimes H_\hbar$ and $\epsilon : H_\hbar \to \mathbb{C}[\hbar]$ called the *coproduct* and *counit* such that the following holds:

(i) (Coassociativity) $(\Delta \otimes 1) \circ \Delta = (1 \otimes \Delta) \circ \Delta$.
(ii) (Counitality) $(\epsilon \otimes 1) \circ \Delta = (1 \otimes \epsilon) \circ \Delta = 1$,
(iii) (Bialgebra property) $\Delta$ and $\epsilon$ are algebra homomorphisms.
(iv) (Existence of antipode) There exists an algebra antihomomorphism $S : H_\hbar \to H_\hbar$ such that

$$\mu \circ (S \otimes 1) \circ \Delta = \eta \circ \epsilon = \mu \circ (1 \otimes S) \circ \Delta, \tag{2.33}$$

where $\mu : H \otimes H \to H$ and $\eta : \mathbb{C}[\hbar] \to H$ are the multiplication and unit maps of $H$.

In *Sweedler notation*, the coproduct is denoted by

$$h_{(1)} \otimes h_{(2)} \coloneqq \Delta(h), \tag{2.34}$$

where it is understood that the left hand side may generally be a sum over pure tensors.

*Remark* 2.9. If $H_\hbar$ is a deformation quantization of a Poisson algebra $H_0$, then $H_0$ defines an affine complex algebraic group Spec $H_0$ whose algebra of regular functions is the Poisson algebra $H_0$. This is the algebro-geometric analog of a Poisson–Lie group.

*Example* 2.10. (i) The algebra of regular functions $\mathcal{O}(G)$ on an affine complex algebraic group $G$ always defines a Hopf algebra with the coproduct $\Delta$, the antipode $S$, and the counit $\epsilon$ coming from the algebraic group structure, i.e.

$$\operatorname{Spec}\Delta:\ G\times G\to G,\qquad \operatorname{Spec}S:\ G\to G,\qquad \operatorname{Spec}\epsilon:\ \mathrm{pt}\to G \tag{2.35}$$

are the multiplication, inverse, and unit maps. Since the algebra of regular functions $\mathcal{O}(G)$ is commutative, the Poisson algebra it quantizes has trivial Poisson bracket, reproducing example 2.3 (i).

(ii) Let $\mathfrak{g}$ be a complex Lie algebra. Define $\mathfrak{g}_\hbar$ to be the Lie algebra $\mathfrak{g}[\hbar]$ over $\mathbb{C}[\hbar]$ with the modified bracket

$$[\xi,\xi']_{\mathfrak{g}_\hbar}\coloneqq\hbar[\xi,\xi']_{\mathfrak{g}},\qquad \xi,\xi'\in\mathfrak{g}. \tag{2.36}$$

Then $U(\mathfrak{g}_\hbar)$ is a deformation quantization of the Poisson algebra $S(\mathfrak{g})$ and has the structure of a Hopf algebra over $\mathbb{C}[\hbar]$ with

$$\Delta:\ U(\mathfrak{g}_\hbar)\to U(\mathfrak{g}_\hbar)\otimes U(\mathfrak{g}_\hbar),\qquad \xi\in\mathfrak{g}\mapsto\xi\otimes 1+1\otimes\xi, \tag{2.37}$$

$$S:\ U(\mathfrak{g}_\hbar)\to U(\mathfrak{g}_\hbar),\qquad \xi\in\mathfrak{g}\mapsto-\xi, \tag{2.38}$$

$$\epsilon:\ U(\mathfrak{g}_\hbar)\to\mathbb{C}[\hbar],\qquad \xi\in\mathfrak{g}\mapsto 0. \tag{2.39}$$

(iii) Following [FRT89; AFS92], we may present the Drinfeld–Jimbo quantum group $U_q(\mathfrak{gl}_N)$ (a Hopf algebra over $\mathbb{C}[q^{\pm1}]$) by generators $l_\pm^{ij}$ for $i,j=1,\dots,N$ satisfying

$$l_\pm^{ij}=0,\qquad i\lessgtr j,\qquad l_\pm^{ii}l_\mp^{ii}=1. \tag{2.40}$$

Defining the matrices

$$L_\pm=\sum_{i,j=1}^{N}l_\pm^{ij}\otimes e_{ij}\in U_q(\mathfrak{gl}_N)\otimes\operatorname{End}(\mathbb{C}^N), \tag{2.41}$$

the commutation relations between the generators can then be concisely written in RLL form quantizing the Poisson brackets (2.19) and (2.20):

$$R(L_\pm)_1(L_\pm)_2=(L_\pm)_2(L_\pm)_1R \tag{2.42}$$

$$R(L_-)_1(L_+)_2=(L_+)_2(L_-)_1R \tag{2.43}$$

with

$$R\coloneqq q\sum_{i=1}^{N}e_{ii}\otimes e_{ii}+\sum_{i\neq j}^{N}e_{ii}\otimes e_{jj}+(q-q^{-1})\sum_{i>j}^{N}e_{ij}\otimes e_{ji}. \tag{2.44}$$

The Hopf algebra structure of $U_q(\mathfrak{gl}_N)$ is given by

$$\Delta:\ U_q(\mathfrak{gl}_N)\to U_q(\mathfrak{gl}_N)\otimes U_q(\mathfrak{gl}_N),\qquad l_\pm^{ij}\mapsto\sum_{k=1}^{N}l_\pm^{ik}\otimes l_\pm^{kj} \tag{2.45}$$

$$S:\ U_q(\mathfrak{gl}_N)\to U_q(\mathfrak{gl}_N),\qquad L_\pm\mapsto L_\pm^{-1} \tag{2.46}$$

$$\epsilon:\ U_q(\mathfrak{gl}_N)\to\mathbb{C}[q^{\pm1}],\qquad l_\pm^{ij}\mapsto\delta^{ij} \tag{2.47}$$

Recall from the previous section that a Hamiltonian Poisson–Lie action is characterized by its momentum map $\mathcal{M}: M \to G^*$, which is a map of Poisson manifolds. In algebraic geometry, this corresponds to a map between the Poisson algebras of regular functions $\mathcal{O}(G^*) \to \mathcal{O}(M)$. The analog of this map in the quantum case is a *quantum momentum map*. We now proceed to define the quantum Hamiltonian reduction given a quantum momentum map. A useful review can be found in the lecture notes [Eti09].

**Definition 2.11.** A *quantum momentum map* is an algebra map $\mu: H_\hbar \to \mathfrak{A}_\hbar$ from a Hopf algebra $H_\hbar$ over $\mathbb{C}[\hbar]$ to a possibly non-commutative $\mathbb{C}[\hbar]$-algebra $\mathfrak{A}_\hbar$. The *adjoint action* of $h \in H_\hbar$ on $f \in \mathfrak{A}_\hbar$ is defined by

$$\mathrm{ad}_h f := \mu(h_{(1)}) f \mu(S(h_{(2)})), \tag{2.48}$$

where we have used the antipode $S$ and Sweedler notation for the coproduct. We denote the invariants under the adjoint action by

$$\mathfrak{A}_\hbar^{\mathrm{ad}} := \{f \in \mathfrak{A}_\hbar \mid \mathrm{ad}_h f = \epsilon(h) f,\ \forall h \in H\} \tag{2.49}$$

**Definition 2.12.** Let $\mu: H_\hbar \to \mathfrak{A}_\hbar$ be a quantum momentum map and $I$ a two-sided ideal of $H_\hbar$. The *quantum Hamiltonian reduction* of $\mathfrak{A}_\hbar$ is defined by

$$\mathfrak{A}_\hbar /\!\!/ H_\hbar := (\mathfrak{A}_\hbar / \mathfrak{A}_\hbar \cdot \mu(I))^{\mathrm{ad}}. \tag{2.50}$$

**Proposition 2.13** ([BFG21]). *The algebra structure of $\mathfrak{A}_\hbar$ descends to a well-defined algebra structure of $\mathfrak{A}_\hbar /\!\!/ H_\hbar$.*

The main property that makes quantum Hamiltonian reduction useful to us is that it provides deformation quantizations of Poisson algebras obtained by Poisson reduction. This idea is summarized by the slogan *"Quantization commutes with reduction."* Let us illustrate this principle by quantizing example 2.5 (ii) using quantum Hamiltonian reduction.

*Example* 2.14. Let $\mathfrak{A}_\hbar$ be the algebra of differential operators on $C^\infty(\mathbb{R}^2)$ generated by smooth functions in the coordinates $x, y$ as well as the derivations $p_x := -\mathrm{i}\hbar\partial_x$, $p_y := -\mathrm{i}\hbar\partial_y$ with the commutators

$$[p_x, f(x,y)] = -\mathrm{i}\hbar\partial_x f(x,y), \qquad [p_y, f(x,y)] = -\mathrm{i}\hbar\partial_y f(x,y). \tag{2.51}$$

The quantum momentum map of the action of SO(2) on the plane $\mathbb{R}^2$ is

$$\mu: U(\mathfrak{so}(2)_\hbar) \to \mathfrak{A}_\hbar, \qquad t \mapsto x p_y - y p_x. \tag{2.52}$$

Again, moving to polar coordinates $(r, \theta)$, we find $\mu(t) = p_\theta := -\mathrm{i}\hbar\partial_\theta$, so that the algebra of invariants $\mathfrak{A}_\hbar^{\mathrm{ad}}$ is given by those differential operators whose coefficients do not depend on $\theta$. Taking the two-sided ideal $I$ generated by $t - \eta$ and modding out by it yields the algebra $\mathfrak{A}_\hbar /\!\!/ H_\hbar$ of differential operators on $C^\infty((0,\infty))$ generated by smooth functions in $r$ as well as the derivation $p_r := -\mathrm{i}\hbar\partial_r$ with Hamiltonian

$$H_{\mathrm{red}} = p_r^2 - \frac{\mathrm{i}\hbar}{r} p_r + \frac{\eta^2}{r^2}. \tag{2.53}$$

Conjugating with $r^{1/2}$ gives the Hamiltonian operator of the quantum rational Calogero–Moser model:

$$r^{1/2} H_{\mathrm{red}} r^{-1/2} = p_r^2 + \frac{(\eta - \frac{\hbar}{2})(\eta + \frac{\hbar}{2})}{r^2}. \tag{2.54}$$

This quantizes the classical Hamiltonian we found in example 2.5 (ii). Note the anomalous shift of the coupling constant by $\pm\frac{\hbar}{2}$.

## 2.3 Coulomb branches

As discussed in section 1.1 of the introduction, the Coulomb branch of a 3d $\mathcal{N} = 4$ gauge theory—with compact gauge group $G_c$ and matter in the hypermultiplet representation $T^*V$ for some representation $V$ of the complexification $G$ of $G_c$—is the moduli space of vacua where the scalars inside the vector multiplets acquire non-zero expectation values.

Classically speaking, the Coulomb branch is described by the quotient $(\mathbb{R}^3 \times S^1)^{\mathrm{rk}\, G}/W$, parametrized by the three real scalars of the vector multiplet together with the dual photon of section 1.1 modulo the action of the Weyl group $W$. Quantum mechanically, this description has to be corrected, and finding the correct algebra of functions on the Coulomb branch was a long-standing open problem. A leap of progress was made when Cremonesi–Hanany–Zaffaroni found the *monopole formula* [CHZ14], which gives the Hilbert series of the algebra of functions on the Coulomb branch. However, the monopole formula is only a character, and says little about the Coulomb branch as a variety. Subsequently, the construction of Braverman, Finkelberg, Nakajima (BFN), and others supplied the algebra structure, recovering the monopole formula as a Hilbert series.

The BFN construction employs the machinery of equivariant Borel–Moore homology of certain infinite-dimensional spaces to solve this problem. The following subsections are meant to familiarize the reader with the basics of their construction.

### 2.3.1 Equivariant Borel–Moore homology

Borel–Moore homology [BM60] is a homology theory that extends the properties of singular homology of compact spaces to locally compact spaces. One of its defining features is that it has a cap product with singular cohomology

$$\cap : \; H_n^{\mathrm{BM}}(X) \otimes H^k(X) \to H_{n-k}^{\mathrm{BM}}(X), \tag{2.55}$$

which induces a Poincaré duality isomorphism for (possibly non-compact!) oriented manifolds with $n = \dim_{\mathbb{R}} X$. Importantly, *equivariant* Borel–Moore homology [AF23] defines a useful extension of equivariant cohomology for which pushforwards exist for all proper maps. This makes it possible to construct convolution products in equivariant Borel–Moore homology. A review of Borel–Moore homology, convolution products, and their uses in geometric representation theory can be found in [Gin98]. For applications to Coulomb branches, we will be particularly interested in spaces related to the affine Grassmannian, which is reviewed in [Zhu16].

**Definition 2.15.** Let $G$ be a complex reductive algebraic group acting on a complex variety $X$. The *equivariant cohomology* and *equivariant Borel–Moore homology* are defined by

$$H_G^\bullet(X) \coloneqq H^\bullet(EG \times^G X), \qquad H_G^{\mathrm{BM}}(X) \coloneqq H^{\mathrm{BM}}(EG \times^G X), \tag{2.56}$$

where $EG$ is the universal principal $G$-bundle on which $G$ acts from the right.

*Remark* 2.16. Typically, $EG$ is an infinite-dimensional space that has to be approximated by finite-dimensional spaces to obtain a well-defined Borel–Moore homology. We will sweep this subtlety under the rug for the purposes of a shortened exposition.

**Proposition 2.17** ([AF23]). *Let $G$ be a complex reductive algebraic group acting on a non-singular complex variety $X$.*

(i) *The equivariant Borel–Moore homology of a point is*

$$H_G^{\mathrm{BM}}(\mathrm{pt}) = H^{\mathrm{BM}}(BG) = H^\bullet(BG) = S(\mathfrak{g}^*)^G. \tag{2.57}$$

*The natural ring structure comes from the cup product of equivariant cohomology.*

(ii) *The unique map* $\varpi_X : X \to \mathrm{pt}$ *defines a pullback map in equivariant cohomology*

$$\varpi_X^* : H_G^{\mathrm{BM}}(\mathrm{pt}) = H_G^\bullet(\mathrm{pt}) \to H_G^\bullet(X) \tag{2.58}$$

*that together with the cap product*

$$\cap : H_G^{\mathrm{BM}}(X) \otimes H_G^\bullet(X) \to H_G^{\mathrm{BM}}(X) \tag{2.59}$$

*endows* $H_G^{\mathrm{BM}}(X)$ *with the structure of a right* $H_G^{\mathrm{BM}}(\mathrm{pt})$*-module.*

(iii) *Let* $x \in X^G$ *be a fixed point with embedding* $\iota_x : \mathrm{pt} \hookrightarrow X$*. This defines pullback and pushforward maps in equivariant Borel–Moore homology:*

$$\iota_{x*} : H_G^{\mathrm{BM}}(\mathrm{pt}) \to H_G^{\mathrm{BM}}(X), \qquad \iota_x^* : H_G^{\mathrm{BM}}(X) \to H_G^{\mathrm{BM}}(\mathrm{pt}). \tag{2.60}$$

*The fundamental class of a fixed point is defined by* $[x] \coloneqq \iota_{x*}1$*.*

(iv) *Given a fixed point* $x \in X^G$*, the map* $\iota_x^* \circ \iota_{x*} : H_G^{\mathrm{BM}}(\mathrm{pt}) \to H_G^{\mathrm{BM}}(\mathrm{pt})$ *corresponds to multiplication by an element* $\mathrm{eu}_G(T_xX)$ *of* $H_G^{\mathrm{BM}}(\mathrm{pt})$ *called the* equivariant Euler class *of the tangent bundle at* $x$*.*

(v) *Let* $G = T$ *be a torus and* $x \in X^T$ *a fixed point. Let* $V$ *be a vector bundle over* $X$*. The fiber* $\iota_x^* V$ *of* $V$ *over* $x$ *decomposes into 1-dimensional weight spaces* $L_{\varphi_k}$ *of weight* $\varphi_k$ *for* $k = 1, \dots, r$ *and its equivariant Euler class factorizes accordingly:*

$$\iota_x^* V = \bigoplus_{k=1}^r L_{\varphi_k}, \qquad \mathrm{eu}_T(\iota_x^* V) = \prod_{k=1}^r \varphi_k. \tag{2.61}$$

**Definition 2.18.** Let $G$ be a complex reductive algebraic group acting on a complex variety $X$. The *localized equivariant Borel–Moore homology* is defined by

$$H_G^{\mathrm{BM}}(X)_{\mathrm{loc}} \coloneqq H_G^{\mathrm{BM}}(X) \otimes_{H_G^{\mathrm{BM}}(\mathrm{pt})} \mathrm{Frac}(H_G^{\mathrm{BM}}(\mathrm{pt})), \tag{2.62}$$

where $\mathrm{Frac}(H_G^{\mathrm{BM}}(\mathrm{pt}))$ is the fraction field of $H_G^{\mathrm{BM}}(\mathrm{pt}) = S(\mathfrak{g}^*)^G$.

**Theorem 2.19** (Localization theorem [AF23])**.** *Let* $T$ *be a complex torus acting on a complex variety* $X$*. The inclusion* $\iota : X^T \to X$ *yields a pushforward map*

$$\iota_* : \bigoplus_{x \in X^T} H_T^{\mathrm{BM}}(\{x\}) \to H_T^{\mathrm{BM}}(X) \tag{2.63}$$

*which is an isomorphism after localization. If* $X^T$ *consists of finitely many and non-singular points, then* $\iota^* \circ \iota_* = \mathrm{eu}_T$ *is the componentwise multiplication by the Euler class of the tangent bundle. In particular,* $\mathrm{eu}_T^{-1}\, \iota_*$ *is the inverse of* $\iota^*$ *on localized equivariant Borel–Moore homology:*

$$\iota^* : H_T^{\mathrm{BM}}(X)_{\mathrm{loc}} \xrightarrow{\sim} \bigoplus_{x \in X^T} \mathrm{Frac}(H_T^{\mathrm{BM}}(\{x\})) : \mathrm{eu}_T^{-1}\, \iota_*. \tag{2.64}$$

**Corollary 2.20.** *Let $X$ and $Y$ be complex varieties on which a complex torus $T$ acts with finitely many and non-singular fixed points. Let $f : X \to Y$ be a smooth equivariant map and let $\iota_x$ denote the equivariant inclusions of fixed points. The pullback and pushforward along $f$ may be described in terms of fixed point data as follows:*

*(i) Let $c \in H_T^{\mathrm{BM}}(Y)_{\mathrm{loc}}$. Then $\iota_x^* f^* c = \iota_{f(x)}^* c$ for all fixed points $x \in X^T$.*

*(ii) Let $c \in H_T^{\mathrm{BM}}(X)_{\mathrm{loc}}$. Then $\iota_y^* f_* c = \sum_{x \in f^{-1}(y)^T} \frac{\mathrm{eu}_T(T_y Y)}{\mathrm{eu}_T(T_x X)} \iota_x^* c$ for all fixed points $y \in Y^T$.*

*Proof.* We only use functoriality of pullback and pushforward. Case (i) is immediate. In case (ii), we have

$$\iota_y^* f_* c = \iota_y^* \sum_{x \in X^T} f_* \iota_{x*} \mathrm{eu}(T_x X)^{-1} \iota_x^* c = \iota_y^* \sum_{x \in X^T} \iota_{f(x)*} \mathrm{eu}(T_x X)^{-1} \iota_x^* c \tag{2.65}$$

and

$$\iota_y^* \sum_{x \in X^T} \iota_{f(x)*} \mathrm{eu}(T_x X)^{-1} \iota_x^* c = \iota_y^* \sum_{y' \in Y^T} \iota_{y'*} \sum_{x \in f^{-1}(y')^T} \mathrm{eu}(T_x X)^{-1} \iota_x^* c = \sum_{x \in f^{-1}(y)^T} \frac{\mathrm{eu}(T_y Y)}{\mathrm{eu}(T_x X)} \iota_x^* c, \tag{2.66}$$

which shows (ii). □

*Example* 2.21. Consider a torus $T \coloneqq (\mathbb{C}^\times)^N$ and write $H_T^{\mathrm{BM}}(\mathrm{pt}) = \mathbb{C}[a_1, \dots, a_N]$ as well as $\mathrm{Frac}(H_T^{\mathrm{BM}}(\mathrm{pt})) = \mathbb{C}(a_1, \dots, a_N)$. Let $T$ act on $\mathbb{P}^{N-1}$ by

$$(t_1, \dots, t_N) \cdot [z_1 : \cdots : z_N] \coloneqq [t_1 z_1 : \cdots : t_N z_N]. \tag{2.67}$$

The tautological line bundle $\mathcal{O}(-1)$ over $\mathbb{P}^{N-1}$ is $T$-equivariant. Let $t \in H_T^\bullet(\mathbb{P}^{N-1}) = H_T^{\mathrm{BM}}(\mathbb{P}^{N-1})$ denote the first Chern class of its dual $\mathcal{O}(1)$. The fixed points of $\mathbb{P}^{N-1}$ under the action of $T$ are given by

$$\mathbb{C}e_i = [0 : \cdots : 0 : 1 : 0 : \cdots : 0], \qquad i = 1, \dots, N. \tag{2.68}$$

Since the fiber of $\mathcal{O}(-1)$ over $\mathbb{C}e_i$ is one-dimensional and has weight $a_i$ under $T$, we find that the pullback under the equivariant inclusion $\iota_{\mathbb{C}e_i} : \{\mathbb{C}e_i\} \to \mathbb{P}^{N-1}$ is given by

$$\iota_{\mathbb{C}e_i}^* : H_T^{\mathrm{BM}}(\mathbb{P}^{N-1}) \to H_T^{\mathrm{BM}}(\{\mathbb{C}e_i\}), \qquad t \mapsto -a_i, \tag{2.69}$$

where the minus sign comes from taking the dual. It follows that $\iota_{\mathbb{C}e_i}^*$ takes any polynomial $f(t) \in \mathbb{C}[a_1, \dots, a_N][t]$ to its evaluation at $t = -a_i$. Furthermore, the tangent space $T_{\mathbb{C}e_i}\mathbb{P}^{N-1}$ decomposes into $N-1$ one-dimensional weight spaces of weight $a_k - a_i$ for $k \neq i$, so

$$\mathrm{eu}_T(T_{\mathbb{C}e_i}\mathbb{P}^{N-1}) = \prod_{k \neq i}^{N} (a_k - a_i). \tag{2.70}$$

Combining these two facts, we see that $\iota_{\mathbb{C}e_i}^* \prod_{k \neq i} (t + a_k) = \mathrm{eu}_T(T_{\mathbb{C}e_i}\mathbb{P}^{N-1})$. The localization theorem thus implies $[\mathbb{C}e_i] = \iota_{\mathbb{C}e_i *} 1 = \prod_{k \neq i} (t + a_k)$. The isomorphism

$$\iota^* : H_T^{\mathrm{BM}}(\mathbb{P}^{N-1})_{\mathrm{loc}} \xrightarrow{\sim} \bigoplus_{i=1}^{N} \mathrm{Frac}(H_T^{\mathrm{BM}}(\mathbb{C}e_i)) : \ \mathrm{eu}_T^{-1}\, \iota_*. \tag{2.71}$$

is therefore nothing but the Lagrange interpolation isomorphism

$$\iota^* : \mathbb{C}(a_1, \dots, a_N)[t]/(\prod_{i=1}^{N}(t + a_i)) \xrightleftarrows{\sim} \bigoplus_{i=1}^{N} \mathbb{C}(a_1, \dots, a_N) : \ \mathrm{eu}_T^{-1}\, \iota_*, \tag{2.72}$$

where $\iota^*$ evaluates at the points $t = -a_1, \dots, -a_N$ and

$$\mathrm{eu}_T^{-1}\, \iota_*(f_1, \cdots, f_N) = \sum_{i=1}^{N} f_i \prod_{k \neq i}^{N} \frac{-t - a_k}{a_i - a_k}. \tag{2.73}$$

### 2.3.2 BFN construction of Coulomb branches

Let $G$ be a complex reductive algebraic group (the complexified gauge group) with maximal torus $T$ and $V$ a representation of $G$ (the matter representation). Let $\mathcal{O} \coloneqq \mathbb{C}[\![t]\!]$ and $\mathcal{K} \coloneqq \mathbb{C}(\!(t)\!)$. We further let $G_{\mathcal{O}} \coloneqq G(\mathcal{O})$ and $G_{\mathcal{K}} \coloneqq G(\mathcal{K})$ as well as $V_{\mathcal{O}} \coloneqq V \otimes \mathcal{O}$ and $V_{\mathcal{K}} \coloneqq V \otimes \mathcal{K}$. Note that $G_{\mathcal{K}}$ comes equipped with actions of $G_{\mathcal{O}}$ from the left and from the right.

**Definition 2.22.** The *affine Grassmannian* is the quotient $\mathrm{Gr}_G \coloneqq G_{\mathcal{K}}/G_{\mathcal{O}}$ on which $G_{\mathcal{O}}$ acts from the left.

*Example* 2.23. Let $G = \mathrm{GL}_N$. The columns of any matrix inside $G_{\mathcal{K}}$ define a set of $N$ elements of $\mathcal{K}^N$ which is linearly independent over $\mathcal{K}$, therefore linearly independent over $\mathcal{O}$, generating a free rank $N$ $\mathcal{O}$-submodule of $\mathcal{K}^N$. Free rank $N$ $\mathcal{O}$-submodules of $\mathcal{K}^N$ are also called *lattices* in $\mathcal{K}^N$. Acting with a matrix in $G_{\mathcal{O}}$ on the generating set of a lattice preserves the lattice. This establishes a bijection between the $\mathbb{C}$-points of $\mathrm{Gr}_G$ and the set of lattices in $\mathcal{K}^N$.

**Proposition 2.24** ([Zhu16]). *Let $X_*^+(T)$ be the set of dominant coweights. Then $\mathrm{Gr}_G$ splits into left $G_{\mathcal{O}}$-orbits indexed by dominant coweights:*

$$\mathrm{Gr}_G = \bigsqcup_{\lambda \in X_*^+(T)} \mathrm{Gr}_G^{\lambda}, \qquad \mathrm{Gr}_G^{\lambda} \coloneqq G_{\mathcal{O}} \cdot x^{\lambda}, \qquad x^{\lambda} \coloneqq t^{\lambda} G_{\mathcal{O}}, \tag{2.74}$$

*where $t^{\lambda}$ is the image of $t \in \mathcal{K}$ under the coweight $\lambda : \mathcal{K}^{\times} \to G_{\mathcal{K}}$.*

**Definition 2.25.** The *Schubert varieties* are defined by

$$\overline{\mathrm{Gr}_G^{\lambda}} \coloneqq \bigsqcup_{\mu \leq \lambda} \mathrm{Gr}_G^{\mu}, \tag{2.75}$$

where $\mu \leq \lambda$ denotes the dominance order on $X_*^+(T)$. In particular, for minuscule $\lambda$ (minimal elements with respect to the dominance order), we have $\overline{\mathrm{Gr}_G^{\lambda}} = \mathrm{Gr}_G^{\lambda}$.

**Proposition 2.26** ([Zhu16]). *The Schubert varieties $\overline{\mathrm{Gr}_G^{\lambda}}$ are projective varieties and the inductive limit $\varinjlim_{\lambda} \overline{\mathrm{Gr}_G^{\lambda}}$ exhibits $\mathrm{Gr}_G$ as an ind-projective scheme.*

*Remark* 2.27. Proposition 2.26 is the basis for any finite-dimensional approximations that are needed to define the Borel–Moore homological constructions rigorously. For the purposes of this thesis, we will sweep these technicalities under the rug. The interested reader can go back to the original reference [BFN18b] for details.

**Definition 2.28.** Let $G'$ be a group such that $G$ is a normal subgroup and $V$ extends to a representation of $G'$. The *flavor symmetry* is the group $F := G'/G$, which we assume to be a torus. Let

$$\tilde{G} := G'_{\mathcal{O}} \rtimes \mathbb{C}^\times_\hbar, \tag{2.76}$$

where $s \in \mathbb{C}^\times_\hbar$ acts on $\mathcal{O}$ by $t \mapsto st$ and $\hbar$ is the generator of $H^{\mathrm{BM}}_{\mathbb{C}^\times_\hbar}(\mathrm{pt}) = \mathbb{C}[\hbar]$. Then $\tilde{G}$ acts on $V_{\mathcal{O}}$ where $\mathbb{C}^\times_\hbar$ acts on $V$ with weight $\frac{\hbar}{2}$.

**Definition 2.29** ([BFN18b]). Equip $T_{G,V} := G_{\mathcal{K}} \times^{G_{\mathcal{O}}} V_{\mathcal{O}}$ with the natural action of $\tilde{G}$ from the left. There are $\tilde{G}$-equivariant maps

$$\rho : T_{G,V} \to V_{\mathcal{K}}, \qquad [g, v] \mapsto gv, \tag{2.77}$$
$$\pi : T_{G,V} \to \mathrm{Gr}_G, \qquad [g, v] \mapsto gG_{\mathcal{O}}. \tag{2.78}$$

The *BFN space* $R_{G,V}$ for gauge group $G$ and matter representation $V$ is the pullback

$$\begin{array}{ccc} R_{G,V} & \dashrightarrow & T_{G,V} \\ \vdots & & \downarrow \rho \\ V_{\mathcal{O}} & \hookrightarrow & V_{\mathcal{K}} \end{array}$$

We let $R^\lambda_{G,V} := \pi^{-1}(\mathrm{Gr}^\lambda_G) \cap R_{G,V}$ and $R^{\leq\lambda}_{G,V} := \pi^{-1}(\overline{\mathrm{Gr}^\lambda_G}) \cap R_{G,V}$.

*Remark* 2.30. The spaces $R^{\leq\lambda}_{G,V}$ serve as approximations for the full space $R_{G,V}$ in the same way $\overline{\mathrm{Gr}^\lambda_G}$ serve as approximations for $\mathrm{Gr}_G$.

**Proposition 2.31** ([BFN18b]). *The equivariant Borel–Moore homology $H^{\mathrm{BM}}_{\tilde{G}}(R_{G,V})$ comes equipped with a convolution product.*

*Proof sketch.* Concretely, one considers the convolution diagram

$$\begin{array}{ccccc} & & G_{\mathcal{K}} \times R_{G,V} & \xrightarrow{q} & G_{\mathcal{K}} \times^{G_{\mathcal{O}}} R_{G,V} \\ & \swarrow^{p} & & & \searrow^{m} \\ T_{G,V} \times R_{G,V} & & & & T_{G,V} \end{array}$$

where $T_{G,V} \times R_{G,V}$ comes equipped with the natural product $\tilde{G} \times \tilde{G}$-action and $G_{\mathcal{K}} \times^{G_{\mathcal{O}}} R_{G,V}$ and $R_{G,V}$ come equipped with the natural $\tilde{G}$-action from the left, while $G_{\mathcal{K}} \times R_{G,V}$ comes equipped with the $\tilde{G} \times \tilde{G}$-action

$$(h_1, h_2) \cdot (g_1, [g_2, v]) := (h_1 g_1 h_2^{-1}, [h_2 g_2, v]). \tag{2.79}$$

The maps in the diagram are defined by

$$p : G_{\mathcal{K}} \times R_{G,V} \to T_{G,V} \times R_{G,V}, \qquad (g_1, [g_2, v]) \mapsto ([g_1, g_2 v], [g_2, v]), \tag{2.80}$$
$$q : G_{\mathcal{K}} \times R_{G,V} \to G_{\mathcal{K}} \times^{G_{\mathcal{O}}} R_{G,V}, \qquad (g_1, [g_2, v]) \mapsto [g_1, [g_2, v]], \tag{2.81}$$
$$m : G_{\mathcal{K}} \times^{G_{\mathcal{O}}} R_{G,V} \to T_{G,V}, \qquad [g_1, [g_2, v]] \mapsto [g_1 g_2, v]. \tag{2.82}$$

Note that $p$ and $m$ are equivariant on the nose, while $q$ is equivariant over the left projection $\tilde{G} \times \tilde{G} \to \tilde{G}$. This turns $q^*$ into the descent isomorphism

$$q^* : H^{\mathrm{BM}}_{\tilde{G}}(G_{\mathcal{K}} \times^{G_{\mathcal{O}}} R_{G,V}) \xrightarrow{\sim} H^{\mathrm{BM}}_{\tilde{G}\times\tilde{G}}(G_{\mathcal{K}} \times R_{G,V}). \tag{2.83}$$

Importantly, $m$ maps $q(p^{-1}(R_{G,V} \times R_{G,V}))$ to $R_{G,V}$, which makes the convolution product well-defined:

$$c_1 * c_2 := m_*(q^*)^{-1}p^*(c_1 \boxtimes c_2), \qquad c_1, c_2 \in H^{\mathrm{BM}}_{\tilde{G}}(R_{G,V}). \tag{2.84}$$

□

*Remark* 2.32. In the following, we will often suppress the data $G, V$ from the notation and write for example $R$ instead of $R_{G,V}$.

**Definition 2.33.** The *quantized Coulomb branch algebra* is $\mathfrak{C}_\hbar := H^{\mathrm{BM}}_{\tilde{G}}(R)$ equipped with the convolution product, which is an algebra over $\mathbb{C}[\hbar]$. The *(classical) Coulomb branch algebra* is its specialization $\mathfrak{C} := \mathfrak{C}_\hbar/\hbar\mathfrak{C}_\hbar$.

**Proposition 2.34** ([BFN18b]). *The Coulomb branch algebra $\mathfrak{C}$ is commutative and has the structure of a Poisson algebra.*

**Definition 2.35** ([BFN18b]). The *Coulomb branch* is the affine complex Poisson variety

$$\mathfrak{M} := \operatorname{Spec} \mathfrak{C}. \tag{2.85}$$

*Example* 2.36. Let $G = \mathbb{C}^\times$ and $V = 0$. Then $R_{G,V} = \mathrm{Gr}_{\mathbb{C}^\times} = \bigsqcup_{\lambda\in\mathbb{Z}}\{x^\lambda\} \cong \mathbb{Z}$ is discrete and all points are fixed points under $\mathcal{O}^\times$ with trivial equivariant Euler class. By the localization theorem, the fundamental classes $[x^\lambda]$ thus generate the Coulomb branch algebra over $\mathrm{Frac}(H^{\mathrm{BM}}_{\tilde{G}}(\mathrm{pt})) = \mathbb{C}(\hbar, a)$ and they multiply according to

$$(f_1(a)[x^{\lambda_1}]) * (f_2(a)[x^{\lambda_2}]) = f_1(a)f_2(a + \lambda_1\hbar)[x^{\lambda_1+\lambda_2}]. \tag{2.86}$$

To see this, we apply the localization theorem to the convolution diagram, which in this case reduces to

$$\begin{array}{ccccc} & & \mathcal{K}^\times \times \mathrm{Gr}_{\mathbb{C}^\times} & \xrightarrow{\ q\ } & \mathcal{K}^\times \times^{\mathcal{O}^\times} \mathrm{Gr}_{\mathbb{C}^\times} & & \\ & \swarrow^{p} & & & & \searrow^{m} & \\ \mathrm{Gr}_{\mathbb{C}^\times} \times \mathrm{Gr}_{\mathbb{C}^\times} & & & & & & \mathrm{Gr}_{\mathbb{C}^\times} \end{array}$$

To compute the convolution product $(f_1(a)[x^{\lambda_1}]) * (f_2(a)[x^{\lambda_2}])$, we need to find the homology class $(q^*)^{-1}p^*((f_1(a_1)[x^{\lambda_1}])\boxtimes(f_2(a_2)[x^{\lambda_2}]))$ and push it forward along $m$. Here, $a_1$ and $a_2$ denote the equivariant parameters of the first and second factor of $\tilde{G} \times \tilde{G}$. Notice that $\mathcal{K}^\times \times \mathrm{Gr}_{\mathbb{C}^\times}$ does not have fixed points, but all points $[g_1, t^{\lambda_2}\mathcal{O}^\times] \in \mathcal{K}^\times \times^{\mathcal{O}^\times} \mathrm{Gr}_{\mathbb{C}^\times}$ are fixed points. Moreover, there exist $\lambda_1 \in X_*^+(T)$ and $h_2 \in \mathcal{O}^\times$ such that $g_1 = t^{\lambda_1}h_2$, so

$$[g_1, t^{\lambda_2}\mathcal{O}^\times] = [t^{\lambda_1}h_2, t^{\lambda_2}\mathcal{O}^\times] = [t^{\lambda_1}, t^{\lambda_2}\mathcal{O}^\times] =: x^{\lambda_1,\lambda_2} \tag{2.87}$$

We choose $(t^{\lambda_1}, t^{\lambda_2}\mathcal{O}^\times) \in \mathcal{K}^\times \times \mathrm{Gr}_{\mathbb{C}^\times}$ to be the canonical lift of $x^{\lambda_1,\lambda_2}$. Next, we need to work out the descent condition. Being fixed under the $\tilde{G} \times \tilde{G}$-action means

$$(t^{\lambda_1}, t^{\lambda_2}\mathcal{O}^\times) = (h_1, h_2) \cdot (t^{\lambda_1}, t^{\lambda_2}\mathcal{O}^\times) = (h_1 t^{\lambda_1} h_2^{-1}, h_2 t^{\lambda_2}\mathcal{O}^\times), \tag{2.88}$$

which is equivalent to $h_2 = t^{-\lambda_1}h_1 t^{\lambda_1}$, fixing $a_2 = a_1 + \lambda_1\hbar$. It follows that

$$(q^*)^{-1}p^*((f_1(a_1)[x^{\lambda_1}]) \boxtimes (f_2(a_2)[x^{\lambda_2}])) = f_1(a)f_2(a + \lambda_1\hbar)[x^{\lambda_1,\lambda_2}]. \tag{2.89}$$

Since the fixed point $x^{\lambda_1,\lambda_2}$ maps to $x^{\lambda_1+\lambda_2}$ under $m$, we find

$$m_*(q^*)^{-1}p^*((f_1(a_1)[x^{\lambda_1}]) \boxtimes (f_2(a_2)[x^{\lambda_2}])) = f_1(a)f_2(a + \lambda_1\hbar)[x^{\lambda_1+\lambda_2}]. \tag{2.90}$$

More generally, we obtain the following:

**Proposition 2.37** ([BFN18b]). *Suppose $G' = G = T = (\mathbb{C}^\times)^N$ and $V = 0$. Then $\mathfrak{C}_{\hbar,\mathrm{loc}} \coloneqq H^{\mathrm{BM}}_{\tilde{G}}(R)_{\mathrm{loc}}$ is generated by the fixed point fundamental classes $[x^\lambda]$ for $\lambda \in X_*(T)$ as a right module over $\mathrm{Frac}(H^{\mathrm{BM}}_{\tilde{G}}(pt)) = \mathbb{C}(\hbar, a_1, \dots, a_N)$. The convolution product is given by*

$$(f_1(a)[x^{\lambda_1}]) * (f_2(a)[x^{\lambda_2}]) = f_1(a) f_2(a + \lambda_1 \hbar)[x^{\lambda_1+\lambda_2}]. \tag{2.91}$$

*In particular, the isomorphism*

$$H^{\mathrm{BM}}_{\tilde{G}}(R)_{\mathrm{loc}} \to \mathbb{C}(\hbar, a_1, \dots, a_N)\langle e^{\pm\hbar\partial_1}, \dots, e^{\pm\hbar\partial_N}\rangle, \tag{2.92}$$

$$f(a_1, \dots, a_N) \mapsto f(a_1, \dots, a_N) \tag{2.93}$$

$$[x^\lambda] \mapsto \prod_{i=1}^{N} e^{\hbar\lambda_i\partial_i} \tag{2.94}$$

*exhibits $\mathfrak{C}_{\hbar,\mathrm{loc}}$ as an algebra of difference operators on $\mathbb{C}(\hbar, a_1, \dots, a_N)$.*

**Proposition 2.38** ([BFN18b; BFN18a]). *Let $\mathbf{z} : R_{T,0} \to R_{T,V}$ be the zero section and let $\iota : R_{T,V} \to R_{G,V}$ be the abelianization map. Then*

$$H^{\mathrm{BM}}_{\tilde{G}}(R_{G,V})_{\mathrm{loc}} \cong H^{\mathrm{BM}}_{\tilde{T}}(R_{G,V})^W_{\mathrm{loc}} \xrightarrow{(\iota_*)^{-1}} H^{\mathrm{BM}}_{\tilde{T}}(R_{T,V})^W_{\mathrm{loc}} \xrightarrow{\mathbf{z}^*} H^{\mathrm{BM}}_{\tilde{T}}(R_{T,0})^W_{\mathrm{loc}} \hookrightarrow H^{\mathrm{BM}}_{\tilde{T}}(R_{T,0})_{\mathrm{loc}} \tag{2.95}$$

*is a well-defined algebra map into an algebra of difference operators. For a dominant minuscule coweight $\lambda$ and an element $f \in \mathbb{C}[\mathfrak{t}]^{W_\lambda}$, where $W_\lambda$ is the stabilizer of $\lambda$, we have*

$$\mathbf{z}^*(\iota_*)^{-1}(f[R^\lambda_{G,V}]) = \sum_{w \in W/W_\lambda} wf \frac{\mathrm{eu}_T(t^{w\lambda}V_{\mathcal{O}}/(t^{w\lambda}V_{\mathcal{O}} \cap V_{\mathcal{O}}))}{\mathrm{eu}_T(T_{x^{w\lambda}}\mathrm{Gr}^\lambda_G)}[x^{w\lambda}]. \tag{2.96}$$

### 2.3.3 Coulomb branches of ADE quiver gauge theories

We now restrict our attention to quiver gauge theories whose quiver is an ADE quiver. In this case, the BFN Coulomb branch has a particularly nice description, which we will recount here.

**Definition 2.39.** (i) We say that a quiver $Q$ is *of finite or affine type ADE* if its diagram is a simply laced Dynkin diagram or its affinization.

(ii) Let $Q$ be a quiver and $\boldsymbol{N}$ a dimension vector. Let $G$ be the associated complexified gauge group and $V$ the associated matter representation. We denote the corresponding Coulomb branch by $\mathfrak{M}_Q$.

(iii) If the quiver $Q$ is of finite or affine type ADE, then we associate to it the complex adjoint simple group $G_Q$ of the same type.

**Definition 2.40.** Let $Q$ be a quiver of finite or affine type ADE with dimension vector $\boldsymbol{N}$ and associated complex adjoint simple group $G_Q$. Let $B_\pm$ be a pair of opposite Borel subgroups for a given maximal torus $T_Q \subseteq G_Q$. The *rational open zastava* is the moduli space $\mathring{Z}^{\mathrm{rat}}_Q$ of based rational maps into the flag variety $G_Q/B_+$

$$\varphi : \mathbb{P}^1 \to G_Q/B_+, \qquad \varphi(\infty) = B_- \tag{2.97}$$

of degree $\boldsymbol{N}$.

*Remark* 2.41. The word *zastava* means *flag* in various Slavic languages, explaining the terminology.

**Theorem 2.42** ([BFN18a]). *Let $Q$ be a quiver of finite type ADE or affine type A and $\mathbf{N}$. a dimension vector. Then*

$$\mathfrak{M}_Q \cong \mathring{Z}_Q^{\mathrm{rat}}. \tag{2.98}$$

**Definition 2.43.** The *necklace quiver* with $\ell$ nodes of rank $N$ is the quiver $Q$ of affine type $A^{(1)}_{\ell-1}$ with constant dimension vector $\mathbf{N} = N$, see figure 1.1.

**Proposition 2.44** ([FR14; NT25]). *Let $Q$ be the necklace quiver. Then $\mathring{Z}_Q^{\mathrm{rat}}$ has étale rational coordinates $\varphi_i^\alpha$ and $u_i^\alpha$ for $\alpha \in \mathbb{Z}_\ell$ and $i = 1, \dots, N$ and their Poisson brackets are given by*

$$\{\varphi_i^\alpha, \varphi_j^\beta\} = 0, \qquad \{\varphi_i^\alpha, u_j^\beta\} = \delta_{ij}\delta^{\alpha\beta} u_j^\beta, \qquad \{u_i^\alpha, u_j^\beta\} = G_{\mathrm{rat}}^{\alpha\beta}(\varphi_i^\alpha - \varphi_j^\beta) u_i^\alpha u_j^\beta, \tag{2.99}$$

*where the rational structure function $G_{\mathrm{rat}}^{\alpha\beta}(z)$ decomposes according to the adjacency of the nodes $\alpha$ and $\beta$ as*

$$G_{\mathrm{rat}}^{\alpha\beta}(z) = G_{\mathrm{rat}}^{\alpha\to\beta}(z) + G_{\mathrm{rat}}^{\alpha=\beta}(z) + G_{\mathrm{rat}}^{\alpha\leftarrow\beta}(z), \tag{2.100}$$

*where*

$$G_{\mathrm{rat}}^{\alpha\to\beta}(z) = \frac{1}{z - m^{\alpha\to\beta}}, \qquad G_{\mathrm{rat}}^{\alpha=\beta}(z) = \frac{1}{z} \qquad G_{\mathrm{rat}}^{\alpha\leftarrow\beta}(z) = \frac{1}{z + m^{\beta\to\alpha}}, \tag{2.101}$$

*and we define $\frac{1}{z}$ at $z = 0$ by its regular part, which is zero.*

*Remark* 2.45. These Poisson brackets are exactly the generalization of the Poisson brackets obtained in [BDG15] for Coulomb branches of linear quivers using the abelianization method. We will therefore refer to the Poisson algebra of the étale rational coordinates as the *abelianized Coulomb branch algebra*.

### 2.3.4 $K$-theoretic and elliptic Coulomb branches

The BFN construction of Coulomb branches of 3d $\mathcal{N} = 4$ gauge theories has natural trigonometric and elliptic deformations. Namely, instead of taking equivariant Borel–Moore homology of the BFN space $R_{G,V}$, we can take equivariant $K$-theory, or equivariant elliptic Borel–Moore homology. The former case is well-known, while the latter case is less explored. However, the description of Coulomb branches of quivers of (finite) type ADE in terms of the open zastava is known to generalize to the trigonometric and elliptic cases [Fin+18; FMP22].

Physically speaking, the trigonometric and elliptic deformations describe the Coulomb branches of 4d $\mathcal{N} = 2$ and 5d $\mathcal{N} = 1$ gauge theories on the compactified spacetimes $\mathbb{R}^3 \times S^1$ and $\mathbb{R}^3 \times T^2$. The fields in the effective three-dimensional theory then consist of infinite towers of Kaluza–Klein modes (Fourier modes of the fields along $S^1$ or $T^2$) and summing over them yields periodic or doubly quasi-periodic—*i.e.* trigonometric or elliptic—functions.

**Proposition 2.46** ([Fin+18; FMP22]). *Let $Q$ be a quiver of finite type ADE with dimension vector $\mathbf{N}$. and $G, V$ the corresponding complexified gauge group and matter representation.*

(i) *Then*

$$\operatorname{Spec} K_{G_{\mathcal{O}}}(R_{G,V}) \cong \mathring{Z}^{\mathrm{tri}}_{Q}, \qquad \underline{\operatorname{Spec}}\, E_{G_{\mathcal{O}}}(R_{G,V}) \cong \mathring{Z}^{\mathrm{ell}}_{Q}, \tag{2.102}$$

*where $K_{G_{\mathcal{O}}}(-)$ and $E_{G_{\mathcal{O}}}(-)$ denote equivariant $K$-theory and elliptic Borel–Moore homology,* $\underline{\operatorname{Spec}}$ *denotes the relative spectrum, and $\mathring{Z}^{\mathrm{tri}}_{Q}$ and $\mathring{Z}^{\mathrm{ell}}_{Q}$ are the trigonometric open zastava and the elliptic (Coulomb) open zastava.*

(ii) *$\mathring{Z}^{\mathrm{tri}}_{Q}$ and $\mathring{Z}^{\mathrm{ell}}_{Q}$ have étale rational coordinates $\varphi_i^\alpha, u_i^\alpha$ with $\alpha \in Q_0$ and $i = 1, \dots, N_\alpha$ with Poisson brackets exactly as in equation (2.99), except that the rational structure functions $G_{\mathrm{rat}}^{\alpha\to\beta}(z)$, $G_{\mathrm{rat}}^{\alpha=\beta}(z)$, and $G_{\mathrm{rat}}^{\alpha\leftarrow\beta}(z)$ are replaced by their trigonometric and elliptic analogs*

$$G_{\mathrm{tri}}^{\alpha\to\beta}(z) = \cot(z - m^{\alpha\to\beta}),\ G_{\mathrm{tri}}^{\alpha=\beta}(z) = \cot(z),\ G_{\mathrm{tri}}^{\alpha\leftarrow\beta}(z) = \cot(z + m^{\beta\to\alpha}), \tag{2.103}$$

*and*

$$G_{\mathrm{ell}}^{\alpha\to\beta}(z) = E_1(z - m^{\alpha\to\beta}), \quad G_{\mathrm{ell}}^{\alpha=\beta}(z) = E_1(z), \quad G_{\mathrm{ell}}^{\alpha\leftarrow\beta}(z) = E_1(z + m^{\beta\to\alpha}), \tag{2.104}$$

*where we define $\cot(z)$ and $E_1(z)$ at $z = 0$ by their regular parts, which are zero. [TODO: Unify normalizations: $\cot(z)$ here, $\frac{1}{2}\coth(z/2)$ in chapter 4, and $E_1 \to \pi\cot(\pi z)$ in the appendix; they agree up to rescaling $z$ by $2\pi\mathrm{i}$, which should be stated once.]*

*Remark* 2.47. Since the generators and relations of the Poisson algebras of the étale rational coordinates are local in the quiver (depending only on a single node and its adjacent nodes), the Poisson algebras for linear quivers (finite type A) are easily generalized to necklace quivers (affine type A). In this thesis, we will assume that the description of $K$-theoretic and elliptic Coulomb branches of quivers of finite type ADE in terms of the étale rational coordinates on open zastavas and their étale rational coordinates extends to quivers of affine type A as in the rational case.

# 3

# Spin Ruijsenaars–Schneider models from Hamiltonian reductions

**This chapter constructs the quantum rational spin Ruijsenaars–Schneider model, describing $N$ particles that each carry $\ell$ spin states, by quantum Hamiltonian reduction. After reviewing the classical Hamiltonian reduction of $T^*\mathrm{GL}_N \times T^*\mathbb{C}^{N\times\ell}$, we tensor the quantum cotangent bundle of $\mathrm{GL}_N$ with an algebra of canonical oscillators, identify the quantum moment map, and reduce at the value fixed by the coupling constant $\eta$. The resulting algebra $\mathfrak{A}_{N,\ell}$ of observables is described in terms of the quantum current $S(z)$, whose commutation relations take $R$-matrix form and reduce to the classical Poisson brackets in the limit $\hbar \to 0$, and which determines all higher modes through a recurrence relation. We further show that $\mathfrak{A}_{N,\ell}$ contains the Yangian of $\mathfrak{gl}_\ell$ and compute its cross relations with the quantum current, which we expect to identify $\mathfrak{A}_{N,\ell}$ with a truncated affine Yangian. Finally, we use the Yangian generators as creation operators to propose a method for computing the spectrum of the lowest Hamiltonian at non-critical coupling. At $\eta = -\hbar$, this reduces the eigenvalue problem to a difference equation, which for the sector without excitations is given by the spinless Ruijsenaars–Schneider Hamiltonian, and we find that the energy levels are independent of the spin configuration.**

## 3.1 Quantum rational spin Ruijsenaars–Schneider models from quantum Hamiltonian reduction

### 3.1.1 Review of the classical case

We start this section with a brief review of [AF98], where the Poisson structure of the rational spin Ruijsenaars–Schneider model was obtained via Hamiltonian reduction with the unreduced phase space given by the cotangent bundle $T^*\mathrm{GL}_N \times T^*\mathbb{C}^{N\times\ell}$. This is

parametrized by a tuple of matrices $(g, x, a, c) \in \mathrm{GL}_N \times \mathfrak{gl}_N \times \mathbb{C}^{N\times\ell} \times \mathbb{C}^{\ell\times N}$ with coefficients $g_{ij}, x_{ij}, a_i^\alpha, c_i^\alpha$ for $i, j = 1, \dots, N$ and $\alpha = 0, \dots, \ell - 1$ equipped with the Poisson structure

$$\{x_1, x_2\} = C_{12}x_1 - x_1C_{12}, \quad \{x_1, g_2\} = g_2C_{12}, \quad \{a_1^\alpha, c_2^\beta\} = -\delta^{\alpha\beta}\Delta_{12}, \tag{3.1}$$

where $C_{12} = \sum_{i,j=1}^N e_{ij} \otimes e_{ji} \in \mathrm{End}(\mathbb{C}^N)^{\otimes 2}$ and $\Delta_{12} = \sum_{i=1}^N e_i \otimes e_i^t \in \mathbb{C}^N \otimes \mathbb{C}^{N*}$, and all other pairs Poisson-commuting. Here, $e_{ij}$ are the standard matrix units and $e_i$ are the standard unit vectors. The reduction is performed with respect to the $\mathrm{GL}_N$-action

$$h \cdot (g, x, a, c) := (hgh^{-1}, hxh^{-1}, ha, ch^{-1}) \tag{3.2}$$

and the moment map equation amounts to

$$x - gxg^{-1} - ac = \eta, \tag{3.3}$$

where $\eta \in \mathbb{C}$ later becomes the coupling constant of the model.

To perform the reduction, one restricts to the open dense locus where the matrix $x$ has distinct eigenvalues, allowing for a unique diagonalization $x = TQT^{-1}$, where $Q = \mathrm{diag}(x_1, \dots, x_N)$ with ordered and pairwise distinct $x_1, \dots, x_N$, and $T$ satisfies the mirabolic condition $Te = e$, where

$$e = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \tag{3.4}$$

is the column vector only containing ones. While $Q$ uniquely labels orbits, $T$ gives a parameterization of each orbit and satisfies the Kostant–Kirillov bracket. Similarly, one diagonalizes $gxg^{-1} = UQU^{-1}$ with $Ue = e$. It follows that $g = UPT^{-1}$ for some diagonal matrix $P = \mathrm{diag}(P_1, \dots, P_N)$. The transformation properties of these variables under an element $h \in \mathrm{GL}_N$ are as follows:

$$h \cdot T = hTh[T], \quad h \cdot U = hUh[U], \quad h \cdot P = h[U]^{-1}Ph[T], \tag{3.5}$$

where $h[T]$ is the unique diagonal matrix satisfying $h[T]e = T^{-1}h^{-1}e$ and similarly $h[U]$ is the unique diagonal matrix satisfying $h[U]e = U^{-1}h^{-1}e$. We note that $h[T] = 1$ and $h[U] = 1$ whenever $h$ satisfies the mirabolic condition $he = e$, i.e. $h$ lies in the *mirabolic subgroup*

$$\mathrm{M}_N := \{h \in \mathrm{GL}_N \mid he = e\}. \tag{3.6}$$

Its Lie algebra is the *mirabolic subalgebra*

$$\mathfrak{m}_N := \{h \in \mathfrak{gl}_N \mid he = 0\}. \tag{3.7}$$

Introducing $t$ and $u$ to be the unique diagonal matrices satisfying $te = \sum_{\rho=0}^{\ell-1} T^{-1}a^\rho$ and $ue = \sum_{\rho=0}^{\ell-1} U^{-1}a^\rho$, they transform as

$$h \cdot t = h[T]^{-1}t, \quad h \cdot u = h[U]^{-1}u. \tag{3.8}$$

Combining the variables $T, U, Q, P, t, u, a, c$ appropriately yields a complete set of invariants given by the coefficients $x_1, \dots, x_N$ of $Q$ as well as

$$\mathbf{a} := t^{-1}T^{-1}a, \quad \mathbf{c} := cUPt, \quad \mathbf{L} := t^{-1}T^{-1}UPt, \quad \mathbf{P} := u^{-1}Pt. \tag{3.9}$$

In terms of these variables, the fixing of the moment map reads

$$Q\mathbf{L} - \mathbf{L}Q - \eta\mathbf{L} = \mathbf{ac} \qquad \Leftrightarrow \qquad \mathbf{L}_{ij} = \frac{\mathbf{a}_i\mathbf{c}_j}{x_{ij} - \eta}. \tag{3.10}$$

Here $x_1, \dots, x_N$ denote the diagonal entries of $Q$ and $x_{ij} := x_i - x_j$. The fully reduced Poisson algebra is thus generated by the variables $Q, \mathbf{P}, \mathbf{a}$, and $\mathbf{c}$, while $\mathbf{L}$ can be expressed in terms of these generators, and the Poisson bracket is induced from the Poisson bracket on the cotangent bundle.

### 3.1.2 Quantum cotangent bundle of $\mathrm{GL}_N$

Moving towards the quantization, let us describe the *quantum cotangent bundle of* $\mathrm{GL}_N$, which is the algebra that quantizes the Poisson structure on the cotangent bundle $T^*\mathrm{GL}_N$ in the parameterization via $T, U, Q$, and $P$ discussed above. It was introduced in [AF96] to derive the commutation relations satisfied by the quantum Lax operator of the rational *spinless* ($\ell = 1$) Ruijsenaars–Schneider model. In the following, the notation for the variables will be the same as in the classical case, but should be interpreted as quantum operators without further comment.

**Definition 3.1.** Let $\hbar \in \mathbb{C}^\times$. The *quantum cotangent bundle of* $\mathrm{GL}_N$, denoted $\mathcal{O}_\hbar(T^*\mathrm{GL}_N)$, is the unital associative algebra generated by the coefficients of matrices $T^{\pm 1}, U^{\pm 1}, Q, P^{\pm 1}$ with $T^{-1}T = TT^{-1} = 1, U^{-1}U = UU^{-1} = 1, P^{-1}P = PP^{-1} = 1$ and $Q, P^{\pm 1}$ diagonal, subject to the commutation relations

$$\begin{aligned} T_1T_2 &= T_2T_1R_{12}, \qquad & U_1U_2 &= U_2U_1R_{21}, \\ T_1P_2 &= P_2T_1\bar{R}_{12}, \qquad & U_1P_2 &= P_2U_1\bar{R}_{12}, \end{aligned} \qquad P_2Q_1 = (Q_1 - \hbar Y_{12})P_2, \tag{3.11}$$

and all other pairs commuting. Here, we have introduced

$$R = 1 + \sum_{i\neq j} \frac{\hbar}{x_{ij}} f_{ij} \otimes f_{ji}, \quad \bar{R} = 1 + \sum_{i\neq j} \frac{\hbar}{x_{ij} - \hbar} f_{ij} \otimes e_{jj}, \quad Y := \sum_{i=1}^N e_{ii} \otimes e_{ii}, \tag{3.12}$$

where $f_{ij} := e_{ii} - e_{ij}$ is a basis for the mirabolic subalgebra $\mathfrak{m}_N$, and subscripts label auxiliary spaces. It is important to note that $R$ satisfies the Yang-Baxter equation and is unitary:

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}, \quad R^{-1} = R_{21}, \tag{3.13}$$

ensuring the consistency of the commutation relations (3.11). Further, it was shown in [AF96] that the combinations $x := TQT^{-1}$ and $g := UPT^{-1}$ satisfy the commutation relations

$$[x_1, x_2] = \hbar(C_{12}x_1 - x_1C_{12}), \quad [x_1, g_2] = \hbar g_2 C_{12}, \quad [g_1, g_2] = 0, \tag{3.14}$$

which is the canonical quantization of the canonical Poisson structure on $T^*\mathrm{GL}_N$.

*Remark* 3.2. Let us note that the mirabolic subgroup originally appeared under the name *Frobenius subgroup* in [AF96]. The name *mirabolic subgroup* came to our attention through the work [Kal25]. The $R$-matrix recovered in *loc. cit.* are closely related to the $R$-matrix $R$ in this chapter. The more recent work [Kal26] building on [Kal25] contains a quantum Hamiltonian reduction similar to ours that results in the quantum Toda lattice, whose classical limit is known [FT97] to arise from the classical Hamiltonian reduction $N_+ \sslash T^*\mathrm{GL}_N /\!\!/ N_-$ where $N_\pm$ are the subgroups of upper and lower unipotent matrices. The quantum Hamiltonian reduction is furthermore identified with a quotient of the $(-2N)$-shifted Yangian of $\mathfrak{sl}_2$.

**Quantum cotangent bundle of $\mathrm{GL}_N$ via functional operators**

A useful conceptual shift for this work is the reinterpretation of the presentation of the quantum cotangent bundle of $\mathrm{GL}_N$ in terms of functional operators. To this end, let $\mathbb{F} \coloneqq \mathbb{C}(x_1, \dots, x_N)$ be the field of rational functions in the variables $x_1, \dots, x_N$. Consider the space $\mathbb{F}[z]_N$ of polynomials in $z$ with coefficients in $\mathbb{F}$ and degree strictly less than $N$. Such polynomials are uniquely determined by their values on $x_1, \dots, x_N$. A convenient basis of $\mathbb{F}[z]_N$ for our purposes is given by the Lagrange interpolation polynomials

$$\ell_i(z) = \prod_{j(\neq i)} \frac{z - x_j}{x_i - x_j}, \quad i = 1, \dots, N, \tag{3.15}$$

which satisfy the Lagrange interpolation formula:

$$f(z) = \sum_{i=1}^{N} f(x_i)\ell_i(z), \quad \forall f(z) \in \mathbb{F}[z]_N. \tag{3.16}$$

Consequently, in this basis, the special vector $e$ defining the mirabolic condition is nothing but the constant polynomial 1:

$$1 = \sum_{i=1}^{N} \ell_i(z). \tag{3.17}$$

Let us set $\chi(z) \coloneqq \det(z - Q)$ and introduce the inner product

$$\langle -, - \rangle : \mathbb{F}[z]_N \otimes_{\mathbb{F}} \mathbb{F}[z]_N \to \mathbb{F}, \quad f(z) \otimes g(z) \mapsto \oint \frac{dz}{\chi(z)} f(z)g(z), \tag{3.18}$$

where $\oint dz$ takes the negative residue at infinity, i.e. the sum of all residues away from infinity. This inner product is evidently bilinear and symmetric as well as non-degenerate with the dual basis to $\ell_i(z)$ given by

$$\ell_i^*(z) \coloneqq \prod_{j(\neq i)} (z - x_j), \tag{3.19}$$

due to the identity

$$\langle \ell_i^*(z), \ell_j(z) \rangle = \oint \frac{dz}{z - x_i} \ell_j(z) = \ell_j(x_i) = \delta_{ij}. \tag{3.20}$$

With this in hand, we may express $R_{12}$ as a functional operator on $\mathbb{F}[z_1]_N \otimes_{\mathbb{F}} \mathbb{F}[z_2]_N$. Let

$$\mathbf{R}_{12}^* f(z_1, z_2) = f(z_1, z_2) + \hbar \partial_{12} f(z_1, z_2), \tag{3.21}$$

where we have introduced Newton's divided difference operator

$$\partial_{12} f(z_1, z_2) \coloneqq \frac{f(z_1, z_2) - f(z_2, z_1)}{z_1 - z_2}, \tag{3.22}$$

which preserves the space $\mathbb{F}[z_1]_N \otimes_{\mathbb{F}} \mathbb{F}[z_2]_N$. The operator $\mathbf{R}_{12}^*$ satisfies the Yang-Baxter equation when acting on $\mathbb{F}[z_1]_N \otimes_{\mathbb{F}} \mathbb{F}[z_2]_N \otimes_{\mathbb{F}} \mathbb{F}[z_3]_N$ and it is quickly checked that

$$\langle \mathbf{R}_{12}^* \ell_i^*(z_1)\ell_j^*(z_2), \ell_k(z_1)\ell_l(z_2) \rangle = \langle R_{12}^t e_i \otimes e_j, e_k \otimes e_l \rangle, \tag{3.23}$$

where on the right hand side we use the standard inner product on $\mathbb{C}^N \otimes \mathbb{C}^N$. This parallels the construction of [FP94], where $R$-matrices are obtained as the restriction of functional operators to a finite-dimensional subspace. The adjoint of $\partial_{12}$ with respect to the inner product defined above is given by

$$\partial_{12}^* f(z_1, z_2) := \frac{f(z_1, z_2) + f(z_2, z_1) - f(z_1, z_1) - f(z_2, z_2)}{z_1 - z_2}, \tag{3.24}$$

The operator $\partial_{12}^*$ again preserves the space $\mathbb{F}[z_1]_N \otimes_{\mathbb{F}} \mathbb{F}[z_2]_N$. In fact, $\partial_{12}^* f(z_1, z_2)$ always lies in the ideal generated by $z_1 - z_2$ and fulfills $(z_1 - z_2)^{-1}\partial_{12}^*(z_1 - z_2) = \partial_{12}$. Letting $\mathbf{R}_{12} := 1 + \hbar\partial_{12}^*$ we hence obtain

$$\langle \ell_i^*(z_1)\ell_j^*(z_2), \mathbf{R}_{12}\ell_k(z_1)\ell_l(z_2)\rangle = \langle e_i \otimes e_j, R_{12}e_k \otimes e_l\rangle, \tag{3.25}$$

giving the square of equalities

$$\begin{array}{ccc} \langle \ell_i^*(z_1)\ell_j^*(z_2), \mathbf{R}_{12}\ell_k(z_1)\ell_l(z_2)\rangle & = & \langle e_i \otimes e_j, R_{12}e_k \otimes e_l\rangle \\ \| & & \| \\ \langle \mathbf{R}_{12}^*\ell_i^*(z_1)\ell_j^*(z_2), \ell_k(z_1)\ell_l(z_2)\rangle & = & \langle R_{12}^t e_i \otimes e_j, e_k \otimes e_l\rangle. \end{array} \tag{3.26}$$

**Definition 3.3.** Let us write $\ell^*(z) := \chi(z)(z - Q)^{-1}$ and $\ell(z) = \chi'(Q)^{-1}\ell^*(z)$. These are the diagonal matrices whose diagonal entries are $\ell_i^*(z)$ and $\ell_i(z)$. Define the combinations

$$\begin{aligned} \mathbf{T}(z) &:= T\ell^*(z)e, \quad & \mathbf{T}^{-1}(z) &:= e^t\ell(z)T^{-1}, \\ \mathbf{U}(z) &:= U\ell^*(z)e, \quad & \mathbf{U}^{-1}(z) &:= e^t\ell(z)U^{-1}, \\ \mathbf{P}(z) &:= P\ell^*(z)e, \quad & \mathbf{P}^{-1}(z) &:= e^t\ell(z)P^{-1}. \end{aligned} \tag{3.27}$$

Then $\mathbf{T}(z), \mathbf{U}(z)$ and $\mathbf{T}^{-1}(z), \mathbf{U}^{-1}(z)$ transform as vectors and covectors under $\mathrm{M}_N$, respectively, and both depend polynomially on $z$.

*Remark* 3.4. The combinations $\mathbf{T}(z), \mathbf{U}(z), \mathbf{P}(z)$ are invertible in the following sense:

$$\begin{aligned} \mathbf{T}^{-1}(z)\mathbf{T}(z') &= \delta(z, z'), \qquad & \langle \mathbf{T}(z), \mathbf{T}^{-1}(z)\rangle &= 1, \\ \mathbf{U}^{-1}(z)\mathbf{U}(z') &= \delta(z, z'), \qquad & \langle \mathbf{U}(z), \mathbf{U}^{-1}(z)\rangle &= 1, \\ \mathbf{P}^{-1}(z)\mathbf{P}(z') &= \delta(z, z'), \qquad & \langle \mathbf{P}(z), \mathbf{P}^{-1}(z)\rangle &= 1. \end{aligned} \tag{3.28}$$

Here, the left equalities are equalities of polynomials and the right equalities are equalities of $N \times N$ matrices and we make use of the $\delta$-polynomial

$$\delta(z, z') := \sum_{i=1}^{N} \ell_i(z)\ell_i^*(z'). \tag{3.29}$$

**Proposition 3.5.** *The relations of $\mathcal{O}_\hbar(T^*\mathrm{GL}_N)$ may equivalently be written as $P_2Q_1 = (Q_1 - \hbar Y_{12})P_2$ together with*

$$\begin{aligned} \mathbf{T}_1(z_1)\mathbf{T}_2(z_2) &= \mathbf{R}_{12}^*\mathbf{T}_2(z_2)\mathbf{T}_1(z_1), & \mathbf{U}_1(z_1)\mathbf{U}_2(z_2) &= \mathbf{R}_{21}^*\mathbf{U}_2(z_2)\mathbf{U}_1(z_1), \\ \mathbf{T}_1^{-1}(z_1)\mathbf{T}_2^{-1}(z_2) &= \mathbf{R}_{12}\mathbf{T}_2^{-1}(z_2)\mathbf{T}_1^{-1}(z_1), & \mathbf{U}_1^{-1}(z_1)\mathbf{U}_2^{-1}(z_2) &= \mathbf{R}_{21}\mathbf{U}_2^{-1}(z_2)\mathbf{U}_1^{-1}(z_1), \\ \mathbf{P}_1^{-1}(z_1)\mathbf{T}_2(z_2) &= \mathbf{T}_2(z_2)\mathbf{P}_1^{-1}(z_1), & \mathbf{P}_1^{-1}(z_1)\mathbf{U}_2(z_2) &= \mathbf{U}_2(z_2)\mathbf{P}_1^{-1}(z_1), \\ \mathbf{P}_1^{-1}(z_1)\mathbf{T}_2^{-1}(z_2) &= \mathbf{R}_{21}^*[\mathbf{T}_2^{-1}(z_2)]\mathbf{P}_1^{-1}(z_1), & \mathbf{P}_1^{-1}(z_1)\mathbf{U}_2^{-1}(z_2) &= \mathbf{R}_{21}^*[\mathbf{U}_2^{-1}(z_2)]\mathbf{P}_1^{-1}(z_1). \end{aligned} \tag{3.30}$$

*Proof.* The first and second lines are derived from the fact that $\mathbf{R}^*_{12}$ and $\mathbf{R}_{12}$ are represented by $R^t_{12}$ and $R_{12}$ in the Lagrange basis. For the third line, we have the identity $P_1^{-1}\ell_2^*(z_2)e_2 = \bar{R}_{21}^{-1}\ell_2^*(z_2)e_2P_1^{-1}$, which then implies

$$P_1^{-1}T_2\ell_2^*(z_2)e_2 = T_2\bar{R}_{21}P_1^{-1}\ell_2^*(z_2)e_2 = T_2\ell_2^*(z_2)e_2P_1^{-1}$$
$$P_1^{-1}U_2\ell_2^*(z_2)e_2 = U_2\bar{R}_{21}P_1^{-1}\ell_2^*(z_2)e_2 = U_2\ell_2^*(z_2)e_2P_1^{-1}.$$

For the fourth line, we use the identity

$$P_1^{-1}e_2^t\ell_2(z_2) = e_2^t\left(\ell_2(z_2) - \frac{\hbar}{z_2 - Q_1}(\ell_2(z_2) - \ell_2(Q_1))\right)\bar{R}_{21}P_1^{-1}. \qquad \square$$

### 3.1.3 Quantum Hamiltonian reduction

Quantum Hamiltonian reduction systematically quantizes the classical Hamiltonian reduction procedure and can be viewed as a special case of BRST quantization. We give an overview over the method of quantum Hamiltonian reduction in section 2.2. To perform the quantum Hamiltonian reduction determining the algebra $\mathfrak{A}_{N,\ell}$ which describes the observables of the rational spin Ruijsenaars–Schneider model, we extend the quantum cotangent bundle $\mathcal{O}_\hbar(T^*\mathrm{GL}_N)$ by tensoring with an algebra of canonical oscillators, in analogy with the classical case:

**Definition 3.6.** We let $\mathcal{O}_\hbar(T^*\mathbb{C}^{N\times\ell})$ be the unital associative algebra generated by the respective $N$ coefficients of $\ell$ column vectors $a^\alpha$ and of $\ell$ row vectors $c^\alpha$ ($\alpha = 0, \dots, \ell-1$), subject to the canonical commutation relations $[a_1^\alpha, c_2^\beta] = -\hbar\delta^{\alpha\beta}\Delta_{12}$ and $[a_1^\alpha, a_2^\beta] = [c_1^\alpha, c_2^\beta] = 0$, where $\Delta = \sum_{i=1}^N e_i \otimes e_i^t$ and the subscripts label auxiliary spaces.

We thus take our algebra of observables before reduction to be the tensor product of algebras $\mathcal{O}_\hbar(T^*\mathrm{GL}_N) \otimes \mathcal{O}_\hbar(T^*\mathbb{C}^{N\times\ell})$, which is generated by the coefficients of $T^{\pm1}, U^{\pm1}, Q, P^{\pm1}, a^\alpha$, and $c^\alpha$. In particular, taking the tensor product implies that the generators $T^{\pm1}, U^{\pm1}, Q, P^{\pm1}$ from the first factor commute with the generators $a^\alpha, c^\alpha$ from the second factor. The next step in the reduction procedure is to identify the quantum moment map.

#### Quantum moment map and induced action

**Proposition 3.7.** *Let $E = (e_{ij})$ denote the matrix of matrix unit generators of $\mathfrak{gl}_N$. Then*

$$\mu : U(\mathfrak{gl}_N) \to \mathcal{O}_\hbar(T^*\mathrm{GL}_N) \otimes \mathcal{O}_\hbar(T^*\mathbb{C}^{N\times\ell}), \quad E \mapsto TQT^{-1} - UQU^{-1} - ac \tag{3.31}$$

*is a homomorphism.*

*Proof.* The variables $x := TQT^{-1}$ and $x' := -UQU^{-1}$ satisfy the commutation relations of $\mathfrak{gl}_N$:

$$[x_1, x_2] = \hbar(C_{12}x_1 - x_1C_{12}), \quad [x'_1, x'_2] = \hbar(C_{12}x'_1 - x'_1C_{12}).$$

This is due to

$$\begin{aligned}(T_1Q_1T_1^{-1})(T_2Q_2T_2^{-1}) &= T_2T_1R_{12}Q_1R_{21}Q_2R_{12}T_2^{-1}T_1^{-1}\\ &= (T_2Q_2T_2^{-1})(T_1Q_1T_1^{-1}) + \hbar C_{12}((T_1Q_1T_1^{-1}) - (T_2Q_2T_2^{-1})),\\ (U_1Q_1U_1^{-1})(U_2Q_2U_2^{-1}) &= U_2U_1R_{21}Q_1R_{12}Q_2R_{21}U_2^{-1}U_1^{-1}\\ &= (U_2Q_2U_2^{-1})(U_1Q_1U_1^{-1}) - \hbar C_{12}((U_1Q_1U_1^{-1}) - (U_2Q_2U_2^{-1})),\end{aligned}$$

where we have used the identities

$$R_{12}Q_1R_{21}Q_2 - Q_2R_{12}Q_1R_{21} = \hbar C_{12}Q_1R_{21} - \hbar R_{12}Q_1C_{12},$$
$$R_{21}Q_1R_{12}Q_2 - Q_2R_{21}Q_1R_{12} = -\hbar C_{12}Q_1R_{12} + \hbar R_{21}Q_1C_{12}.$$

Furthermore, it is easily checked that $-ac$ also satisfies the commutation relations of $\mathfrak{gl}_N$ due to the Jordan-Schwinger representation:

$$[(ac)_1, (ac)_2] = -\hbar(C_{12}(ac)_1 - (ac)_1C_{12}).$$

Finally, we note that $x = TQT^{-1}$ and $x' = -UQU^{-1}$ as well as $ac$ all commute among each other, making $\mu$ into a homomorphism. □

We now take $\mu$ to be our quantum moment map and perform quantum Hamiltonian reduction at the ideal generated by the coefficients of $E - \eta I$. The reduced algebra $\mathfrak{A}_{N,\ell}$ resulting from quantum Hamiltonian reduction is then given by taking invariants and modding out the moment map equation:

$$\mathfrak{A}_{N,\ell} := \Big((\mathcal{O}_\hbar(T^*\mathrm{GL}_N) \otimes \mathcal{O}_\hbar(T^*\mathbb{C}^{N\times\ell}))/(\mu(E) - \eta)\Big)^{\mathfrak{gl}_N}. \tag{3.32}$$

This means that that next step is to determine the algebra of invariants. We begin by determining the induced action:

**Lemma 3.8.** *The induced action of $U(\mathfrak{gl}_N)$ on $\mathcal{O}_\hbar(T^*\mathrm{GL}_N) \otimes \mathcal{O}_\hbar(T^*\mathbb{C}^{N\times\ell})$ is given by*

$$\begin{aligned}
\mathrm{ad}_{E_1} T_2 &= -\hbar C_{12}T_2 + \hbar T_2(T_1X_{12}T_1^{-1}), & \mathrm{ad}_{E_1} a_2 &= -\hbar C_{12}a_2,\\
\mathrm{ad}_{E_1} U_2 &= -\hbar C_{12}U_2 + \hbar U_2(U_1X_{12}U_1^{-1}), & \mathrm{ad}_{E_1} c_2 &= \hbar c_2C_{12},\\
\mathrm{ad}_{E_1} P_2 &= \hbar P_2(T_1X_{12}T_1^{-1}) - \hbar(U_1X_{12}U_1^{-1})P_2, & \mathrm{ad}_{E_1} Q_2 &= 0,
\end{aligned} \tag{3.33}$$

*where $X_{12} := \sum_{i,j=1}^N e_{ij} \otimes e_{jj}$.*

*Proof.* Any homomorphism from a Hopf algebra into another algebra ('quantum moment map') gives rise to an action on the codomain via the adjoint action, see section 2.2. In our case, this means that the action of the generating matrix $E$ of $U(\mathfrak{gl}_N)$ acts on $O \in \mathcal{O}_\hbar(T^*\mathrm{GL}_N) \otimes \mathcal{O}_\hbar(T^*\mathbb{C}^{N\times\ell})$ via

$$\mathrm{ad}_E O := [\mu(E), O] = [TQT^{-1}, O] - [UQU^{-1}, O] - [ac, O].$$

The rest is a straightforward calculation using the identities

$$Q_1R_{12} - R_{12}Q_1 = \hbar(C_{12} - X_{12}), \tag{3.34}$$
$$Q_1\bar{R}_{12}^{-1} - \bar{R}_{12}^{-1}Q_1 = \hbar(X_{12} - Y_{12}), \tag{3.35}$$

as well as $Y_{12}\bar{R}_{12} = Y_{12}$, and

$$\begin{aligned}
&R_{12}X_{12} = X_{12}, \quad R_{21}X_{12} = X_{12}, \quad X_{12}R_{12} = X_{12}, \quad X_{12}R_{21} = X_{12},\\
&\bar{R}_{12}X_{12} = X_{12}, \quad X_{12}\bar{R}_{12} = X_{12}, \quad X_{12}\bar{R}_{21} = X_{12}, \quad X_{12}C_{12} = X_{12}.
\end{aligned}$$

For example, for $T$, we can use equation (3.34) to derive

$$\begin{aligned}
\mathrm{ad}_{E_1} T_2 &= [T_1Q_1T_1^{-1}, T_2]\\
&= -\hbar T_2T_1(C_{12} - X_{12})R_{21}T_1^{-1}\\
&= -\hbar C_{12}T_2 + \hbar T_2(T_1X_{12}T_1^{-1}).
\end{aligned}$$

□

*Remark* 3.9. The action of a Lie algebra element $M \in \mathfrak{gl}_N$ can be computed as follows

$$\mathrm{ad}_M\, T = \mathrm{Tr}_1(M_1\, \mathrm{ad}_{E_1}\, T_2) = -\hbar MT + \hbar T \sum_j e_{jj}(e_j^t T^{-1} M e) \tag{3.36}$$

We remark that for torus elements of the form $M = TDT^{-1}$ with $D$ diagonal, the action simplifies drastically due to

$$\hbar T \sum_j e_{jj}(e_j^t T^{-1} M e) = \hbar T D. \tag{3.37}$$

Furthermore, the action of any $M \in \mathfrak{gl}_N$ indeed preserves the mirabolic condition for $T$ as

$$\mathrm{ad}_M\, Te = -\hbar Me + \hbar T \sum_j e_j e_j^t T^{-1} M e = 0. \tag{3.38}$$

In this sense, we may say that the $\mathfrak{m}_N$-gauge symmetry is preserved at the quantum level. Furthermore, when $M$ lies in the mirabolic subalgebra, i.e. $Me = 0$, then $\mathrm{ad}_M\, T$ becomes equal to $\hbar MT$ as expected. More generally, the terms in lemma 3.8 that include $T_1 X_{12} T_1^{-1}$ and $U_1 X_{12} U_1^{-1}$ drop out when we restrict to the action of $\mathfrak{m}_N$ alone. In particular, $P$ is invariant under the action of the mirabolic subalgebra and $T, U, a$ transform in the vector representation of $\mathfrak{m}_N$, while $T^{-1}, U^{-1}, c$ transform in the covector representation.

### Algebra of mirabolic invariants

The next step is to determine the algebra of $\mathfrak{gl}_N$-invariants, i.e. the commutant of the moment map. To this end, we factorize the group $\mathrm{GL}_N$ as $\mathrm{GL}_N = \mathrm{T}_N \mathrm{M}_N$, where $\mathrm{T}_N$ is the maximal torus of diagonal invertible matrices and $\mathrm{M}_N$ is the mirabolic subgroup. This allows us to perform the reduction in two steps, first with respect to the mirabolic subgroup $\mathrm{M}_N$, and then with respect to the torus $\mathrm{T}_N$, and it will turn out to be interesting to consider mirabolic invariants in their own right. To this end, let us now describe the algebra of observables that is invariant under the mirabolic subgroup $\mathrm{M}_N$. Such invariants can be expressed via $Q$ and $P$ as well as the following variables:

**Definition 3.10.** The *dressed oscillators* are defined to be $A := T^{-1}a, B := cU$ and $C := cUP$ while the *Frobenius matrix* and the *Lax matrix* are defined to be $W := T^{-1}U$ and $L := T^{-1}UP$, respectively. We also define $\bar{A} := CL^{-1}$ and $\bar{C} := L^{-1}A$ for later purposes. It will be convenient for us to express these generators as polynomials as follows:

$$A^\alpha(z) := e^t \ell(z) A^\alpha, \quad C^\alpha(z) := C^\alpha \ell^*(z) e, \quad L(z,z') := e^t \ell(z) L \ell^*(z') e. \tag{3.39}$$

**Proposition 3.11.** *The algebra* $\left(\mathcal{O}_\hbar(T^*\mathrm{GL}_N) \otimes \mathcal{O}_\hbar(T^*\mathbb{C}^{N\times\ell})\right)^{\mathfrak{m}_N}$ *of operators invariant under* $\mathfrak{m}_N$ *is generated by the coefficients of* $Q, A, C, P^{\pm 1}$, *and* $L^{\pm 1}$, *subject to the relations* $C_1^\alpha Q_2 = (Q_2 - \hbar Y_{12}) C_1^\alpha$ *and* $L_1 Q_2 = (Q_2 - \hbar Y_{12}) L_1$ *as well as*

$$\begin{aligned}
A_1^\alpha A_2^\beta &= R_{12} A_2^\beta A_1^\alpha, & A_1^\alpha C_2^\beta + \hbar \delta^{\alpha\beta} \Delta_{12} L_2 &= C_2^\beta \bar{R}_{12}^{-1} A_1^\alpha, \\
C_1^\alpha C_2^\beta \bar{R}_{12}^{-1} &= C_2^\beta C_1^\alpha \bar{R}_{21}^{-1} R_{21}, & L_1 \bar{R}_{21}^{-1} L_2 \bar{R}_{12}^{-1} R_{12} &= R_{12} L_2 \bar{R}_{12}^{-1} L_1 \bar{R}_{21}^{-1}, \\
L_1 A_2^\alpha &= R_{12} A_2^\alpha L_1, & C_1^\alpha L_2 \bar{R}_{12}^{-1} &= L_2 C_1^\alpha \bar{R}_{21}^{-1} R_{21},
\end{aligned} \tag{3.40}$$

*with all other pairs commuting. In terms of polynomials, we can write*

$$A^\alpha(z_1)A^\beta(z_2) = \mathbf{R}_{12}A^\beta(z_2)A^\alpha(z_1), \tag{3.41}$$

$$A^\alpha(z_1)C^\beta(z_2) + \hbar\delta^{\alpha\beta}L(z_1,z_2) = \chi(z_2)\mathbf{R}^*_{12}[\chi(z_2)^{-1}C^\beta(z_2)]A^\alpha(z_1) + \hbar\chi(z_2)\partial_{12}S^{\alpha\beta}(z_1), \tag{3.42}$$

$$C^\alpha(z_1)C^\beta(z_2) = \mathbf{R}^*_{12}C^\beta(z_2)C^\alpha(z_1), \tag{3.43}$$

*where* $S^{\alpha\beta}(z) \coloneqq C^\alpha(z-Q)^{-1}A^\beta$.

*Proof.* The first six relations then follow from a straightforward application of the commutation relations of $\mathcal{O}_\hbar(T^*\mathrm{GL}_N)$, and we can rewrite them in terms of polynomials using the identities

$$R_{21}\ell^*_1(z_1)e_1\ell^*_2(z_2)e_2 = \mathbf{R}^*_{21}\ell^*_1(z_1)e_1\ell^*_2(z_2)e_2, \quad \ell^*_2(z)e_2C^\alpha_1 = C^\alpha_1\bar{R}^{-1}_{21}\ell^*_2(z)e_2,$$

and

$$e^t_2\ell_2(z_2)C^\alpha_1\bar{R}^{-1}_{21} = C^\alpha_1e^t_2\ell_2(z_2) - \hbar C^\alpha_1(z_2-Q_1)^{-1}e^t_2(\ell_2(z_2)-Y_{12}).$$

With this, we can derive the cross relation in polynomial form using:

$$e^t_2\ell_2(z_2)C^\alpha_1\bar{R}^{-1}_{21}A^\beta_2\ell^*_1(z_1)e_1 = \mathbf{R}^*_{21}[C^\alpha_1(z_1-Q_1)^{-1}e_1]\chi(z_1)e^t_2\ell_2(z_2)A^\beta_2 + \hbar\partial_{12}[S^{\alpha\beta}(z_2)]\chi(z_1),$$

while the relations for $C^\alpha$ in the dual Lagrange basis become

$$C^\alpha_1C^\beta_2\bar{R}^{-1}_{12}\ell^*_1(z_1)e_1\ell^*_2(z_2)e_2 = C^\beta_2C^\alpha_1\bar{R}^{-1}_{21}R_{21}\ell^*_1(z_1)e_1\ell^*_2(z_2)e_2,$$

which using the relations above simplifies to

$$C^\alpha(z_1)C^\beta(z_2) = \mathbf{R}^*_{12}C^\beta(z_2)C^\alpha(z_1). \qquad \square$$

*Remark* 3.12. We briefly note that the commutation relations for $L^{-1}$ follow from the commutation relations of $L$. In addition, we have $L^{-1}e = P^{-1}U^{-1}Te = P^{-1}e$ and $P^{-1}$ is diagonal, the coefficients of $P^{-1}e$ are the diagonal entries of $P^{-1}$, which means that the commutation relations for $P^{\pm1}$ also follow from the commutation relations of $L$.

An important property of the dressed oscillators $C^\alpha(z)$ is that they satisfy the exchange relation in spin space:

**Proposition 3.13.** *The commutation relations*

$$C^\alpha(z_1)C^\beta(z_2) = \mathbf{R}^*_{12}C^\beta(z_2)C^\alpha(z_1) \tag{3.44}$$

*may equivalently be written as*

$$C^1(z_1)C^2(z_2) = \hat{R}^{12}(z_1-z_2)C^2(z_2)C^1(z_1), \tag{3.45}$$

*where* $\hat{R}^{12}(z) = \frac{z}{z+\hbar} + \frac{\hbar}{z+\hbar}P^{12}$ *is Yang's unitary R-matrix acting in spin space* $\mathbb{C}^\ell \otimes \mathbb{C}^\ell$. *The upper indices in equation* (3.45) *denote auxiliary spin spaces.*

*Remark* 3.14. Equation (3.45) forms one half of a Faddeev–Zamolodchikov algebra [ZZ79; Fad95]. By Faddeev–Zamolodchikov algebra, we mean the algebra generated by a vector $C$ and covector $C^\dagger$ satisfying a generalization of canonical commutation relations. These have the form (3.45) together with its adjoint for $C^\dagger$ and the cross relation

$$C^1(z_1)C^{2\dagger}(z_2) = C^{2\dagger}(z_2)\hat{R}^{12}(z_1 - z_2)C^1(z_1) + \Delta^{12}\delta(z_1 - z_2) \tag{3.46}$$

where $\Delta^{12} = \sum_\alpha e_\alpha^t \otimes e_\alpha$ and $\delta(z_1 - z_2)$ is the Dirac delta function. A word of caution: While the dressed oscillators $A^\alpha(z)$ are natural candidates to play the role of $C^\dagger$, they do not satisfy the exchange relation for any $R$-matrix acting in spin space, as the operator $\mathbf{R}_{12}A^\beta(z_2)A^\alpha(z_1)$ includes terms with coincident arguments such as $A^\beta(z_1)A^\alpha(z_1)$ and $A^\beta(z_2)A^\alpha(z_2)$. Thus, unfortunately $A^\alpha(z)$ and $C^\alpha(z)$ do not form a Faddeev–Zamolodchikov algebra together.

### Fixing the residual torus gauge

In the preceding section, we have found the subalgebra of invariants under the action of the mirabolic subgroup. It remains to consider the residual gauge symmetry of the torus. There are several ways to go about finding invariant variables under the torus action. Let us try to proceed in analogy with the classical case and adjoin the inverses $t^{-1}$ to the algebra, with $t$ the unique diagonal matrices satisfying $te = \sum_{\rho=0}^{\ell-1} A^\rho$. In this case, we would want to build the fully gauge invariant combinations

$$\mathbf{a}^\alpha := t^{-1}A^\alpha, \quad \mathbf{c}^\alpha := C^\alpha t, \quad \mathbf{L} := t^{-1}Lt. \tag{3.47}$$

However, when we try to compute the commutation relations for these variables, we see that the components $t_i$ of $t$ commute with their inverses according to

$$[t_i^{-1}, t_j] = \frac{\hbar}{x_{ij}}(1 - t_j t_i^{-1} - t_i^{-1}t_j + t_i^{-1}t_j^2 t_i^{-1}) \tag{3.48}$$

and notice that this cannot be rearranged in a way that allows a reordering making $t_i$ and $t_i^{-1}$ into generators of a Poincaré–Birkhoff–Witt basis. The failure of the Poincaré–Birkhoff–Witt property is a strong indication that the variables $\mathbf{a}$ that contain $t^{-1}$ are problematic for quantization. In particular, there is no canonical ordering that makes cancelations manifest when computing commutators containing $\mathbf{a}$.

We can alternatively go the route of imposing a gauge fixing constraint via the Dirac method, and there are two interesting constraints to consider. Firstly, we note that the commutation relations for $A$ may be written in component form as

$$[A_i^\alpha, A_j^\beta] = -\frac{\hbar}{x_{ij}}(A_i^\alpha - A_j^\alpha)(A_i^\beta - A_j^\beta). \tag{3.49}$$

From this it is clear that $\sum_{\rho=0}^{\ell-1} A_i^\rho = 1$ for $i = 1, \dots, N$ is in fact a system of first-class constraints and that they give a complete gauge fixing of the torus action on $A$. However, the computations become slightly more transparent if we instead impose the gauge fixing $A_i^{\ell-1} = 1$, i.e. we impose that the $\ell$th spin components are fixed to one. [TODO: Chapters 1 and 4 instead fix $a^0 = 1$; consider fixing $A_i^0 = 1$ here and adapting $\mathfrak{m}_\ell$ accordingly.] Let $J$ be the right ideal generated by $A_i^{\ell-1} - 1$ for $i = 1, \dots, N$. This is also a system of first-class constraints, i.e. $[J, J] \subseteq J$, and importantly satisfies

$$[J, A_i^\alpha] \subseteq J, \quad [J, C_i^\alpha] \subseteq J - \hbar\delta^{\ell-1,\alpha}(\cdots), \tag{3.50}$$

so as long as $\alpha < \ell - 1$ we also have $[J, C_i^\alpha] \subseteq J$. According to the Dirac method, physical observables $O$ are characterized by the condition $[J, O] \subseteq J$. It follows that $x_i, A_i^\alpha, C_i^\alpha$ for $i = 1, \dots, N$ and $\alpha = 0, \dots, \ell - 2$ are all physical observables under this gauge fixing, i.e. we have essentially eliminated one unphysical polarization from the spins. We can solve for the unphysical $B_i^{\ell-1}$ via the moment map equation and the mirabolic condition for $W$, which gives

$$1 = \sum_j W_{ij} = A_i \sum_j \frac{1}{x_{ij} - \eta} B_j = \sum_j \frac{1}{x_{ij} - \eta} B_j^{\ell-1} + \sum_{\rho=0}^{\ell-2} A_i^\rho \sum_j \frac{1}{x_{ij} - \eta} B_j^\rho. \tag{3.51}$$

Letting $\Gamma_{ij} = \frac{1}{x_{ij} - \eta}$ be the Cauchy matrix, we can write

$$B_i^{\ell-1} = \sum_j \Gamma_{ij}^{-1} - \sum_{\rho=0}^{\ell-2} \sum_{jk} \Gamma_{ij}^{-1} A_j^\rho \Gamma_{jk} B_k^\rho, \tag{3.52}$$

which implies

$$W_{ij} = \bar{W}_{ij} + W_{ij}^{\mathrm{red}} - \bar{W}_{ij} (\bar{W}^{-1} W^{\mathrm{red}} e)_j, \tag{3.53}$$

where

$$\bar{W}_{ij} := \frac{\prod_{a(\neq i)} (x_{aj} - \eta)}{\prod_{a(\neq j)} x_{aj}}, \qquad W_{ij}^{\mathrm{red}} := \sum_{\rho=0}^{\ell-2} A_i^\rho \Gamma_{ij} B_j^\rho. \tag{3.54}$$

Note that $\bar{W}$ coincides with the spinless case [AF96]. The commutation relations between $A$ and $B$ can then be written as

$$\begin{aligned} [A_i^\alpha, B_j^\beta] &= -\hbar \delta^{\alpha\beta} W_{ij} \\ &= -\hbar \delta^{\alpha\beta} \Gamma_{ij} \sum_k \Gamma_{jk}^{-1} - \hbar \delta^{\alpha\beta} \sum_{\rho=0}^{\ell-2} \sum_{kl} (\delta_{ik} \delta_{jl} - \Gamma_{ij} \Gamma_{jk}^{-1}) A_k^\rho \Gamma_{kl} B_l^\rho \\ &= -\hbar \delta^{\alpha\beta} \bar{W}_{ij} - \hbar \delta^{\alpha\beta} W_{ij}^{\mathrm{red}} + \hbar \delta^{\alpha\beta} \bar{W}_{ij} (\bar{W}^{-1} W^{\mathrm{red}} e)_j \end{aligned} \tag{3.55}$$

and a sanity check reveals that the Poisson bracket that is the classical limit of these commutators does indeed satisfy the Jacobi identity.

We would like to rewrite the commutation relations (3.55) in polynomial form, and we can see that

$$\sum_{i,j=1}^N \ell_i(z_1) \bar{W}_{ij} \ell_j^*(z_2 - \eta) = \delta(z_1, z_2). \tag{3.56}$$

However, it is unclear to us how the other terms in (3.55) can be simplified in polynomial form. Instead, we can consider the alternative gauge fixing that imposes $e^t \chi'(Q)^{-1} W = e^t \chi'(Q)^{-1}$. This is indeed a first-class constraint. To see this, note that $W$ satisfies the commutation relation

$$R_{12} W_1 W_2 = W_2 W_1 R_{12} \tag{3.57}$$

and that $(e^t \chi'(Q)^{-1} \otimes e^t \chi'(Q)^{-1}) R_{12} = (e^t \chi'(Q)^{-1} \otimes e^t \chi'(Q)^{-1})$. The advantage of this constraint is that the moment map equation, which in terms of $W$ reads $QW - WQ -$

$\eta W = AB$, can be cast into a particularly nice polynomial form. Specifically, we use the identities

$$\ell(z)Q = z\ell(z) - \chi(z)\chi'(Q)^{-1}, \quad Q\ell^*(z) = z\ell^*(z) - \chi(z) \tag{3.58}$$

to obtain

$$\begin{aligned} e^t\ell(z)AB\ell^*(z')e &= e^t\ell(z)(QW - WQ - \eta)\ell^*(z')e \\ &= (z - z' - \eta)e^t\ell(z)W\ell^*(z')e - \chi(z)e^t\chi'(Q)^{-1}W\ell^*(z')e + \chi(z')e^t\ell(z)We \\ &= (z - z' - \eta)e^t\ell(z)W\ell^*(z')e - \chi(z) + \chi(z') \end{aligned} \tag{3.59}$$

Introducing $B(z) \coloneqq B\ell^*(z)e$ and $W(z,z') \coloneqq e^t\ell(z)W\ell^*(z')e$ and noticing that $\chi(z) - \chi(z') = (z - z')\delta(z,z')$ with $\delta(z,z') \coloneqq \sum_{i=1}^N \ell_i(z)\ell_i^*(z')$, we may thus write

$$A(z)B(z') = (z - z' - \eta)W(z,z') - (z - z')\delta(z,z'), \tag{3.60}$$

which is a particularly elegant form of the moment map equation. In particular, it implies

$$A(z)B(z - \eta) = -\eta\delta(z, z - \eta). \tag{3.61}$$

Further, noting that $A$ and $B$ commute according to $A_1^\alpha B_2^\beta + \hbar\delta^{\alpha\beta}\Delta_{12}W_2 = B_2^\beta A_1^\alpha$, we can write down their commutation relation in polynomial form:

$$A^\alpha(z_1)B^\beta(z_2) + \hbar\delta^{\alpha\beta}\sum_{\rho=0}^{\ell-1}\frac{A^\rho(z_1)B^\rho(z_2)}{z_1 - z_2 - \eta} = B^\beta(z_2)A^\alpha(z_1) + \frac{z_1 - z_2}{z_1 - z_2 - \eta}\delta(z_1,z_2), \tag{3.62}$$

which may be written in auxiliary space notation as a Faddeev–Zamolodchikov-type relation:

$$A^1(z_1)\hat{R}^{12}(z_1 - z_2 - \eta)B^2(z_2) = B^2(z_2)A^1(z_1) + \frac{z_1 - z_2}{z_1 - z_2 - \eta}\delta(z_1,z_2), \tag{3.63}$$

where we used $\hat{R}^{12}(z) = 1 + \frac{\hbar}{z}P^{12}$. However, despite these suggestive formulas, it is at this moment unclear to us how to find a good set of generators of the algebra of physical operators corresponding to this gauge fixing.

### 3.1.4 The reduced algebra

Besides the gauge fixing procedures above, we can also perform the reduction with respect to the residual torus in a fully gauge covariant way by simply building combinations that are products of strings of $L^{\pm1}$ and $Q$ sandwiched between $C$ and $A$, e.g. $CL^{-1}QLLACQA$. Such combinations are manifestly invariant under the residual torus symmetry (and by construction also under the mirabolic subgroup). Importantly, we can use the moment map equation, which may be cast as $[Q,L] = AC + \eta L$, to bring these strings into a canonical form where $L^{\pm1}$ always stands to the left of $Q$. A generating function for these canonical forms may be given by

$$S[n](z) \coloneqq CL^{n-1}(z - Q)^{-1}A = \sum_{r=0}^{\infty} z^{-r-1}CL^{n-1}Q^rA. \tag{3.64}$$

We can further use the moment map equation to obtain a recurrence relation expressing $S[n](z)$ in terms of $S[n-1](z)$ and $S[1](z)$ for any positive $n$:

**Proposition 3.15.** *The generating functions $S[n](z)$ with $n > 1$ satisfy the recurrence relation*

$$S[n](z) = \oint_C d\tau S[n-1](\tau)\partial_{\tau-\eta,z}S[1](z), \tag{3.65}$$

*where $C$ is a contour encircling $x_1, \dots, x_N$ counterclockwise, but no other poles. Here we have used the notation $\partial_{w,z}f(z) := \frac{f(w)-f(z)}{w-z}$.*

*Proof.* We first note that the moment map equation in components reads

$$x_iL_{ij} - L_{ij}x_j - \eta L_{ij} = A_iC_j,$$

which implies $(x_{ij} + \hbar - \eta)L_{ij} = A_iC_j$, so

$$L_{ij} = A_i\frac{1}{x_{ij}+\hbar-\eta}C_j.$$

Due to $\frac{1}{x_{ij}+\hbar-\eta} = \oint_C d\tau(\tau - x_i)^{-1}(\tau - x_j + \hbar - \eta)^{-1}$, we find

$$L = \oint_C d\tau(\tau - Q)^{-1}AC(\tau - \eta - Q)^{-1}.$$

Using $(\tau - \eta - Q)^{-1}(z - Q)^{-1} = \partial_{\tau-\eta,z}(z - Q)^{-1}$, it follows that

$$\begin{aligned} S[n](z) &= CL^{n-1}(z - Q)^{-1}A \\ &= \oint_C d\tau CL^{n-2}(\tau - Q)^{-1}AC(\tau - \eta - Q)^{-1}(z - Q)^{-1}A \\ &= \oint_C d\tau S[n-1](\tau)\partial_{\tau-\eta,z}S[1](z). \end{aligned}$$

□

**Quantum current**

The recurrence relation above implies that the positive modes $S[n](z)$ can be expressed in terms of $S(z) := S[1](z)$, which in turn can be expressed in terms of its residues

$$S_i := C_iA_i. \tag{3.66}$$

We call $S(z)$ the *quantum current*, stemming from the fact that the $S_i$ satisfy the following commutation relations that first appeared in [RS90] under that name. It can be checked that the classical limit of these commutation relations coincides with the Poisson bracket determined in [AF98] as expected.

**Proposition 3.16.** *For any $\epsilon \in \mathbb{C}^\times$, we have*

$$\hat{R}^{21}(x_{ji})S_i^1\hat{R}^{12}(x_{ij} - \eta)S_j^2 = S_j^2\hat{R}^{21}(x_{ji} - \eta)S_i^1\hat{R}^{12}(x_{ij}), \quad (i \neq j) \tag{3.67}$$

$$\hat{R}^{21}(\epsilon)S_i^1\hat{R}^{12}(-\eta)S_i^2 = S_i^2\hat{R}^{21}(-\eta)S_i^1\hat{R}^{12}(\epsilon), \tag{3.68}$$

*where $\hat{R}^{12}(z) = 1 + \frac{\hbar}{z}P^{12}$ is Yang's rational R-matrix acting in spin space $\mathbb{C}^\ell$ and we have used auxiliary space notation.*

*Proof.* Looking at the coefficients individually, we derive

$$[S_i^{\alpha\beta}, S_j^{\mu\nu}] = \hbar \sum_{\rho=0}^{\ell-1} S_j^{\mu\rho} \frac{\delta^{\alpha\nu}}{x_{ji} - \eta} S_i^{\rho\beta} - \hbar \sum_{\rho=0}^{\ell-1} S_i^{\alpha\rho} \frac{\delta^{\mu\beta}}{x_{ij} - \eta} S_j^{\rho\nu} \tag{3.69}$$

$$+ \frac{\hbar}{x_{ij} + \epsilon\delta_{ij}} \left( S_i^{\mu\beta} S_j^{\alpha\nu} + \hbar \sum_{\rho=0}^{\ell-1} S_i^{\mu\rho} \frac{\delta^{\alpha\beta}}{x_{ij} - \eta} S_j^{\rho\nu} \right) \tag{3.70}$$

$$- \frac{\hbar}{x_{ji} + \epsilon\delta_{ij}} \left( S_j^{\mu\beta} S_i^{\alpha\nu} + \hbar \sum_{\rho=0}^{\ell-1} S_j^{\mu\rho} \frac{\delta^{\alpha\beta}}{x_{ji} - \eta} S_i^{\rho\nu} \right) \tag{3.71}$$

making use of the identity

$$\begin{aligned} \bar{R}_{21}^{-1} R_{21} \bar{R}_{12} (e_{ii})_1 \bar{R}_{12}^{-1} (e_{jj})_2 R_{12} = (e_{jj})_2 \bar{R}_{21}^{-1} (e_{ii})_1 + \frac{\hbar}{x_{ij} + \epsilon\delta_{ij}} (e_{ii})_2 \bar{R}_{21}^{-1} (e_{jj})_1 C_{12} R_{12} \\ - \frac{\hbar}{x_{ji} + \epsilon\delta_{ij}} (e_{jj})_2 \bar{R}_{21}^{-1} (e_{ii})_1 C_{12} R_{12}, \end{aligned}$$

which holds for any $\epsilon \in \mathbb{C}^\times$. The relation can then be rewritten in $R$-matrix form as claimed. □

*Remark* 3.17. Let us examine skew-symmetry in the formula for the commutator $[S_i^{\alpha\beta}, S_j^{\mu\nu}]$. The line (3.69) gives something manifestly skew-symmetric, but the lines (3.70) and (3.71) do not. However, we have the identity

$$\begin{aligned} &\left( (e_{jj})_2 \bar{R}_{21}^{-1} (e_{ii})_1 + (e_{ii})_2 \bar{R}_{21}^{-1} (e_{jj})_1 \right) C_{12} R_{12} \\ &= \bar{R}_{21}^{-1} R_{21} \bar{R}_{12} C_{12} \left( (e_{ii})_2 \bar{R}_{21}^{-1} (e_{jj})_1 + (e_{jj})_2 \bar{R}_{21}^{-1} (e_{ii})_1 \right), \end{aligned} \tag{3.72}$$

giving the additional relation

$$\begin{aligned} &S_j^{\mu\beta} S_i^{\alpha\nu} + \hbar \sum_{\rho=0}^{\ell-1} S_j^{\mu\rho} \frac{\delta^{\alpha\beta}}{x_{ji} - \eta} S_i^{\rho\nu} + S_i^{\mu\beta} S_j^{\alpha\nu} + \hbar \sum_{\rho=0}^{\ell-1} S_i^{\mu\rho} \frac{\delta^{\alpha\beta}}{x_{ij} - \eta} S_j^{\rho\nu} \\ &= S_i^{\alpha\nu} S_j^{\mu\beta} + \hbar \sum_{\rho=0}^{\ell-1} S_i^{\alpha\rho} \frac{\delta^{\mu\nu}}{x_{ij} - \eta} S_j^{\rho\beta} + S_j^{\alpha\nu} S_i^{\mu\beta} + \hbar \sum_{\rho=0}^{\ell-1} S_j^{\alpha\rho} \frac{\delta^{\mu\nu}}{x_{ji} - \eta} S_i^{\rho\beta}, \end{aligned} \tag{3.73}$$

which is equivalent to skew-symmetry of the commutator.

*Remark* 3.18. We can now present the algebra of positive modes via $x_i$ and $S_i^{\alpha\beta}$ subject to the commutation relations $[x_i, S_j^{\alpha\beta}] = \hbar\delta_{ij} S_j^{\alpha\beta}$ as well as (3.67) and (3.68). We remark that the moment map equation for the variables $\mathbf{a}, \mathbf{c}, \mathbf{L}$ can be written as $Q\mathbf{L} - \mathbf{L}Q - \eta\mathbf{L} = \mathbf{ac}$, which can be solved in the same way as before to express $\mathbf{L}$ in terms of $\mathbf{a}$ and $\mathbf{c}$:

$$\mathbf{L} = \oint_C d\tau (\tau - Q)^{-1} \mathbf{a}\mathbf{c} (\tau - \eta - Q)^{-1}, \tag{3.74}$$

where again $C$ is a contour encircling $x_1, \ldots, x_N$ counterclockwise. It follows that the variables $\mathbf{a}, \mathbf{c}$, and $\mathbf{L}$ can all be expressed in terms of $S_i^{\alpha\beta}$ after imposing the moment map equation and adjoining certain inverses to the algebra, since

$$\mathbf{c}_i^\alpha = C_i^\alpha t_i = \sum_{\rho=0}^{\ell-1} S_i^{\alpha\rho}, \quad \mathbf{a}_i^\alpha = t_i^{-1} A_i^\alpha = (\mathbf{c}_i^\alpha)^{-1} S_i^{\alpha\alpha}. \tag{3.75}$$

Here, we notice again that the variables $\mathbf{a}_i^\alpha$ are problematic for quantization because they can only be expressed via inverses that have to be adjoined to the algebra, which generally destroys the Poincaré–Birkhoff–Witt property.

### Loop algebra and Yangian

Besides the quantum current, which may be seen as the basic building block from which we can build up the positive modes of the algebra $\mathfrak{A}_{N,\ell}$ of fully gauge-invariant observables, we can also find two important symmetry algebras within $\mathfrak{A}_{N,\ell}$: The loop algebra $L(\mathfrak{gl}_\ell)$ and the Yangian $Y(\mathfrak{gl}_\ell)$. The center of the loop algebra turns out to supply us with an infinite tower of commuting Hamiltonians, proving the integrability of the model, while the rest of the loop algebra equips the model with additional non-abelian integrals of motion. In contrast, the Yangian does not commute with the loop algebra and may be used to build up the spectrum of the commuting Hamiltonians via raising and lowering operators.

**Definition 3.19.** Introduce the full invariants

$$J[n] \coloneqq cg^n a = CL^{n-1}A = \oint dz S[n](z), \tag{3.76}$$

$$T(z) \coloneqq 1 + c(z-x)^{-1}a = 1 + \bar{A}(z-Q)^{-1}A, \tag{3.77}$$

$$\bar{T}(z) \coloneqq 1 - c(z+\hbar - gxg^{-1})^{-1}a = 1 - C(z-Q)^{-1}\bar{C}. \tag{3.78}$$

**Proposition 3.20.** *The coefficients $J^{\alpha\beta}[n]$ satisfy the relations for the loop algebra $L(\mathfrak{gl}_\ell)$.*

*Proof.* It may be quickly checked that $J^{\alpha\beta}[0] = c^\alpha a^\beta$ satisfy the commutation relations for the Lie algebra $\mathfrak{gl}_\ell$. Furthermore,

$$\begin{aligned}
&J^{\alpha\beta}[n]J^{\mu\nu}[1] \\
&\quad = c_1^\alpha (UP)_1 (T^{-1}UP)_1^{n-1} T_1^{-1} a_1^\beta c_2^\mu (UP)_2 T_2^{-1} a_2^\nu \\
&\quad = c_2^\mu c_1^\alpha (UP)_2 (UP)_1 \bar{R}_{21}^{-1} R_{21} \bar{R}_{12} (\bar{R}_{12}^{-1} T_1^{-1} U_1 P_1 \bar{R}_{21}^{-1} R_{21} \bar{R}_{12})^{n-1} \bar{R}_{12}^{-1} R_{12} T_2^{-1} T_1^{-1} a_2^\nu a_1^\beta \\
&\qquad + c_1^\alpha (UP)_1 (T_1^{-1} U_1 P_1)^{n-1} T_1^{-1} [a_1^\beta, c_2^\mu] (UP)_2 T_2^{-1} a_2^\nu \\
&\quad = J^{\mu\nu}[1]J^{\alpha\beta}[n] + \hbar\delta^{\alpha\nu} J^{\mu\beta}[n+1] - \hbar\delta^{\mu\beta} J^{\alpha\nu}[n+1].
\end{aligned}$$

□

**Proposition 3.21.** *The coefficients $T^{\alpha\beta}(z)$ satisfy the relations for the Yangian $Y(\mathfrak{gl}_\ell)$:*

$$T^{\alpha\beta}(z_1)T^{\mu\nu}(z_2) = T^{\mu\nu}(z_2)T^{\alpha\beta}(z_1) + \hbar\partial_{12} T^{\mu\beta}(z_2)T^{\alpha\nu}(z_1) \tag{3.79}$$

*Furthermore, since $\prod_{i=1}^N (z-x_i)T(z)$ is polynomial of degree $N$, we obtain an algebra homomorphism from the $N$-truncated Yangian $Y^N(\mathfrak{gl}_\ell)$ into $\mathfrak{A}_{N,\ell}$.*

*Proof.* For $T^{\alpha\beta}(z)$, we derive

$$\begin{aligned}
T^{\alpha\beta}(z_1)T^{\mu\nu}(z_2) &= \delta^{\alpha\beta}\delta^{\mu\nu} + \delta^{\alpha\beta}\bar{A}^\mu (z_2-Q)^{-1}A^\nu + \bar{A}^\alpha (z_1-Q)^{-1}A^\beta \\
&\quad + \bar{A}_2^\mu \bar{A}_1^\alpha R_{12}(z_1-Q_1)^{-1}R_{21}(z_2-Q_2)^{-1}R_{12}A_2^\nu A_1^\beta \\
&\quad - \hbar\delta^{\mu\beta}\bar{A}^\alpha (z_1-Q)^{-1}(z_2-Q)^{-1}A^\nu \\
&= T^{\mu\nu}(z_2)T^{\alpha\beta}(z_1) + \hbar\partial_{12}T^{\mu\beta}(z_2)T^{\alpha\nu}(z_1)
\end{aligned}$$

due to $(z_1-Q)^{-1}(z_2-Q)^{-1} = \partial_{12}(z_2-Q)^{-1}$ and the identity

$$\begin{aligned}
R_{12}(z_1-Q_1)^{-1}R_{21}(z_2-Q_2)^{-1}R_{12} &= (z_2-Q_2)^{-1}R_{12}(z_1-Q_1)^{-1} \\
&\quad + \hbar\partial_{12}(z_2-Q_2)^{-1}R_{12}(z_1-Q_1)^{-1}C_{12}R_{12}.
\end{aligned}$$

□

**Proposition 3.22.** *We have* $T(z)^{-1} = \bar{T}(z-\eta)$.

*Proof.* This follows from the moment map equation

$$AC = QL - LQ - \eta L \quad \text{or} \quad \bar{C}\bar{A} = L^{-1}Q - QL^{-1} - \eta L^{-1}$$

using $(z-Q)^{-1}Q = z(z-Q)^{-1} - 1$:

$$\begin{aligned}(z-Q)^{-1}AC(z-\eta-Q)^{-1} &= (z-Q)^{-1}L - L(z-\eta-Q)^{-1},\\ (z-\eta-Q)^{-1}\bar{C}\bar{A}(z-Q)^{-1} &= L^{-1}(z-Q)^{-1} - (z-\eta-Q)^{-1}L^{-1},\end{aligned}$$

hence

$$\begin{aligned}\sum_{\rho=0}^{\ell-1}(\delta^{\mu\rho} + \bar{A}^{\mu}(z-Q)^{-1}A^{\rho})(\delta^{\rho\beta} - C^{\rho}(z-\eta-Q)^{-1}\bar{C}^{\beta}) - \delta^{\mu\beta} = 0,\\ \sum_{\rho=0}^{\ell-1}(\delta^{\alpha\rho} - C^{\alpha}(z-\eta-Q)^{-1}\bar{C}^{\rho})(\delta^{\rho\nu} + \bar{A}^{\rho}(z-Q)^{-1}A^{\nu}) - \delta^{\alpha\nu} = 0,\end{aligned}$$

which means $T(z)^{-1} = \bar{T}(z-\eta)$. □

*Remark* 3.23. In particular, we notice that the poles of $\bar{T}(z-\eta)$ must be the zeros of $T(z)$ and vice versa. To compute the position of the poles, we must be careful about the ordering of the $x_i$ inside $\bar{T}(z)$, since $C$ and $\bar{C}$ do not commute with $x_i$. We follow the convention that the $x_i$ must be ordered to the left of $C$. In this case, the poles of $\bar{T}(z)$ arise at $z = x_i - \hbar$, where the shift by $\hbar$ comes about by reordering. It follows that

$$T(x_i - \hbar + \eta) = 0. \tag{3.80}$$

**Theorem 3.24.** *The generator matrices* $T(z)$ *and* $\bar{T}(z)$ *satisfy the relations*

$$\hat{R}^{12}(z_1-z_2)T^1(z_1)T^2(z_2) = T^2(z_2)T^1(z_1)\hat{R}^{12}(z_1-z_2) \tag{3.81}$$

$$\bar{T}^1(z_1)\bar{T}^2(z_2)\hat{R}^{12}(z_1-z_2) = \hat{R}^{12}(z_1-z_2)\bar{T}^2(z_2)\bar{T}^1(z_1) \tag{3.82}$$

$$\bar{T}^1(z_1)\hat{R}^{21}(z_2-z_1-\eta)T^2(z_2) = T^2(z_2)\hat{R}^{21}(z_2-z_1-\eta)\bar{T}^1(z_1). \tag{3.83}$$

*with* $\bar{T}(z-\eta) = T(z)^{-1}$ *and* $\hat{R}^{12}(z) = \frac{z}{z+\hbar} + \frac{\hbar}{z+\hbar}P^{12}$.

Having established the loop algebra and Yangian relations, we my now determine their cross relation via the commutator between the Yangian and the quantum current:

**Proposition 3.25.** *The Yangian acts on the quantum current according to:*

$$\begin{aligned}[S^{\alpha\beta}(z_1), T^{\mu\nu}(z_2)] = \hbar T^{\mu\beta}(z_2)\partial_{12}S^{\alpha\nu}(z_1) + \hbar\delta^{\alpha\nu}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(z_2)\partial_{z_2-\eta,z_1}S^{\rho\beta}(z_1)\\ + \frac{\hbar^2\delta^{\alpha\beta}}{z_1-z_2}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(z_2)\big(\partial_{z_2-\eta,z_1}S^{\rho\nu}(z_1) - \partial_{z_2-\eta,z_2}S^{\rho\nu}(z_2)\big).\end{aligned} \tag{3.84}$$

*Proof.* We calculate

$$\begin{aligned} S^{\alpha\beta}(z_1)T^{\mu\nu}(z_2) &= S^{\alpha\beta}(z_1)\delta^{\mu\nu} + \bar{A}_2^{\mu}C_1^{\alpha}\bar{R}_{21}^{-1}(z_1-Q_1)^{-1}R_{12}(z_2-Q_2)^{-1}R_{12}A_2^{\nu}A_1^{\beta} \\ &\quad - \hbar\delta^{\mu\beta}C_1^{\alpha}(z_1-Q_1)^{-1}\partial_{12}(z_2-Q_2)^{-1}A_2^{\nu} \\ &= T^{\mu\nu}(z_2)S^{\alpha\beta}(z_1) + \hbar T^{\mu\beta}(z_2)\partial_{12}S^{\alpha\nu}(z_1) + \hbar\delta^{\alpha\nu}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(z_2)\partial_{z_2-\eta,z_1}S^{\rho\beta}(z_1) \\ &\quad + \frac{\hbar^2\delta^{\alpha\beta}}{z_1-z_2}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(z_2)\big(\partial_{z_2-\eta,z_1}S^{\rho\nu}(z_1) - \partial_{z_2-\eta,z_2}S^{\rho\nu}(z_2)\big), \end{aligned}$$

where we have used

$$\begin{aligned} &\bar{R}_{21}^{-1}(z_1-Q_1)^{-1}R_{21}(z_2-Q_2)^{-1}R_{12} \\ &\quad = (z_2-Q_2-\hbar Y)^{-1}\bar{R}_{21}^{-1}\big((z_1-Q_1)^{-1} + \hbar\partial_{12}(z_1-Q_1)^{-1}C_{12}R_{12},\big), \end{aligned}$$

and that the moment map equation implies

$$\bar{A}^{\mu}(z_2-Q)^{-1}L(z_1-Q)^{-1}A^{\beta} = \sum_{\rho=0}^{\ell-1}T^{\mu\rho}(z_2)\frac{S^{\rho\beta}(z_1)-S^{\rho\beta}(z_2-\eta)}{z_2-z_1-\eta},$$

which comes from considering

$$\bar{A}^{\mu}(z_2-Q)^{-1}AC(z_2-Q-\eta)^{-1}(z_1-Q)^{-1}A^{\beta}$$

using $(z_2-Q)^{-1}Q = (z_2-Q)^{-1}z_2 - 1$ as well as the identity $(z_2-Q-\eta)^{-1}(z_1-Q)^{-1} = (z_2-z_1-\eta)^{-1}\left((z_1-Q)^{-1}-(z_2-Q-\eta)^{-1}\right)$. □

*Remark* 3.26. This cross relation between $S(z_1)$ and $T(z_2)$ determines the cross relations between the Yangian and the loop algebra in $\mathfrak{A}_{N,\ell}$, which we expect to be identified with a truncated affine Yangian. However, carefully matching this cross relation to the relations of affine Yangians is a very non-trivial task, as the known presentations of affine Yangians are highly complicated and presented in very different generators.

**Corollary 3.27.** *It follows that $J[1] = \oint dz S(z)$ commutes with $T(z)$ according to*

$$[J[1]^{\alpha\beta}, T^{\mu\nu}(z)] = -\hbar T^{\mu\beta}(z)S^{\alpha\nu}(z) + \hbar\delta^{\alpha\nu}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(z)S^{\rho\beta}(z-\eta) + \hbar^2\delta^{\alpha\beta}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(z)\Delta S^{\rho\nu}(z), \tag{3.85}$$

*where we have introduced the finite difference $\Delta f(z) \coloneqq \frac{f(z)-f(z-\eta)}{\eta}$.*

*Remark* 3.28. Let us examine the pole structure of (3.85), where we again have to be careful to order all the $x_i$ to the left. Indeed, we see that the right-hand side has simple poles at $x_i - \hbar + \eta$ that do not appear on the left-hand side. The vanishing of the residue yields the following additional relation:

$$0 = -\eta\hbar\delta^{\alpha\nu}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(x_i-\hbar+\eta)S_i^{\rho\beta} + \hbar^2\delta^{\alpha\beta}\sum_{\rho=0}^{\ell-1}T^{\mu\rho}(x_i-\hbar+\eta)S_i^{\rho\nu}, \tag{3.86}$$

but this is trivially satisfied, since we already saw that $T(x_i-\hbar+\eta) = 0$. Besides the poles at $x_i-\hbar+\eta$, we find poles at $z = x_i$ as well as $z = x_i - \hbar$, but these are matched with poles on the left-hand side.

At this point we can use (3.85) to check whether the action of $\mathfrak{A}_{1,\ell}$ coincides with the action of the loop Yangian in the simplest non-trivial representation—the vector representation $\mathbb{C}^\ell[x]$. Concretely, the action of the Yangian and loop algebra on $\mathbb{C}^\ell[x]$ are

$$T^{\alpha\beta}(z) \mapsto \delta_{\alpha\beta} - \frac{\hbar}{z-x} e_{\alpha\beta}, \qquad J[1]^{\alpha\beta} \mapsto e_{\alpha\beta} e^{-\hbar\partial}. \tag{3.87}$$

Due to our specification $N = 1$, we have $J[1] = S_1$ and $S(z) = \frac{1}{z-x_1+\hbar} S_1$, and indeed we find that (3.85) is satisfied for $\eta = (\ell+1)\hbar$. More generally, we conjecture that $\mathfrak{A}_{N,\ell}$ can be represented on the Schur–Weyl bimodule $(\mathbb{C}^\ell)^{\otimes N} \otimes \mathbb{C}[x_1, \dots, x_N]$, which also appears in the quantum spin Ruijsenaars–Schneider models of [KL24].

### 3.1.5 Calculation of the spectrum

We now come to a proposal to calculate eigenvalues and eigenvectors of the lowest Hamiltonian $H := \operatorname{Tr} J[1] = \sum_{i=1}^N \operatorname{Tr} S_i$. While it is a very ambitious goal to find eigenfunctions for the rational potential even in the spinless case $\ell = 1$, requiring heavy methods of functional analysis [HR12; HR17; HR19], we can hope to at least discover the spectrum and Fock space structure of an interesting subsector of the model with the help of the Yangian. Indeed, from (3.85), we derive

$$[H, T(z)] = \hbar(\ell\hbar - \eta) T(z) \Delta S(z), \tag{3.88}$$

which closely resembles the way creation operators should commute with the Hamiltonian. We remark that the Yangian becomes a symmetry of the Hamiltonian at $\eta = \ell\hbar$, which is the critical level in the language of [RS90]. The appearance of a critical level should be compared with the Gaudin model, where the commutator between the Segal–Sugawara element and the current of the affine Kac–Moody algebra is proportional to $\kappa - \kappa_c$ (where $\kappa$ is the level and $\kappa_c$ is the critical level) [Fre10]. On the other hand, we conclude that in the case of non-critical level $\eta \neq \ell\hbar$, we can use the generators of the Yangian as creation operators to move between energy levels and build up the spectrum of the Hamiltonian.

To this end, let $E := \oint dz T(z)$, which is the generator of $\mathfrak{gl}_\ell$-symmetry and commutes with $S(z)$ according to

$$[S^{\alpha\beta}(z), E^{\mu\nu}] = -\hbar(\delta^{\mu\beta} S^{\alpha\nu}(z) - \delta^{\alpha\nu} S^{\mu\beta}(z)). \tag{3.89}$$

While it is at the moment unclear to us how to formulate a highest weight condition for both the Yangian and the Hamiltonian $H$, we may consider a vacuum vector $|0\rangle$ satisfying

$$S^{\alpha\beta}(z)|0\rangle = \delta^{\alpha\beta} \sum_{i=1}^N \frac{1}{z - x_i + \hbar} \prod_{j(\neq i)} \frac{x_{ij} - \hbar}{x_{ij}} |0\rangle, \quad E^{\alpha\alpha}|0\rangle = \delta^{\alpha,\ell-1} N|0\rangle, \quad E^{\alpha\beta}|0\rangle = 0, \quad \alpha > \beta. \tag{3.90}$$

It is easily checked that these conditions are compatible with the commutation relations of $S(z)$ and $E$ when $\eta = -\hbar$, so we will set $\eta = -\hbar$ in the following. It follows that the vacuum energy is given by

$$H|0\rangle = \ell \sum_{i=1}^N \prod_{j(\neq i)} \frac{x_{ij} - \hbar}{x_{ij}} |0\rangle = N\ell|0\rangle. \tag{3.91}$$

We now consider the creation operator $\bar{\mathbf{A}}^{\mu}(w) \coloneqq T^{\mu,\ell-1}(w)$ for $\mu = 0,\dots,\ell-1$, which is equal to $\bar{A}^{\mu}(w-Q)^{-1}e$ under the gauge condition $A_i^{\ell-1} = 1$, and build up states of the form

$$\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle, \tag{3.92}$$

which are subject to the exchange relation

$$\begin{aligned}&\bar{\mathbf{A}}^{1}(w_1)\cdots\bar{\mathbf{A}}^{a}(w_a)\bar{\mathbf{A}}^{a+1}(w_{a+1})\cdots\bar{\mathbf{A}}^{k}(w_k)|0\rangle\\ &\quad= \hat{R}^{a,a+1}(w_a-w_{a+1})\bar{\mathbf{A}}^{1}(w_1)\cdots\bar{\mathbf{A}}^{a+1}(w_{a+1})\bar{\mathbf{A}}^{a}(w_a)\cdots\bar{\mathbf{A}}^{k}(w_k)|0\rangle,\end{aligned} \tag{3.93}$$

where we have used auxiliary space notation in spin space. Acting with $S^{\alpha\alpha}(z)$ for $\alpha = 0,\dots,\ell-2$, we obtain

$$\begin{aligned}&S^{\alpha\alpha}(z)\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle\\ &\quad= \frac{1}{2}\sum_{i=1}^{N}\frac{1}{z-x_i+\hbar}\left(1+\prod_{a=1}^{k}(1-2\hbar V(w_a-x_i+\hbar))\right)\prod_{j(\neq i)}\frac{x_{ij}-\hbar}{x_{ij}}\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle\end{aligned} \tag{3.94}$$

as well as

$$\begin{aligned}&S^{\ell-1,\ell-1}(z)\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle\\ &\quad= \sum_{i=1}^{N}\frac{1}{z-x_i+\hbar}\prod_{a=1}^{k}(1-2\hbar V(w_a-x_i+\hbar))\prod_{j(\neq i)}\frac{x_{ij}-\hbar}{x_{ij}}\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle,\end{aligned} \tag{3.95}$$

where we have introduced the potential $V(z) \coloneqq \frac{1}{z} - \frac{1}{z+\hbar}$. This implies

$$\begin{aligned}&H\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle\\ &\quad= \left(\frac{N(\ell-1)}{2}+\frac{\ell+1}{2}\sum_{i=1}^{N}\prod_{a=1}^{k}(1-2\hbar V(w_a-x_i+\hbar))\prod_{j(\neq i)}\frac{x_{ij}-\hbar}{x_{ij}}\right)\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle.\end{aligned} \tag{3.96}$$

Importantly, this is not an eigenvalue equation since the term in front depends on $x_1,\dots,x_N$, which do not act as a scalar on the vacuum. However, we can define the rescaled and shifted Hamiltonian $\tilde{H} \coloneqq \frac{2}{\ell+1}H - \frac{\ell-1}{\ell+1}N$ and multiply the state with an arbitrary function $\psi(x_1,\dots,x_N)$ giving

$$\begin{aligned}&\tilde{H}\psi(x_1,\dots,x_N)\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle\\ &= \sum_{i=1}^{N}\psi(x_1,\dots,x_i-\hbar,\dots,x_N)\prod_{a=1}^{k}(1-2\hbar V(w_a-x_i+\hbar))\prod_{j(\neq i)}\frac{x_{ij}-\hbar}{x_{ij}}\bar{\mathbf{A}}^{\mu_1}(w_1)\cdots\bar{\mathbf{A}}^{\mu_k}(w_k)|0\rangle.\end{aligned} \tag{3.97}$$

This becomes an eigenvalue equation when $\psi$ fulfills the difference equation

$$\sum_{i=1}^{N}\prod_{a=1}^{k}(1-2\hbar V(w_a-x_i+\hbar))\prod_{j(\neq i)}\frac{x_{ij}-\hbar}{x_{ij}}e^{-\hbar\partial_i}\psi = \lambda\psi. \tag{3.98}$$

It should be remarked that the energy eigenvalues of the eigenstates determined by solutions of (3.98) do not depend on the spin configuration given by $\mu_1,\dots,\mu_k$. This is a phenomenon observed more generally in spin-type integrable many-body systems. For example, the spectrum of the rational spin Calogero model considered in [BM24] does not

depend on the underlying spin configurations either. Instead, the introduction of spins enters into the degeneracies of the energy spectrum and modifies the exchange statistics.

We note that for $k = 0$ the difference operator in (3.98) reproduces exactly the spinless quantum Ruijsenaars–Schneider Hamiltonian with shift parameter $\eta = -\hbar$. Its simplest entire solution is given by plane waves multiplied with an attractive and symmetric Vandermonde-type factor:

$$\psi(x_1, \dots, x_N) := \exp\left(\sum_{i=1}^{N} \frac{p_i x_i}{\hbar}\right) \Delta(x_1, \dots, x_N), \quad \Delta(x_1, \dots, x_N) := \prod_{i<j} \operatorname{sinc}\left(2\pi \frac{x_i - x_j}{\hbar}\right) \tag{3.99}$$

with the unnormalized $\operatorname{sinc}(x) := \sin(x)/x$. The corresponding eigenvalue is $\lambda = \sum_{i=1}^{N} e^{-p_i}$ and its $\mathfrak{gl}_\ell$-degeneracy is $\operatorname{Sym}^N \mathbb{C}^\ell$ coming from the action of $E$ on the vacuum. We also know that $J[1]^{\alpha\alpha}$ for $\alpha = 0, \dots, \ell - 1$ acts as a symmetry of the Hamiltonian, but we find

$$J[1]^{\alpha\alpha} \psi|0\rangle = \sum_{i=1}^{N} \prod_{j(\neq i)} \frac{x_{ij} - \hbar}{x_{ij}} e^{-\hbar\partial_i} \psi|0\rangle = \lambda \psi|0\rangle, \tag{3.100}$$

so this does not produce more states. We conclude that the $L(\mathfrak{gl}_\ell)$-degeneracy of these eigenstates is just the evaluation module $\operatorname{Sym}^N \mathbb{C}^\ell$. This should be compared with subsection 3.1.4 of [LPS22], where it is found that the ground state of their model has degeneracy $\operatorname{Sym}^N \mathbb{C}^\ell$.

In the case $k = 1$, we find that the mirabolic subalgebra $\mathfrak{m}_\ell = \langle e_{\alpha\beta} \mid \alpha < \ell - 1 \rangle \subseteq \mathfrak{gl}_\ell$ acts as

$$E^{\alpha\beta} \psi \bar{\mathbf{A}}^\mu(w)|0\rangle = -\hbar \delta^{\mu\beta} \psi \bar{\mathbf{A}}^\alpha(w)|0\rangle + \psi \bar{\mathbf{A}}^\mu(w) E^{\alpha\beta}|0\rangle. \tag{3.101}$$

However, the action of the full $\mathfrak{gl}_\ell$ contains extra terms involving Yangian generators that cannot be simplified without imposing further relations on this representation, which is beyond the scope of this thesis. This and a detailed description of the spectrum for higher $k$ will be subject of future work.

## 3.2 A new presentation of the loop Yangian?

## 3.3 Elliptic spin Ruijsenaars–Schneider models from Poisson reduction

# 4

# Spin Ruijsenaars–Schneider models from Coulomb branches

**This chapter derives spin Ruijsenaars–Schneider (RS) models of $N$ particles with $\ell$ spin states from Coulomb branches of quiver gauge theories, where the quiver is the necklace quiver with $\ell$ nodes and gauge group $\mathrm{U}(N)$ at each node. For the rational model, we start from an $\eta$-deformed GKLO representation of the homological Coulomb branch of a 3d $\mathcal{N}=4$ gauge theory, where $\eta$ is the coupling constant, and show that the resulting Poisson algebra carries representations of the $N$-truncated affine Yangian and of the loop algebra $L(\mathfrak{gl}_\ell)$. Using an algebra of $L$-operators, we show that the model is superintegrable and that its equations of motion are those of the spin RS model of Krichever and Zabrodin, and we compute the Poisson brackets of the spin variables, which reproduce the known brackets of the rational spin RS model after a rescaling. The same programme is carried out for the trigonometric model, using the $K$-theoretic Coulomb branch of a 4d $\mathcal{N}=2$ gauge theory, where the quantum toroidal algebra takes the place of the affine Yangian. Finally, we quantize the trigonometric model: we construct a quantum $L$-operator algebra inside the quantized $K$-theoretic Coulomb branch, write the commuting Hamiltonians in terms of generalized Macdonald operators, identify a horizontal quantum loop algebra whose Bethe subalgebra contains the defining Hamiltonian, and determine the commutation relations and Heisenberg equations of the physical spin variables.**

## 4.1 Rational spin Ruijsenaars–Schneider model from 3d $\mathcal{N}=4$ gauge theory

### 4.1.1 GKLO representation of the homological Coulomb branch

Let $\mathfrak{M}_{N,\ell}$ be the Coulomb branch of the 3d $\mathcal{N}=4$ necklace quiver gauge theory with $\ell$ nodes of rank $N$ (see section 2.3), and let $\mathfrak{C}_{N,\ell} \coloneqq \mathbb{C}[\mathfrak{M}_{N,\ell}]$ be its (homological) Coulomb

branch algebra. We proceed along the lines of [Ger+05] to construct an $\eta$-deformed GKLO representation of $\mathfrak{C}_{N,\ell}$. To this end, we give the following

**Definition 4.1.** Let $\mathfrak{A}_{N,\ell}$ be the commutative $\mathbb{C}[\![\eta]\!]$-algebra

$$\mathfrak{A}_{N,\ell} := \mathbb{C}[\![\eta]\!][\varphi_i^\alpha, (P_i^\alpha)^{\pm 1}][(\varphi_i^\alpha - \varphi_j^\beta)^{-1}]/J, \tag{4.1}$$

where the generators $\varphi_i^\alpha, P_i^\alpha$ have indices $i = 1, \dots, N$ and $\alpha \in \mathbb{Z}$, we localize at the elements $\varphi_i^\alpha - \varphi_j^\beta$ for $(i, \alpha \bmod \ell) \neq (j, \beta \bmod \ell)$, and $J$ is the ideal generated by the cyclic relations

$$\varphi_i^{\alpha+\ell} = \varphi_i^\alpha + \eta, \qquad P_i^{\alpha+\ell} = P_i^\alpha. \tag{4.2}$$

We then make $\mathfrak{A}_{N,\ell}$ into a Poisson $\mathbb{C}[\![\eta]\!]$-algebra via the log-canonical Poisson bracket

$$\{\varphi_i^\alpha, P_j^\beta\} = \delta_{ij}\delta^{\alpha\beta} P_j^\beta. \tag{4.3}$$

Here and in the following, Kronecker deltas $\delta^{\alpha\beta}$ of node indices are understood modulo $\ell$. We furthermore write

$$\chi^\alpha(z) := \prod_{k=1}^N (z - \varphi_k^\alpha), \qquad \kappa^{\alpha\beta} := 2\delta^{\alpha\beta} - \delta^{\alpha+1,\beta} - \delta^{\alpha,\beta+1}, \tag{4.4}$$

where $\kappa^{\alpha\beta}$ is the Cartan matrix of the affine Lie algebra $\widehat{\mathfrak{sl}}_\ell$.

**Proposition 4.2.** *There is an injective homomorphism $\psi : \mathfrak{C}_{N,\ell} \to \mathfrak{A}_{N,\ell}/\eta\mathfrak{A}_{N,\ell}$ of Poisson algebras that sends*

$$\psi : \quad \varphi_i^\alpha \mapsto \varphi_i^\alpha, \qquad u_i^{\alpha\pm} \mapsto \pm (P_i^\alpha)^{\pm 1} \frac{\chi^{\alpha\pm 1}(\varphi_i^\alpha)}{\prod_{j\neq i}(\varphi_i^\alpha - \varphi_j^\alpha)} + \eta\mathfrak{A}_{N,\ell}. \tag{4.5}$$

*Remark* 4.3. Because of the existence of $\psi$, we justify abusing notation and writing

$$u_i^{\alpha\pm} := \pm (P_i^\alpha)^{\pm 1} \frac{\chi^{\alpha\pm 1}(\varphi_i^\alpha)}{\prod_{j\neq i}(\varphi_i^\alpha - \varphi_j^\alpha)} \in \mathfrak{A}_{N,\ell} \tag{4.6}$$

as a shorthand. Henceforth we will only be working with these *$\eta$-deformed* monopole operators inside $\mathfrak{A}_{N,\ell}$.

*Proof.* When $\ell > 2$, we compute the Poisson brackets inside $\mathfrak{A}_{N,\ell}$ to be [TODO: The coefficients differ from the brackets (2.99) ($1/z$ on the diagonal, $1/(z \mp m)$ for neighbours) versus $\kappa^{\alpha\beta}/z$ here ($2/z$ and $-1/z$). Check normalization of $u^\pm$ and of the bracket.]

$$\{u_i^{\alpha\pm}, u_j^{\beta\pm}\} = \pm(1 - \delta_{ij}\delta^{\alpha\beta})\kappa^{\alpha\beta} u_i^{\alpha\pm} u_j^{\beta\pm} \begin{cases} \frac{1}{\varphi_i^\ell - \varphi_j^{\ell-1}}, & (\alpha,\beta) = (0, \ell-1), \\ \frac{1}{\varphi_i^{\ell-1} - \varphi_j^\ell}, & (\alpha,\beta) = (\ell-1, 0), \\ \frac{1}{\varphi_i^\alpha - \varphi_j^\beta}, & \text{otherwise.} \end{cases}$$

For $\ell = 2$, we have

$$\{u_i^{\alpha\pm}, u_j^{\beta\pm}\} = \pm(1 - \delta_{ij}\delta^{\alpha\beta})\kappa^{\alpha\beta} u_i^{\alpha\pm} u_j^{\beta\pm} \begin{cases} \frac{1}{2}\left(\frac{1}{\varphi_i^0 - \varphi_j^1} + \frac{1}{\varphi_i^2 - \varphi_j^1}\right), & (\alpha,\beta) = (0,1), \\ \frac{1}{2}\left(\frac{1}{\varphi_i^1 - \varphi_j^0} + \frac{1}{\varphi_i^1 - \varphi_j^2}\right), & (\alpha,\beta) = (1,0), \\ \frac{1}{\varphi_i^\alpha - \varphi_j^\beta}, & \text{otherwise,} \end{cases}$$

and for $\ell = 1$, we have

$$\{u_i^{0\pm}, u_j^{0\pm}\} = \pm(1 - \delta_{ij}) u_i^{0\pm} u_j^{0\pm} \left( \frac{2}{\varphi_i^0 - \varphi_j^0} - \frac{1}{\varphi_i^0 - \varphi_j^1} - \frac{1}{\varphi_i^1 - \varphi_j^0} \right).$$

Since $\frac{1}{\varphi_i^\ell - \varphi_j^{\ell-1}} \equiv \frac{1}{\varphi_i^0 - \varphi_j^{\ell-1}} \mod \eta\mathfrak{A}_{N,\ell}$, the result follows. □

### 4.1.2 Affine Yangian of $\mathfrak{gl}_\ell$

With this in hand, we may define the generating series

$$e^\alpha(z) \coloneqq \sum_{i=1}^N \frac{u_i^{\alpha+}}{z-\varphi_i^\alpha} \in \mathfrak{A}_{N,\ell}[\![z^{-1}]\!], \qquad f^\alpha(z) \coloneqq \sum_{i=1}^N \frac{u_i^{\alpha-}}{z-\varphi_i^\alpha} \in \mathfrak{A}_{N,\ell}[\![z^{-1}]\!], \tag{4.7}$$

as well as

$$h^\alpha(z) \coloneqq \frac{\chi^{\alpha+1}(z)\chi^{\alpha-1}(z)}{\chi^\alpha(z)^2} \in \mathfrak{A}_{N,\ell}[\![z^{-1}]\!], \tag{4.8}$$

and check that they satisfy the relations of the classical limit of the affine Yangian:

**Proposition 4.4.** *The generating series $e^\alpha(z), f^\alpha(z), \chi^\alpha(z)$ define a representation of the classical limit of the $N$-truncated affine Yangian of $\mathfrak{gl}_\ell$ in the sense that $\chi^\alpha(z)$ is a polynomial of degree $N$ and the relations*

$$\{\chi^\alpha(z), \chi^\beta(w)\} = 0, \tag{4.9}$$

$$\{e^\alpha(z), f^\beta(w)\} = -\delta^{\alpha\beta}\frac{h^\alpha(z)-h^\beta(w)}{z-w}, \tag{4.10}$$

$$\{\chi^\alpha(z), e^\beta(w)\} = \delta^{\alpha\beta}\chi^\alpha(z)\frac{e^\beta(z)-e^\beta(w)}{z-w}, \tag{4.11}$$

$$\{\chi^\alpha(z), f^\beta(w)\} = -\delta^{\alpha\beta}\chi^\alpha(z)\frac{f^\beta(z)-f^\beta(w)}{z-w} \tag{4.12}$$

*are satisfied.*

*Remark* 4.5. This representation is nothing but the classical limit of the GKLO representation of the affine Yangian of $\mathfrak{gl}_\ell$ [Ger+05]. We note that $e^\alpha(z), f^\alpha(z)$ for $\alpha = 1, \dots, \ell-1$ and $\chi^\alpha(z)$ for $\alpha = 0, \dots, \ell-1$ generate the (finite) Yangian of $\mathfrak{gl}_\ell$ with quantum determinant

$$\operatorname{qdet}(z) = \prod_{\alpha=0}^{\ell-1}\frac{\chi^{\alpha+1}(z)}{\chi^\alpha(z)} = \frac{\chi^0(z-\eta)}{\chi^0(z)}. \tag{4.13}$$

When $\eta = 0$, it follows that $\operatorname{qdet}(z) = 1$, which reduces us to the Yangian of $\mathfrak{sl}_\ell$. In that sense, $\eta$ is the charge under the center of $\mathfrak{gl}_\ell$.

**Corollary 4.6.** *The zero modes*

$$E^\alpha \coloneqq \frac{1}{2\pi\mathrm{i}}\oint_\infty e^\alpha(z)dz = \sum_{i=1}^N u_i^{\alpha+}, \qquad F^\alpha \coloneqq \frac{1}{2\pi\mathrm{i}}\oint_\infty f^\alpha(z)dz = \sum_{i=1}^N u_i^{\alpha-}, \tag{4.14}$$

*and*

$$H^\alpha \coloneqq \frac{1}{2\pi\mathrm{i}}\oint_\infty h^\alpha(z)dz = \sum_{i=1}^N (2\varphi_i^\alpha - \varphi_i^{\alpha+1} - \varphi_i^{\alpha-1}), \tag{4.15}$$

*define a representation of $\widehat{\mathfrak{sl}}_\ell$ in the sense that*

$$\{H^\alpha, H^\beta\} = 0, \tag{4.16}$$

$$\{H^\alpha, E^\beta\} = \kappa^{\alpha\beta}E^\beta, \tag{4.17}$$

$$\{H^\alpha, F^\beta\} = -\kappa^{\alpha\beta}F^\beta, \tag{4.18}$$

$$\{E^\alpha, F^\beta\} = \delta^{\alpha\beta}H^\alpha. \tag{4.19}$$

### 4.1.3 $L$-operator algebra

Our goal is to exhibit $\mathfrak{A}_{N,\ell}$ as the Poisson algebra of the rational spin RS model. To make such a connection, it is useful to have an $L$-operator algebra at our disposal.

**Definition 4.7.** Introduce the one-site $L$-operators

$$L_{ij}^{\alpha\pm} := \frac{u_j^{\alpha+1,\pm}}{\varphi_j^{\alpha+1} - \varphi_i^{\alpha}} \in \mathfrak{A}_{N,\ell}, \tag{4.20}$$

as well as the total $L$-operator

$$L := L^{0+} \cdots L^{\ell-1,+}. \tag{4.21}$$

*Remark* 4.8. We note that $L^{\alpha+\ell,\pm} = L^{\alpha\pm}$ by the cyclic relations of $\mathfrak{A}_{N,\ell}$.

**Proposition 4.9.** *The one-site $L$-operators satisfy the Poisson brackets*

$$\{L_1^{\alpha\pm}, L_2^{\beta\pm}\} = \pm(\delta^{\alpha\beta} r^{\alpha} L_1^{\alpha\pm} L_2^{\beta\pm} - \delta^{\alpha\beta} L_1^{\alpha\pm} L_2^{\beta\pm} \underline{r}^{\alpha+1} + \delta^{\alpha+1,\beta} L_1^{\alpha\pm} \bar{r}_{21}^{\beta} L_2^{\beta\pm} - \delta^{\alpha,\beta+1} L_2^{\beta\pm} \bar{r}^{\alpha} L_1^{\alpha\pm}), \tag{4.22}$$

*where we have used the matrices from [AF96]:*

$$r^{\alpha} := \sum_{i\neq j} \frac{1}{\varphi_i^{\alpha} - \varphi_j^{\alpha}} (e_{ii} - e_{ij}) \otimes (e_{jj} - e_{ji}), \tag{4.23}$$

$$\bar{r}^{\alpha} := \sum_{i\neq j} \frac{1}{\varphi_i^{\alpha} - \varphi_j^{\alpha}} (e_{ii} - e_{ij}) \otimes e_{jj}, \tag{4.24}$$

$$\underline{r}^{\alpha} := \sum_{i\neq j} \frac{1}{\varphi_i^{\alpha} - \varphi_j^{\alpha}} (e_{ij} \otimes e_{ji} - e_{ii} \otimes e_{jj}). \tag{4.25}$$

**Corollary 4.10.** *The total $L$-operator satisfies*

$$\{L_1, L_2\} = r^0 L_1 L_2 - L_1 L_2 \underline{r}^0 + L_1 \bar{r}_{21}^0 L_2 - L_2 \bar{r}^0 L_1, \tag{4.26}$$

*which reproduces the Poisson bracket of the Lax matrix from [AF98].*

*Proof.* This follows from the Poisson algebra of the one-site $L$-operators and the identity

$$r^{\alpha} + \bar{r}_{21}^{\alpha} - \bar{r}^{\alpha} - \underline{r}^{\alpha} = 0. \qquad \square$$

**Corollary 4.11.** *The Hamiltonians $H[n] := \operatorname{Tr} L^n$ are mutually Poisson-commuting.*

**Proposition 4.12.** *The Hamiltonians $H[n]$ are central with respect to $\widehat{\mathfrak{sl}}_\ell$:*

$$\{H[n], E^{\alpha}\} = 0, \qquad \{H[n], F^{\alpha}\} = 0, \qquad \{H[n], H^{\alpha}\} = 0. \tag{4.27}$$

*Proof.* Let us consider the bracket $\{H[n], E^{\alpha}\}$ as an example. From the $L$-operator algebra, we derive the bracket

$$\{L_1^{\alpha+}, u_2^{\beta+1,+}\} = -\delta^{\alpha\beta} L_1^{\alpha+} u_2^{\beta+1,+} \underline{r}^{\alpha+1} - \delta^{\alpha,\beta+1} u_2^{\beta+1,+} \bar{r}^{\alpha} L_1^{\alpha+} - \delta^{\alpha\beta} Z L_1^{\alpha+} L_2^{\beta+} + \delta^{\alpha+1,\beta} L_1^{\alpha+} Z L_2^{\beta+}.$$

with $Z = \sum_{i=1}^{N} e_{ii} \otimes e_i^t$. Then

$$\begin{aligned}
\{H[n], E^{\alpha}\} &= \sum_{\mu=0}^{n\ell-1} \operatorname{Tr}_1 L_1^{0,+} \cdots L_1^{\mu-1,+} \{L_1^{\mu}, u_2^{\alpha,+}\} L_1^{\mu+1,+} \cdots L_1^{n\ell-1,+} e_2 \\
&= -\sum_{\mu=1}^{n\ell-1} \delta^{\mu\alpha} \operatorname{Tr}_1 L_1^{0,+} \cdots L_1^{\alpha-1,+} u_2^{\alpha,+} (\underline{r}^{\alpha} + \bar{r}^{\alpha}) L_1^{\alpha,+} \cdots L_1^{n\ell-1,+} e_2 - \delta^{\ell\alpha} \operatorname{Tr}_1 L_1^n u_2^{\alpha,+} (\underline{r}^0 + \bar{r}^0) e_2 \\
&\quad + \delta^{0,\alpha-1} \operatorname{Tr}_1 (L_1^n Z L_2^{\alpha-1,+} - Z L_2^{\alpha-1,+} L_1^n) e_2 \\
&= 0,
\end{aligned}$$

where we have used $(\underline{r}^{\alpha} + \bar{r}^{\alpha}) e_2 = 0$. $\square$

### 4.1.4 Superintegrability

We have already seen that the Hamiltonians $H[n]$ Poisson-commute with the generators $E^\alpha, F^\alpha, H^\alpha$, which define a representation of the loop algebra $\widehat{\mathfrak{sl}}_\ell$. It turns out that this representation factors through a representation of the loop algebra $L(\mathfrak{gl}_\ell)$ with generators $J^{\alpha\beta}[n] \in \mathfrak{A}_{N,\ell}$, which can be expressed in terms of the $L$-operators. To see this, let

$$V^\pm_{\alpha,\beta} := u^{\alpha\pm} L^{\alpha\pm} \cdots L^{\beta-1,\pm} e \tag{4.28}$$

with $u^{\alpha\pm} = (u_1^{\alpha\pm}, \dots, u_N^{\alpha\pm})$ and $e = (1, \dots, 1)^t$. Then we can define

$$J^{\alpha\beta}[0] := \begin{cases} V^-_{\beta,\alpha-1}, & \alpha > \beta \\ \sum_{i=1}^N (\varphi_i^\alpha - \varphi_i^{\alpha-1}), & \alpha = \beta \\ V^+_{\alpha,\beta-1}, & \alpha < \beta \end{cases}, \tag{4.29}$$

as well as

$$J^{\alpha\beta}[-n] := V^-_{\beta,\alpha+n\ell-1}, \qquad J^{\alpha\beta}[n] := V^+_{\alpha,\beta+n\ell-1}. \tag{4.30}$$

for $n > 0$. The coefficients $J^{\alpha\beta}[n]$ assemble into an $\ell \times \ell$ matrix $J[n]$.

**Proposition 4.13.** *The generators $J^{\alpha\beta}[n]$ define a representation of the loop algebra $L(\mathfrak{gl}_\ell)$:*

$$\{J^{\alpha\beta}[n], J^{\mu\nu}[m]\} = \delta^{\mu\beta} J^{\alpha\nu}[n+m] - \delta^{\alpha\nu} J^{\mu\beta}[n+m]. \tag{4.31}$$

*Remark* 4.14. It was already clear from [BDG15, §6.6] that $J^{\alpha\beta}[0]$ satisfies the relations of $\mathfrak{gl}_\ell$.

*Remark* 4.15. The loop algebra generators $J[n] = cg^n a$ of chapters 1 and 3 satisfy $\{J^{\alpha\beta}[n], J^{\mu\nu}[m]\} = \delta^{\alpha\nu} J^{\mu\beta}[n+m] - \delta^{\mu\beta} J^{\alpha\nu}[n+m]$, which differs from the bracket above by an overall sign. The two conventions are related by $J[n] \mapsto -J[n]$, or equivalently by transposition $J^{\alpha\beta}[n] \mapsto J^{\beta\alpha}[n]$.

**Lemma 4.16.** *We have* $\operatorname{Tr} J[n] = \eta H[n]$, *where* Tr *is the operation of taking the trace of an $\ell \times \ell$ matrix.*

*Remark* 4.17. In particular, the generators $J^{\alpha\beta}[n]$ define a representation of $L(\mathfrak{sl}_\ell)$ when $\eta = 0$, which again exhibits $\eta$ as the charge under the center of $\mathfrak{gl}_\ell$.

*Proof.* This follows from a telescopic argument similar to lemma 4.18. □

From this, we see that the Hamiltonians $H[n]$ are in the center of the subalgebra of $\mathfrak{A}_{N,\ell}$ generated by $J^{\alpha\beta}[n]$ and are thus part of a larger Poisson-commutative subalgebra given by the $N\ell$ algebraically independent Hamiltonians

$$J^{\alpha\alpha}[n], \qquad n = 1, \dots, N, \quad \alpha = 0, \dots, \ell - 1. \tag{4.32}$$

Since $\mathfrak{A}_{N,\ell}$ has $2N\ell$ algebraically independent generators as a $\mathbb{C}[\![\eta]\!]$-algebra, we conclude that $\mathfrak{A}_{N,\ell}$ describes an integrable model in the sense of the Liouville theorem. In fact, $\mathfrak{A}_{N,\ell}$ describes a *superintegrable* model, since the central Hamiltonians $H[n]$ do not just commute among each other, but also commute with the loop algebra $L(\mathfrak{gl}_\ell)$.

### 4.1.5 Equations of motion

In this section, we show that the superintegrable model defined by the Poisson algebra $\mathfrak{A}_{N,\ell}$ and the Hamiltonians $J^{\alpha\alpha}[n]$ is the spin Ruijsenaars–Schneider model introduced by Krichever and Zabrodin [KZ95]. To show the identification, we introduce

$$a^\alpha := L^{0+}\cdots L^{\alpha-1,+}e, \qquad c^\alpha := u^{\alpha+1,+}L^{\alpha+1,+}\cdots L^{\ell-1,+}, \tag{4.33}$$

for $\alpha = 0,\dots,\ell-1$. Notice that $a^0 = e$, in other words, we have the constraints $a_i^0 = 1$ for $i = 1,\dots,N$. This should be contrasted with [AF98; CF20; AO19; Fai26], where the spin vectors satisfy the alternative constraint $\sum_\rho a_i^\rho = 1$. However, they differ only by an overall rescaling of the spin variables.

**Lemma 4.18.** *The total L-operator can be written as*

$$L_{ij} = -\frac{\sum_{\rho=0}^{\ell-1} a_i^\rho c_j^\rho}{\varphi_i^0 - \varphi_j^\ell}. \tag{4.34}$$

*Proof.* Indeed,

$$\begin{aligned}
\sum_{\rho=0}^{\ell-1} a_i^\rho c_j^\rho &= \sum_{\rho=0}^{\ell-1}\sum_{k,l=1}^{N} (L^0\cdots L^{\rho-1})_{il} u_k^{\rho+1,+} (L^{\rho+1}\cdots L^{\ell-1})_{kj} \\
&= \sum_{\rho=0}^{\ell-1}\sum_{k,l=1}^{N} (L^0\cdots L^{\rho-1})_{il} L_{lk}^\rho (\varphi_k^{\rho+1} - \varphi_l^\rho)(L^{\rho+1}\cdots L^{\ell-1})_{kj} \\
&= \sum_{\rho=0}^{\ell-1}\sum_{k=1}^{N} (L^0\cdots L^{\rho})_{ik} (L^{\rho+1}\cdots L^{\ell-1})_{kj} \varphi_k^{\rho+1} - \sum_{\rho=0}^{\ell-1}\sum_{l=1}^{N} (L^0\cdots L^{\rho-1})_{il} (L^{\rho}\cdots L^{\ell-1})_{lj} \varphi_l^\rho \\
&= L_{ij}(\varphi_j^\ell - \varphi_i^0),
\end{aligned}$$

which yields the result. □

**Theorem 4.19.** *The time evolution under the Hamiltonian $H := \eta H[1]$ reproduces the equations of motion (1.28) with the identification $x_i = \varphi_i^0$ and the rational potential $V(z) = \frac{1}{z} - \frac{1}{z-\eta}$.*

*Proof.* Using the $L$-operator algebra, we find

$$\begin{aligned}
\dot{x}_i &= \{H, \varphi_i^0\} = -\eta L_{ii}, \\
\ddot{x}_i &= \{H, \{H, \varphi_i^0\}\} = \eta^2 \sum_{j(\neq i)} \frac{2}{\varphi_i^0 - \varphi_j^0} L_{ij} L_{ji}, \\
\dot{a}_i^\alpha &= \{H, a_i^\alpha\} = \eta \sum_{j(\neq i)} \frac{1}{\varphi_i^0 - \varphi_j^0} (a_i^\alpha - a_j^\alpha) L_{ij}, \\
\dot{c}_i^\alpha &= \{H, c_i^\alpha\} = -\eta \sum_{j(\neq i)} \frac{1}{\varphi_i^0 - \varphi_j^0} (c_i^\alpha L_{ij} + c_j^\alpha L_{ji}).
\end{aligned}$$

We then use lemma 4.18 and the identities

$$\begin{aligned}
-\eta \frac{1}{\varphi_i^0 - \varphi_j^0} \frac{1}{\varphi_i^0 - \varphi_j^\ell} &= V(x_i - x_j), \\
\eta^2 \frac{2}{\varphi_i^0 - \varphi_j^0} \frac{1}{\varphi_i^0 - \varphi_j^\ell} \frac{1}{\varphi_j^0 - \varphi_i^\ell} &= V(x_i - x_j) - V(x_j - x_i)
\end{aligned}$$

to arrive at the desired equations of motion. □

### 4.1.6 Poisson bracket of the spin variables

**Proposition 4.20.** *The Poisson brackets of the spins can be written as*

$$\{\varphi_i^0, a_j^\alpha\} = 0, \qquad \{\varphi_i^0, c_j^\alpha\} = \delta_{ij} c_j^\alpha, \tag{4.35}$$

$$\{a_i^\alpha, a_j^\beta\} = \frac{1-\delta_{ij}}{\varphi_i^0 - \varphi_j^0}(a_j^\alpha - a_i^\alpha)(a_i^\beta - a_j^\beta), \tag{4.36}$$

$$\{a_i^\alpha, c_j^\beta\} = \frac{1-\delta_{ij}}{\varphi_i^0 - \varphi_j^0}(a_j^\alpha - a_i^\alpha)c_j^\beta - \delta^{0\beta} L_{ij} a_i^\alpha + \delta^{\alpha\beta} L_{ij}, \tag{4.37}$$

$$\{c_i^\alpha, c_j^\beta\} = \frac{1-\delta_{ij}}{\varphi_i^0 - \varphi_j^0}(c_i^\alpha c_j^\beta + c_j^\alpha c_i^\beta) - \delta^{0\alpha} L_{ji} c_j^\beta + \delta^{0\beta} L_{ij} c_i^\alpha. \tag{4.38}$$

*Proof.* Let us consider the Poisson bracket $\{a_1^\alpha, a_2^\beta\}$ as an example, where 1 and 2 label auxiliary spaces. Introduce the partial monodromies $L^{\alpha,\beta} := L^\alpha \cdots L^\beta$. Then

$$\begin{aligned}
\{a_1^\alpha, a_2^\beta\} &= \sum_{\mu=0}^{\alpha-1}\sum_{\nu=0}^{\beta-1} L_1^{0,\mu-1} L_2^{0,\nu-1} \{L_1^\mu, L_2^\nu\} L_1^{\mu+1,\alpha-1} L_2^{\nu+1,\beta-1} e_1 e_2 \\
&= \sum_{\mu=0}^{\alpha-1}\sum_{\nu=0}^{\beta-1} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} r^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 - \sum_{\mu=1}^{\alpha}\sum_{\nu=1}^{\beta} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} \underline{r}^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 \\
&\quad + \sum_{\mu=1}^{\alpha}\sum_{\nu=1}^{\beta-1} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} \bar{r}_{21}^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 - \sum_{\mu=1}^{\alpha-1}\sum_{\nu=1}^{\beta} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} \bar{r}^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 \\
&= \delta_{\alpha>0,\beta>0} r^0 a_1^\alpha a_2^\beta = r^0 a_1^\alpha a_2^\beta,
\end{aligned}$$

where we used the identities

$$\underline{r}^\alpha e_1 e_2 = 0, \qquad (\underline{r}^\alpha + \bar{r}^\alpha) e_2 = 0, \qquad (\underline{r}^\alpha - \bar{r}_{21}^\alpha) e_1 = 0, \qquad r e_1 = 0, \qquad r e_2 = 0.$$

The other brackets follow similarly. In total, we arrive at

$$\begin{aligned}
\{a_1^\alpha, a_2^\beta\} &= r^0 a_1^\alpha a_2^\beta \\
\{a_1^\alpha, c_2^\beta\} &= -c_2^\beta \bar{r}^0 a_1^\alpha - \delta^{0\beta} Z a_1^\alpha L_2 + \delta^{\alpha\beta} Z e_1 L_2 \\
\{c_1^\alpha, c_2^\beta\} &= -c_1^\alpha c_2^\beta \underline{r}^0 + \delta^{0\beta} c_1^\alpha Z L_2 - \delta^{0\alpha} c_2^\beta Z_{21} L_1.
\end{aligned}$$

where $Z = \sum_{i=1}^N e_{ii} \otimes e_i^t$. In components, these are the claimed Poisson brackets. □

*Remark* 4.21. Due to the constraint $a_i^0 = 1$, the "physical" spin degrees of freedom of the rational spin RS model are actually restricted to $a^\alpha, c^\alpha$ for $\alpha > 0$. We note that they form a quadratic Poisson algebra.

**Corollary 4.22.** *The rescaling* $\hat{a}_i^\alpha := a_i^\alpha / \sum_\rho a_i^\rho$, $\hat{c}_i^\alpha := c_i^\alpha \sum_\rho a_i^\rho$, $\hat{L}_{ij} := \frac{\sum_\rho \hat{a}_i^\rho \hat{c}_j^\rho}{\varphi_i^0 - \varphi_j^0 - \eta}$ *reproduces the Poisson brackets from [AF98]:*

$$\{\varphi_i^0, \hat{a}_j^\alpha\} = 0, \qquad \{\varphi_i^0, \hat{c}_j^\alpha\} = \delta_{ij}\hat{c}_j^\alpha, \tag{4.39}$$

$$\{\hat{a}_i^\alpha, \hat{a}_j^\beta\} = \frac{1-\delta_{ij}}{\varphi_i^0 - \varphi_j^0}(\hat{a}_j^\alpha - \hat{a}_i^\alpha)(\hat{a}_i^\beta - \hat{a}_j^\beta), \tag{4.40}$$

$$\{\hat{a}_i^\alpha, \hat{c}_j^\beta\} = \frac{1-\delta_{ij}}{\varphi_i^0 - \varphi_j^0}(\hat{a}_j^\alpha - \hat{a}_i^\alpha)\hat{c}_j^\beta + \hat{a}_i^\alpha \hat{L}_{ij} - \delta^{\alpha\beta}\hat{L}_{ij}, \tag{4.41}$$

$$\{\hat{c}_i^\alpha, \hat{c}_j^\beta\} = \frac{1-\delta_{ij}}{\varphi_i^0 - \varphi_j^0}(\hat{c}_i^\alpha \hat{c}_j^\beta + \hat{c}_j^\alpha \hat{c}_i^\beta) - \hat{c}_i^\alpha \hat{L}_{ij} + \hat{c}_j^\beta \hat{L}_{ji}. \tag{4.42}$$

## 4.2 Trigonometric spin Ruijsenaars–Schneider model from 4d $\mathcal{N} = 2$ gauge theory

### 4.2.1 GKLO representation of the $K$-theoretic Coulomb branch

For the hyperbolic case, we proceed in essentially the same way, starting with a $t$-deformation of the GKLO presentation of the $K$-theoretic Coulomb branch written down in [FT19; Tsy23]. We will use essentially the same notation as in the rational case for this section, as no confusion should arise.

**Definition 4.23.** Let $\mathfrak{A}^K_{N,\ell}$ be the commutative $\mathbb{C}[\![t-1]\!]$-algebra

$$\mathfrak{A}^K_{N,\ell} := \mathbb{C}[\![t-1]\!][(Q_i^\alpha)^{\pm 1/2}, (P_i^\alpha)^{\pm 1}][((Q_i^\alpha/Q_j^\beta)^{1/2} - (Q_j^\beta/Q_i^\alpha)^{1/2})^{-1}]/J, \tag{4.43}$$

where the generators $Q_i^\alpha, P_i^\alpha$ have indices $i = 1, \dots, N$ and $\alpha \in \mathbb{Z}$, we localize at the elements $(Q_i^\alpha/Q_j^\beta)^{1/2} - (Q_j^\beta/Q_i^\alpha)^{1/2}$ for $(i, \alpha \bmod \ell) \neq (j, \beta \bmod \ell)$, and $J$ is the ideal generated by the cyclic relations

$$Q_i^{\alpha+\ell} = tQ_i^\alpha, \qquad P_i^{\alpha+\ell} = P_i^\alpha. \tag{4.44}$$

We then make $\mathfrak{A}^K_{N,\ell}$ into a Poisson $\mathbb{C}[\![t-1]\!]$-algebra via the doubly log-canonical Poisson bracket

$$\{Q_i^\alpha, P_j^\beta\} = \delta_{ij}\delta^{\alpha\beta} Q_i^\alpha P_j^\beta. \tag{4.45}$$

**Proposition 4.24.** *[TODO: The multiplicative $\chi^\alpha(z)$ is not defined (only $\tilde{\chi}^\alpha$ is).] There exists an injective homomorphism $\psi^K : \mathfrak{C}^K_{N,\ell} \to \mathfrak{A}^K_{N,\ell}/(t-1)\mathfrak{A}^K_{N,\ell}$ of Poisson algebras that sends*

$$\psi^K : \quad Q_i^\alpha \mapsto Q_i^\alpha, \qquad u_i^{\alpha\pm} \mapsto \pm (P_i^\alpha)^{\pm 1} \frac{\chi^{\alpha\pm 1}(Q_i^\alpha)}{\prod_{j\neq i}((Q_i^\alpha/Q_j^\alpha)^{1/2} - (Q_j^\alpha/Q_i^\alpha)^{1/2})} + (t-1)\mathfrak{A}^K_{N,\ell}. \tag{4.46}$$

*Remark* 4.25. We again use the existence of $\psi^K$ as a justification for writing

$$u_i^{\alpha\pm} := \pm (P_i^\alpha)^{\pm 1} \frac{\chi^{\alpha\pm 1}(Q_i^\alpha)}{\prod_{j\neq i}((Q_i^\alpha/Q_j^\alpha)^{1/2} - (Q_j^\alpha/Q_i^\alpha)^{1/2})} \in \mathfrak{A}^K_{N,\ell} \tag{4.47}$$

as a shorthand.

*Proof.* We compute the Poisson brackets inside $\mathfrak{A}^K_{N,\ell}$ for $\ell > 2$ to be

$$\{u_i^{\alpha\pm}, u_j^{\beta\pm}\} = \pm(1-\delta_{ij}\delta^{\alpha\beta})\kappa^{\alpha\beta}u_i^{\alpha\pm}u_j^{\beta\pm}\begin{cases}\frac{1}{2}\frac{Q_i^\ell+Q_j^{\ell-1}}{Q_i^\ell-Q_j^{\ell-1}}, & (\alpha,\beta)=(0,\ell-1)\\ \frac{1}{2}\frac{Q_i^{\ell-1}+Q_j^\ell}{Q_i^{\ell-1}-Q_j^\ell}, & (\alpha,\beta)=(\ell-1,0)\\ \frac{1}{2}\frac{Q_i^\alpha+Q_j^\beta}{Q_i^\alpha-Q_j^\beta}, & \text{otherwise,}\end{cases}$$

while for $\ell = 2$, we get

$$\{u_i^{\alpha\pm}, u_j^{\beta\pm}\} = \pm(1-\delta_{ij}\delta^{\alpha\beta})\kappa^{\alpha\beta}u_i^{\alpha\pm}u_j^{\beta\pm}\begin{cases}\frac{1}{4}\frac{Q_i^0+Q_j^{\ell-1}}{Q_i^0-Q_j^{\ell-1}}+\frac{1}{4}\frac{Q_i^\ell+Q_j^{\ell-1}}{Q_i^\ell-Q_j^{\ell-1}}, & (\alpha,\beta)=(0,\ell-1)\\ \frac{1}{4}\frac{Q_i^{\ell-1}+Q_j^0}{Q_i^{\ell-1}-Q_j^0}+\frac{1}{4}\frac{Q_i^{\ell-1}+Q_j^\ell}{Q_i^{\ell-1}-Q_j^\ell}, & (\alpha,\beta)=(\ell-1,0)\\ \frac{1}{2}\frac{Q_i^\alpha+Q_j^\beta}{Q_i^\alpha-Q_j^\beta}, & \text{otherwise,}\end{cases}$$

and for $\ell = 1$, it is

$$\{u_i^{0\pm}, u_j^{0\pm}\} = \pm(1-\delta_{ij})u_i^{0\pm}u_j^{0\pm}\left(\frac{Q_i^0+Q_j^0}{Q_i^0-Q_j^0}-\frac{1}{2}\frac{Q_i^1+Q_j^0}{Q_i^1-Q_j^0}-\frac{1}{2}\frac{Q_i^0+Q_j^1}{Q_i^0-Q_j^1}\right).$$

Since $\frac{Q_i^\ell+Q_j^{\ell-1}}{Q_i^\ell-Q_j^{\ell-1}} \equiv \frac{Q_i^0+Q_j^{\ell-1}}{Q_i^0-Q_j^{\ell-1}} \mod (t-1)\mathfrak{A}^K_{N,\ell}$, the result follows. □

### 4.2.2 Quantum toroidal algebra of $\mathfrak{gl}_\ell$

In the case of the $K$-theoretic Coulomb branch, the affine Yangian is replaced by the quantum toroidal algebra. To see this, we introduce the generating series

$$e^\alpha(z) := \sum_{i=1}^N \frac{u_i^{\alpha+}}{1-Q_i^\alpha/z} \in \mathfrak{A}^K_{N,\ell}[\![z^{-1}]\!], \qquad f^\alpha(z) := \sum_{i=1}^N \frac{u_i^{\alpha-}}{1-Q_i^\alpha/z} \in \mathfrak{A}^K_{N,\ell}[\![z^{-1}]\!], \tag{4.48}$$

as well as

$$h^\alpha(z) := \frac{\tilde\chi^{\alpha+1}(z)\tilde\chi^{\alpha-1}(z)}{\tilde\chi^\alpha(z)^2}\prod_{i=1}^N\frac{Q_i^\alpha}{(Q_i^{\alpha+1}Q_i^{\alpha-1})^{1/2}}, \qquad \tilde\chi^\alpha(z) := \prod_{i=1}^N(1-Q_i^\alpha/z). \tag{4.49}$$

We then expand them according to

$$e^\alpha(z) = \sum_{r\geq 0} e_r^\alpha z^{-r}, \qquad f^\alpha(z) = \sum_{r\geq 0} f_r^\alpha z^{-r}, \qquad h^\alpha(z) = \sum_{r\geq 0} k_r^{\alpha+}z^{-r} = \sum_{r\geq 0} k_r^{\alpha-}z^r. \tag{4.50}$$

**Proposition 4.26.** *The generating series $e^\alpha(z), f^\alpha(z), \tilde\chi^\alpha(z), h^\alpha(z)$ define a representation of the classical limit of the $N$-truncated positive half of the quantum toroidal algebra of $\mathfrak{gl}_\ell$ in the sense that*

$$\{k_r^\alpha, k_s^\beta\} = 0, \tag{4.51}$$

$$\{e_r^\alpha, f_s^\beta\} = \delta^{\alpha\beta}(k_{r+s}^{\alpha+} - k_{r+s}^{\alpha-}), \tag{4.52}$$

$$\{\tilde\chi^\alpha(z), e^\beta(w)\} = \delta^{\alpha\beta}\tilde\chi^\alpha(z)\frac{e^\beta(z)-e^\beta(w)}{z/w-1}, \tag{4.53}$$

$$\{\tilde\chi^\alpha(z), f^\beta(w)\} = -\delta^{\alpha\beta}\tilde\chi^\alpha(z)\frac{f^\beta(z)-f^\beta(w)}{z/w-1}. \tag{4.54}$$

*Remark* 4.27. This is the classical limit of the GKLO representation of the quantum toroidal algebra of $\mathfrak{gl}_\ell$ discussed in [Tsy23].

**Corollary 4.28.** *The zero modes*

$$E^\alpha := e_0^\alpha = \sum_{i=1}^N u_i^{\alpha+}, \quad F^\alpha := f_0^\alpha = \sum_{i=1}^N u_i^{\alpha-}, \quad K^\alpha := k_0^{\alpha+} = (k_0^{\alpha-})^{-1} = \prod_{i=1}^N \frac{Q_i^\alpha}{(Q_i^{\alpha+1}Q_i^{\alpha-1})^{1/2}} \tag{4.55}$$

*define a representation of the classical limit of the quantum affine algebra in the sense that*

$$\{K^\alpha, K^\beta\} = 0, \tag{4.56}$$

$$\{K^\alpha, E^\beta\} = \tfrac{1}{2}\kappa^{\alpha\beta}K^\alpha E^\beta, \tag{4.57}$$

$$\{K^\alpha, F^\beta\} = -\tfrac{1}{2}\kappa^{\alpha\beta}K^\alpha F^\beta, \tag{4.58}$$

$$\{E^\alpha, F^\beta\} = \delta^{\alpha\beta}(K^\alpha - (K^\alpha)^{-1}). \tag{4.59}$$

### 4.2.3 $L$-operator algebra

As in the rational case, we can write down an algebra of $L$-operators:

**Definition 4.29.** Introduce the one-site $L$-operators

$$L_{ij}^\alpha := \frac{u_j^{\alpha+1,+}}{1 - Q_j^{\alpha+1}/Q_i^\alpha} \in \mathfrak{A}_{N,\ell}^K \tag{4.60}$$

as well as the total $L$-operator

$$L := L^0 \cdots L^{\ell-1}. \tag{4.61}$$

**Proposition 4.30.** *The L-operators satisfy the Poisson bracket*

$$\{L_1^\alpha, L_2^\beta\} = \delta^{\alpha\beta} r^\alpha L_1^\alpha L_2^\beta - \delta^{\alpha\beta} L_1^\alpha L_2^\beta \underline{r}^{\alpha+1} + \delta^{\alpha+1,\beta} L_1^\alpha \bar{r}_{21}^\beta L_2^\beta - \delta^{\alpha,\beta+1} L_2^\beta \bar{r}^\alpha L_1^\alpha, \tag{4.62}$$

*where we have used the matrices from [AKO19], except that $\bar{r}^\alpha$ is shifted by $-\frac{1}{2}$:*

$$r^\alpha := \sum_{i\neq j} \left( \frac{1}{Q_i^\alpha/Q_j^\alpha - 1} e_{ii} - \frac{1}{1 - Q_j^\alpha/Q_i^\alpha} e_{ij} \right) \otimes (e_{jj} - e_{ji}), \tag{4.63}$$

$$\bar{r}^\alpha := \sum_{i\neq j} \frac{1}{1 - Q_j^\alpha/Q_i^\alpha} (e_{ii} - e_{ij}) \otimes e_{jj} - \frac{1}{2}, \tag{4.64}$$

$$\underline{r}^\alpha := \sum_{i\neq j} \frac{1}{1 - Q_j^\alpha/Q_i^\alpha} (e_{ij} \otimes e_{ji} - e_{ii} \otimes e_{jj}). \tag{4.65}$$

*Remark* 4.31. The $r$-matrices $r^\alpha, \underline{r}^\alpha$ are not anti-symmetric, but satisfy

$$r^\alpha + r_{21}^\alpha = C - 1, \qquad \underline{r}^\alpha + \underline{r}_{21}^\alpha = C - 1. \tag{4.66}$$

with $C = \sum_{ij} e_{ij} \otimes e_{ji}$.

**Corollary 4.32.** *The total L-operator satisfies*

$$\{L_1, L_2\} = r^0 L_1 L_2 - L_1 L_2 \underline{r}^0 + L_1 \bar{r}_{21}^0 L_2 - L_2 \bar{r}^0 L_1, \tag{4.67}$$

*which reproduces the Poisson bracket of the Lax matrix from [AKO19].*

*Proof.* This follows from the Poisson bracket of the one-site $L$-operators using the identity

$$r^\alpha + \bar{r}^\alpha_{21} - \bar{r}^\alpha - \underline{r}^\alpha = 0. \qquad \square$$

**Corollary 4.33.** *The Hamiltonians $H[n] := \operatorname{Tr} L^n$ are mutually Poisson commuting.*

**Proposition 4.34.** *The Hamiltonians $H[n]$ are central in the classical limit of the quantum affine algebra:*

$$\{H[n], E^\alpha\} = 0, \qquad \{H[n], F^\alpha\} = 0, \qquad \{H[n], K^\alpha\} = 0. \tag{4.68}$$

*Proof.* Let us consider the bracket $\{H[n], E^\alpha\}$ as an example. From the $L$-operator algebra, we derive the bracket

$$\{L_1^\alpha, u_2^{\beta+1,+}\} = -\delta^{\alpha\beta} L_1^\alpha u_2^{\beta+1,+} \underline{r}^{\alpha+1} - \delta^{\alpha,\beta+1} u_2^{\beta+1,+}(\bar{r}^\alpha + \tfrac{1}{2}) L_1^\alpha + \delta^{\alpha\beta} Z L_1^\alpha \tilde{L}_2^\beta - \delta^{\alpha+1,\beta} L_1^\alpha Z \tilde{L}_2^\beta.$$

with $Z = \sum_{i=1}^N e_{ii} \otimes e_i^t$ and the modified $L$-operators $\tilde{L}^\alpha := (Q^\alpha)^{-1} L^\alpha Q^{\alpha+1}$, where $Q^\alpha := \operatorname{diag}(Q_1^\alpha, \dots, Q_N^\alpha)$ (see also section 4.2.4). Then

$$\begin{aligned}
\{H[n], E^\alpha\} &= \sum_{\mu=0}^{n\ell-1} \operatorname{Tr}_1 L_1^0 \cdots L_1^{\mu-1} \{L_1^\mu, u_2^{\alpha,+}\} L_1^{\mu+1} \cdots L_1^{n\ell-1} e_2 \\
&= -\sum_{\mu=1}^{n\ell-1} \delta^{\mu\alpha} \operatorname{Tr}_1 L_1^0 \cdots L_1^{\alpha-1} u_2^{\alpha,+} (\underline{r}^\alpha + \bar{r}^\alpha) L_1^\alpha \cdots L_1^{n\ell-1} e_2 - \delta^{\ell\alpha} \operatorname{Tr}_1 L_1^n u_2^{\alpha,+} (\underline{r}^0 + \bar{r}^0) e_2 \\
&\quad - \sum_{\mu=2}^{n\ell+1} \tfrac{1}{2} \delta^{\mu\alpha} \operatorname{Tr}_1 L_1^n u_2^{\alpha,+} e_2 + \delta^{0,\alpha-1} \operatorname{Tr}_1 (Z \tilde{L}_2^{\alpha-1} L_1^n - L_1^n Z \tilde{L}_2^{\alpha-1}) e_2 \\
&= 0,
\end{aligned}$$

where we have used $(\underline{r}^\alpha + \bar{r}^\alpha) e_2 = -\frac{1}{2} e_2$. $\square$

In analogy with the rational case, this suggests that the central Hamiltonians $H[n]$ together with the rest of the Bethe subalgebra of the classical limit of the quantum affine algebra define a superintegrable system with the $K$-theoretic Coulomb branch as its phase space. In the next section, we identify this superintegrable model with the hyperbolic spin RS model.

### 4.2.4 Equations of motion

Let us study the equations of motion under the first Hamiltonian $H[1]$. To this end, we let $Q^\alpha := \operatorname{diag}(Q_1^\alpha, \dots, Q_N^\alpha)$ and $\tilde{L}^\alpha := (Q^\alpha)^{-1} L^\alpha Q^{\alpha+1}$. Then we define the spin vectors

$$a^\alpha := L^0 \cdots L^{\alpha-1} e, \qquad c^\alpha := u^{\alpha+1,+} \tilde{L}^{\alpha+1} \cdots \tilde{L}^{\ell-1}. \tag{4.69}$$

for $\alpha = 0, \dots, \ell - 1$. Notice again that $a_i^0 = 1$ for $i = 1, \dots, N$.

**Lemma 4.35.** *The total $L$-operator can be written as*

$$L_{ij} = \frac{\sum_{\rho=0}^{\ell-1} a_i^\rho c_j^\rho}{1 - Q_j^\ell / Q_i^0}. \tag{4.70}$$

*Proof.* Indeed,

$$\begin{aligned}
\sum_{\rho=0}^{\ell-1} a_i^\rho c_j^\rho &= \sum_{\rho=0}^{\ell-1}\sum_{k,l=1}^{N} (L^0\cdots L^{\rho-1})_{il} u_k^{\rho+1,+} Q_j^\ell/Q_k^{\rho+1} (L^{\rho+1}\cdots L^{\ell-1})_{kj} \\
&= \sum_{\rho=0}^{\ell-1}\sum_{k,l=1}^{N} (L^0\cdots L^{\rho-1})_{il} L_{lk}^\rho (Q_j^\ell/Q_k^{\rho+1} - Q_j^\ell/Q_l^\rho)(L^{\rho+1}\cdots L^{\ell-1})_{kj} \\
&= \sum_{\rho=0}^{\ell-1}\sum_{k=1}^{N} (L^0\cdots L^{\rho})_{ik} (L^{\rho+1}\cdots L^{\ell-1})_{kj} Q_j^\ell/Q_k^{\rho+1} - \sum_{\rho=0}^{\ell-1}\sum_{l=1}^{N} (L^0\cdots L^{\rho-1})_{il} (L^{\rho}\cdots L^{\ell-1})_{lj} Q_j^\ell/Q_l^\rho \\
&= L_{ij}(1 - Q_j^\ell/Q_i^0),
\end{aligned}$$

which yields the result. □

**Theorem 4.36.** *The time evolution under the Hamiltonian $H := (1-t)H[1]$ reproduces the equations of motion (1.28) with the identification $x_i = \log Q_i^0$, $\eta = \log t$ and the hyperbolic potential $V(z) = \frac{1}{2}\coth\frac{z}{2} - \frac{1}{2}\coth\frac{z-\eta}{2}$.*

*Proof.* We use the $L$-operator algebra to derive the equations of motion

$$\begin{aligned}
\dot{x}_i &= \{H, Q_i^0\}/Q_i^0 = (t-1)L_{ii}, \\
\ddot{x}_i &= \{H, \{H, Q_i^0\}/Q_i^0\} = (t-1)^2 \sum_{j(\neq i)} L_{ij}L_{ji}\frac{Q_i^0 + Q_j^0}{Q_i^0 - Q_j^0}, \\
\dot{a}_i^\alpha &= \{H, a_i^\alpha\} = (t-1)\sum_{j(\neq i)} (a_i^\alpha - a_j^\alpha)\frac{L_{ij}}{1 - Q_i^0/Q_j^0}, \\
\dot{c}_i^\alpha &= \{H, c_i^\alpha\} = -(t-1)\sum_{j(\neq i)} \left( c_i^\alpha \frac{L_{ij}}{1 - Q_i^0/Q_j^0} - c_j^\alpha \frac{L_{ji}}{1 - Q_j^0/Q_i^0}\right).
\end{aligned}$$

We then use lemma 4.35 and the identities

$$\begin{aligned}
(t-1)\frac{1}{1 - Q_i^0/Q_j^0}\frac{1}{1 - Q_j^\ell/Q_i^0} &= V(x_i - x_j), \\
(t-1)^2\frac{Q_i^0 + Q_j^0}{Q_i^0 - Q_j^0}\frac{1}{1 - Q_j^\ell/Q_i^0}\frac{1}{1 - Q_i^\ell/Q_j^0} &= V(x_i - x_j) - V(x_j - x_i)
\end{aligned}$$

to arrive at the desired equations of motion (1.28). □

### 4.2.5 Poisson bracket of the spin variables

**Proposition 4.37.** *The Poisson brackets of the spins can be written as*

$$\{Q_i^0, a_j^\alpha\} = 0, \qquad \{Q_i^0, c_j^\alpha\} = \delta_{ij} Q_i^0 c_j^\alpha, \tag{4.71}$$

$$\begin{aligned}\{a_i^\alpha, a_j^\beta\} &= (1-\delta_{ij})\left(\frac{a_i^\alpha(a_j^\beta - a_i^\beta)}{Q_i^0/Q_j^0 - 1} - \frac{a_j^\alpha(a_j^\beta - a_i^\beta)}{1 - Q_j^0/Q_i^0}\right) - \delta^{\alpha<\beta} a_j^\alpha a_i^\beta \\ &\quad + \tfrac{1}{2}(2\delta^{0=\alpha<\beta} + \delta^{\alpha>\beta>0} + \delta^{0<\alpha<\beta}) a_i^\alpha a_j^\beta,\end{aligned} \tag{4.72}$$

$$\begin{aligned}\{a_i^\alpha, c_j^\beta\} &= (1-\delta_{ij})\frac{(a_j^\alpha - a_i^\alpha)c_j^\beta}{1 - Q_j^0/Q_i^0} - \delta^{\alpha\beta}\sum_{\rho=0}^{\alpha-1} a_i^\rho c_j^\rho + \delta^{0\beta}\tilde{L}_{ij} a_i^\alpha - \delta^{\alpha\beta}\tilde{L}_{ij} \\ &\quad + \tfrac{1}{2}(1 - \delta^{0=\alpha<\beta} + \delta^{\alpha>\beta=0} - \delta^{\alpha\beta}) a_i^\alpha c_j^\beta,\end{aligned} \tag{4.73}$$

$$\begin{aligned}\{c_i^\alpha, c_j^\beta\} &= (1-\delta_{ij})\left(\frac{c_j^\alpha c_i^\beta}{Q_i^0/Q_j^0 - 1} + \frac{c_i^\alpha c_j^\beta}{1 - Q_j^0/Q_i^0}\right) + \delta^{\alpha<\beta} c_j^\alpha c_i^\beta - \delta^{0\beta} c_i^\alpha \tilde{L}_{ij} + \delta^{0\alpha} c_j^\beta \tilde{L}_{ji} \\ &\quad - \tfrac{1}{2}(2\delta^{\alpha>\beta=0} + \delta^{\alpha>\beta>0} + \delta^{0<\alpha<\beta}) c_i^\alpha c_j^\beta,\end{aligned} \tag{4.74}$$

*where*

$$\tilde{L}_{ij} = \frac{\sum_{\rho=0}^{\ell-1} a_i^\rho c_j^\rho}{Q_i^0/Q_j^\ell - 1}. \tag{4.75}$$

*Proof.* We again consider the bracket $\{a_1^\alpha, a_2^\beta\}$ as an example and make use of the partial monodromies $L^{\alpha,\beta} := L^\alpha \cdots L^\beta$. Here 1 and 2 again label auxiliary spaces. Then

$$\begin{aligned}\{a_1^\alpha, a_2^\beta\} &= \sum_{\mu=0}^{\alpha-1}\sum_{\nu=0}^{\beta-1} L_1^{0,\mu-1} L_2^{0,\nu-1} \{L_1^\mu, L_2^\nu\} L_1^{\mu+1,\alpha-1} L_2^{\nu+1,\beta-1} e_1 e_2 \\ &= \sum_{\mu=0}^{\alpha-1}\sum_{\nu=0}^{\beta-1} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} r^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 - \sum_{\mu=1}^{\alpha}\sum_{\nu=1}^{\beta} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} \underline{r}^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 \\ &\quad + \sum_{\mu=1}^{\alpha}\sum_{\nu=1}^{\beta-1} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} \bar{r}_{21}^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 - \sum_{\mu=1}^{\alpha-1}\sum_{\nu=1}^{\beta} \delta^{\mu\nu} L_1^{0,\mu-1} L_2^{0,\nu-1} \bar{r}^\mu L_1^{\mu,\alpha-1} L_2^{\nu,\beta-1} e_1 e_2 \\ &= (r^0 - \delta_{0<\alpha<\beta} P + \tfrac{1}{2}\delta_{0<\alpha<\beta} + \tfrac{1}{2}\delta_{0<\beta<\alpha}) a_1^\alpha a_2^\beta\end{aligned}$$

[TODO: The last line of this computation disagrees with the statement for $\alpha = 0$: since $a^0 = e$, the bracket must vanish, which the statement satisfies but this line does not (difference $(1-P)(e \otimes a^\beta)$). Replace by the expression in the statement.] where we used the identities

$$\underline{r}^\alpha e_1 e_2 = 0, \qquad (\underline{r}^\alpha + \bar{r}^\alpha) e_2 = -\tfrac{1}{2} e_2, \qquad (\underline{r}^\alpha - \bar{r}_{21}^\alpha - P) e_1 = -\tfrac{1}{2} e_1.$$

The other brackets follow similarly, except that we have to use the additional identity

$$L^0 \cdots L^{\alpha-1} \tilde{L}^\alpha \cdots \tilde{L}^{\ell-1} = \tilde{L} + \sum_{\rho=0}^{\alpha-1} a^\rho c^\rho,$$

which can be proven by a similar telescopic argument as in the proof of lemma 4.35. All in all, we obtain

$$\{a_1^\alpha, a_2^\beta\} = (r^0 - \delta_{\alpha<\beta}P)a_1^\alpha a_2^\beta + (\delta_{0=\alpha<\beta} + \tfrac{1}{2}\delta_{\alpha>\beta>0} + \tfrac{1}{2}\delta_{0<\alpha<\beta})a_1^\alpha a_2^\beta, \tag{4.76}$$

$$\begin{aligned}\{a_1^\alpha, c_2^\beta\} = &-c_2^\beta \bar{r}^0 a_1^\alpha - (\tfrac{1}{2}\delta^{\alpha\beta} + \tfrac{1}{2}\delta_{0=\alpha<\beta} - \tfrac{1}{2}\delta_{\alpha>\beta=0})a_1^\alpha c_2^\beta - \delta^{\alpha\beta}\sum_{\rho=0}^{\alpha-1} a_1^\rho c_2^\rho \\ &+ \delta^{0\beta} Z a_1^\alpha \tilde{L}_2 - \delta^{\alpha\beta} Z e_1 \tilde{L}_2,\end{aligned} \tag{4.77}$$

$$\begin{aligned}\{c_1^\alpha, c_2^\beta\} = &-c_1^\alpha c_2^\beta(\underline{r}^0 - \delta_{\alpha<\beta}P) - (\delta_{\alpha>\beta=0} + \tfrac{1}{2}\delta_{\alpha>\beta>0} + \tfrac{1}{2}\delta_{0<\alpha<\beta})c_1^\alpha c_2^\beta \\ &- \delta^{0\beta} c_1^\alpha Z\tilde{L}_2 + \delta^{0\alpha} c_2^\beta Z_{21}\tilde{L}_1,\end{aligned} \tag{4.78}$$

where $Z = \sum_{i=1}^N e_{ii} \otimes e_i^t$ and $P = \sum_{ij} e_{ij} \otimes e_{ji}$. Writing this in components, we arrive at the claimed brackets. □

*Remark* 4.38. We again have the constraint $a_i^0 = 1$, which restricts the "physical" spin degrees of freedom of the hyperbolic spin RS model to $a^\alpha, c^\alpha$ for $\alpha > 0$ and notice that they form a quadratic Poisson algebra.

**Corollary 4.39.** *The rescaling* $\hat{a}_i^\alpha := a_i^\alpha / \sum_\rho a_i^\rho, \hat{c}_i^\alpha := c_i^\alpha \sum_\rho a_i^\rho, \hat{\tilde{L}}_{ij} := \frac{\sum_\rho \hat{a}_i^\rho \hat{c}_j^\rho}{Q_i^0/Q_j^{\ell-1}}$ *reproduces the Poisson brackets from [AO19; Fai26]:*

$$\{Q_i^0, \hat{a}_j^\alpha\} = 0, \qquad \{Q_i^0, \hat{c}_j^\alpha\} = \delta_{ij} Q_i^0 \hat{c}_j^\alpha,$$

$$\begin{aligned}\{\hat{a}_i^\alpha, \hat{a}_j^\beta\} = &-(1-\delta_{ij})\frac{1}{2}\frac{Q_i^0 + Q_j^0}{Q_i^0 - Q_j^0}(\hat{a}_i^\alpha - \hat{a}_j^\alpha)(\hat{a}_i^\beta - \hat{a}_j^\beta) - \frac{\operatorname{sgn}(\beta-\alpha)}{2}\hat{a}_j^\alpha \hat{a}_i^\beta \\ &- \sum_{\rho=0}^{\ell-1}\frac{\operatorname{sgn}(\alpha-\rho)}{2}\hat{a}_j^\alpha \hat{a}_j^\beta \hat{a}_i^\rho + \sum_{\rho=0}^{\ell-1}\frac{\operatorname{sgn}(\beta-\rho)}{2}\hat{a}_i^\alpha \hat{a}_i^\beta \hat{a}_j^\rho - \sum_{\mu,\nu=0}^{\ell-1}\frac{\operatorname{sgn}(\mu-\nu)}{2}\hat{a}_i^\mu \hat{a}_j^\nu \hat{a}_i^\alpha \hat{a}_j^\beta,\end{aligned}$$

$$\begin{aligned}\{\hat{a}_i^\alpha, \hat{c}_j^\beta\} = &\hat{a}_i^\alpha \hat{\tilde{L}}_{ij} - (1-\delta_{ij})\frac{1}{2}\frac{Q_i^0 + Q_j^0}{Q_i^0 - Q_j^0}(\hat{a}_i^\alpha - \hat{a}_j^\alpha)\hat{c}_j^\beta + \hat{a}_i^\alpha \sum_{\rho=0}^{\beta-1}\hat{a}_i^\rho \hat{c}_j^\rho + \frac{1}{2}\hat{a}_i^\alpha \hat{a}_i^\beta \hat{c}_j^\beta \\ &- \delta_{\alpha\beta}\left(\hat{\tilde{L}}_{ij} + \frac{1}{2}\hat{a}_i^\alpha \hat{c}_j^\beta + \sum_{\rho=0}^{\beta-1}\hat{a}_i^\rho \hat{c}_j^\rho\right) + \sum_{\rho=0}^{\ell-1}\frac{\operatorname{sgn}(\alpha-\rho)}{2}\hat{c}_j^\beta \hat{a}_j^\alpha \hat{a}_i^\rho + \sum_{\mu,\nu=0}^{\ell-1}\frac{\operatorname{sgn}(\mu-\nu)}{2}\hat{a}_i^\mu \hat{a}_j^\nu \hat{a}_i^\alpha \hat{c}_j^\beta,\end{aligned}$$

$$\{\hat{c}_i^\alpha, \hat{c}_j^\beta\} = \frac{1}{2}(1-\delta_{ij})\frac{Q_i^0+Q_j^0}{Q_i^0-Q_j^0}(\hat{c}_i^\alpha \hat{c}_j^\beta + \hat{c}_j^\alpha \hat{c}_i^\beta) - \hat{c}_i^\alpha \hat{\tilde{L}}_{ij} + \hat{c}_j^\beta \hat{\tilde{L}}_{ji} + \frac{\mathrm{sgn}(\beta-\alpha)}{2}\hat{c}_j^\alpha \hat{c}_i^\beta$$
$$- \hat{c}_i^\alpha \sum_{\rho=0}^{\beta-1} \hat{a}_i^\rho \hat{c}_j^\rho + \hat{c}_j^\beta \sum_{\rho=0}^{\alpha-1} \hat{a}_j^\rho \hat{c}_i^\rho - \sum_{\mu,\nu=0}^{\ell-1} \frac{\mathrm{sgn}(\mu-\nu)}{2}\hat{a}_i^\mu \hat{a}_j^\nu \hat{c}_i^\alpha \hat{c}_j^\beta + \frac{1}{2}\hat{a}_j^\alpha \hat{c}_i^\alpha \hat{c}_j^\beta - \frac{1}{2}\hat{a}_i^\beta \hat{c}_i^\alpha \hat{c}_j^\beta\,. \tag{4.79}$$

*Remark* 4.40. The fact that our Poisson brackets coincide with the ones in [AO19; Fai26] can be viewed as an instance of mirror symmetry. The principle of mirror symmetry posits a correspondence between the multiplicative quiver variety of the Jordan quiver considered in [Fai26] and the $K$-theoretic Coulomb branch of the necklace quiver considered here.

### 4.2.6 Supplementary Poisson brackets

In this section, we give supplementary Poisson brackets of certain variables introduced in [AF98]:

$$S_i^{\alpha\beta} := c_i^\alpha a_i^\beta, \qquad f_{ij} := \sum_{\rho=0}^{\ell-1} a_i^\rho c_j^\rho. \tag{4.80}$$

The variables $S_i^{\alpha\beta}$ were dubbed *quantum current* in [AH26b] due to the form of their quantum commutation relations, which are similar to the commutation relations of the quantum current defined in [RS90]. The variables $f_{ij}$ are the *collective spin variables*. They Poisson-commute with $Q_i^0$ according to

$$\{Q_i^0, S_j^{\alpha\beta}\} = \delta_{ij} Q_i^0 S_j^{\alpha\beta}, \qquad \{Q_i^0, f_{jk}\} = \delta_{ik} Q_i^0 f_{jk}. \tag{4.81}$$

We now present the Poisson brackets of the quantum current and collective spin variables of the hyperbolic spin RS model.

**Proposition 4.41.** *The hyperbolic quantum current satisfies the Poisson bracket*

$$\{S_i^{\alpha\beta}, S_j^{\mu\nu}\} = (1-\delta_{ij})\frac{1}{2}\frac{Q_i^0+Q_j^0}{Q_i^0-Q_j^0}(S_i^{\mu\beta}S_j^{\alpha\nu} + S_i^{\alpha\nu}S_j^{\mu\beta}) + \frac{\mathrm{sgn}(\mu-\alpha)}{2}S_i^{\mu\beta}S_j^{\alpha\nu} - \frac{\mathrm{sgn}(\nu-\beta)}{2}S_i^{\alpha\nu}S_j^{\mu\beta}$$
$$- \delta^{\beta\mu}\left(\tfrac{1}{2}S_i^{\alpha\beta}S_j^{\mu\nu} + \sum_{\rho=0}^{\beta-1} S_i^{\alpha\rho}S_j^{\rho\nu} + \frac{\sum_{\rho=0}^{\ell-1} S_i^{\alpha\rho}S_j^{\rho\nu}}{Q_i^0/Q_j^\ell - 1}\right) + \delta^{\alpha\nu}\left(\tfrac{1}{2}S_i^{\alpha\beta}S_j^{\mu\nu} + \sum_{\rho=0}^{\alpha-1} S_j^{\mu\rho}S_i^{\rho\beta} + \frac{\sum_{\rho=0}^{\ell-1} S_j^{\mu\rho}S_i^{\rho\beta}}{Q_j^0/Q_i^\ell - 1}\right). \tag{4.82}$$

**Proposition 4.42.** *The collective spin variables satisfy the Poisson bracket*

$$\{f_{ij}, f_{kl}\} = (1-\delta_{ik})\left(\frac{(f_{kl}-f_{il})f_{ij}}{Q_i^0/Q_k^0 - 1} - \frac{(f_{kl}-f_{il})f_{kj}}{1-Q_k^0/Q_i^0}\right) + (1-\delta_{jl})\left(\frac{f_{il}f_{kj}}{Q_j^0/Q_l^0 - 1} + \frac{f_{ij}f_{kl}}{1-Q_l^0/Q_j^0}\right)$$
$$- (1-\delta_{kj})\frac{f_{ij}(f_{jl}-f_{kl})}{1-Q_j^0/Q_k^0} + (1-\delta_{il})\frac{f_{kl}(f_{lj}-f_{ij})}{1-Q_l^0/Q_i^0} \tag{4.83}$$
$$+ \frac{f_{ij}f_{il}}{Q_i^0/Q_l^\ell - 1} - \frac{f_{ij}f_{jl}}{Q_j^0/Q_l^\ell - 1} + \frac{f_{kl}f_{lj}}{Q_l^0/Q_j^\ell - 1} - \frac{f_{kl}f_{kj}}{Q_k^0/Q_j^\ell - 1} + \frac{f_{il}f_{kj}}{Q_k^0/Q_j^\ell - 1} - \frac{f_{kj}f_{il}}{Q_i^0/Q_l^\ell - 1}.$$

*Remark* 4.43. It can be checked that the rational degeneration of these Poisson brackets yields the Poisson bracket found in [AF98] and that the form of the Poisson brackets of the collective spin variables matches the conjecture put forth in *loc. cit.*

## 4.3 Quantization of the trigonometric spin Ruijsenaars–Schneider model

### 4.3.1 The $K$-theoretic Coulomb branch algebra

In this section, we construct generators of the abelianized quantized $K$-theoretic Coulomb branch algebra of the necklace quiver following [Tsy23, remark 2.10]. The generators can be expressed in terms of the exponentiated vacuum expectation values of abelianized gauge multiplet scalars $Q_i^\alpha$, where $\alpha \in \mathbb{Z}_\ell$ labels a node of the quiver and $i = 1, \dots, N$ is a color index for the corresponding gauge group factor $U(N)_\alpha$, as well as the $q$-difference operators $P_i^\alpha$, which satisfy the commutation relations

$$Q_i^\alpha Q_j^\beta = Q_j^\beta Q_i^\alpha, \qquad P_i^\alpha Q_j^\beta = q^{\delta_{ij}\delta^{\alpha\beta}} Q_j^\beta P_i^\alpha, \qquad P_i^\alpha P_j^\beta = P_j^\beta P_i^\alpha. \tag{4.84}$$

The abelianized monopole operators of minuscule coweight/magnetic charge, which together with the exponentiated vacuum expectation values $Q_i^\alpha$ generate the abelianized quantized $K$-theoretic Coulomb branch algebra, can then be expressed in terms of the $q$-difference operators as [TODO: Define the multiplicative masses $\mu^\alpha \in \mathbb{C}^\times$ and set $t := \prod_{\alpha\in\mathbb{Z}_\ell} \mu^\alpha$ (checked in the notebook QuantizationOfTheTrigonometricRSModel.nb); state their relation to the masses $m_\alpha$ of figure 1.1 and $m^{\alpha\to\beta}$ of chapter 2.]

$$u_i^{\alpha+} := \prod_{k=1}^N (Q_k^\alpha / Q_k^{\alpha+1})^{1/2} \frac{\prod_{k=1}^N (1 - (\mu^\alpha)^{-1} Q_k^{\alpha+1}/Q_i^\alpha)}{\prod_{k(\neq i)} (1 - Q_k^\alpha / Q_i^\alpha)} P_i^\alpha, \tag{4.85}$$

$$u_i^{\alpha-} := \prod_{k=1}^N (Q_k^\alpha / Q_k^{\alpha-1})^{1/2} \frac{\prod_{k=1}^N (1 - \mu^{\alpha-1} Q_k^{\alpha-1}/Q_i^\alpha)}{\prod_{k(\neq i)} (1 - Q_k^\alpha / Q_i^\alpha)} (P_i^\alpha)^{-1}. \tag{4.86}$$

We now describe their commutation relations. To this end, we define the structure function

$$g_\kappa^\pm(z) := \frac{q^{\mp\kappa/2} z - 1}{z - q^{\mp\kappa/2}}. \tag{4.87}$$

Given $(i,\alpha) \neq (j,\beta)$, we then find

$$u_i^{\alpha\pm} u_j^{\beta\pm} = g^\pm_{\kappa^{\alpha\beta}}\big((q^{-1/2}\mu^\alpha)^{\delta^{\alpha+1,\beta}} (q^{-1/2}\mu^\beta)^{-\delta^{\alpha,\beta+1}} Q_{ij}^{\alpha\beta}\big) u_j^{\beta\pm} u_i^{\alpha\pm}, \qquad (\ell > 2) \tag{4.88}$$

$$u_i^{\alpha\pm} u_j^{\beta\pm} = g_2^\pm\big(Q_{ij}^{\alpha\beta}\big) u_j^{\beta\pm} u_i^{\alpha\pm}, \qquad (\alpha = \beta, \ell = 2) \tag{4.89}$$

$$u_i^{\alpha\pm} u_j^{\beta\pm} = g_{-1}^\pm\big(q^{-1/2}\mu^\alpha Q_{ij}^{\alpha\beta}\big) g_{-1}^\pm\big(q^{1/2}(\mu^\beta)^{-1} Q_{ij}^{\alpha\beta}\big) u_j^{\beta\pm} u_i^{\alpha\pm}, \qquad (\alpha \neq \beta, \ell = 2) \tag{4.90}$$

$$u_i^{0\pm} u_j^{0\pm} = g_2^\pm\big(Q_{ij}^{00}\big) g_{-1}^\pm\big(q^{-1/2}\mu^0 Q_{ij}^{00}\big) g_{-1}^\pm\big(q^{1/2}(\mu^0)^{-1} Q_{ij}^{00}\big) u_j^{0\pm} u_i^{0\pm}, \qquad (\ell = 1) \tag{4.91}$$

where we have used the abbreviation $Q_{ij}^{\alpha\beta} := Q_i^\alpha / Q_j^\beta$ and $\kappa^{\alpha\beta} := 2\delta^{\alpha\beta} - \delta^{\alpha+1,\beta} - \delta^{\alpha,\beta+1}$ is the Cartan matrix of the necklace quiver.

### 4.3.2 The $L$-operator algebra

The main tool for obtaining the classical trigonometric spin RS model from the $K$-theoretic Coulomb branch algebra is the $L$-operator algebra introduced in [AH26c]. In

this section we quantize this algebra. To every arrow $\alpha \to \alpha+1$ in the necklace quiver, we associate two $L$-operators $L^{\alpha\pm}$, which are $N \times N$ matrices with coefficients

$$L_{ij}^{\alpha+} := \frac{q^{1/2}}{1-(\mu^\alpha)^{-1}Q_i^{\alpha+1}/Q_j^\alpha} u_j^{\alpha+}, \tag{4.92}$$

$$L_{ij}^{\alpha-} := \frac{q^{-1/2}}{\mu^\alpha Q_i^\alpha/Q_j^{\alpha+1}-1} u_j^{\alpha+1,-}. \tag{4.93}$$

[TODO: Sign of $L^{\alpha-}$: the notebook uses $L_{ij}^{\alpha-} = q^{-1/2}u_j^{\alpha+1,-}/(1-\mu^\alpha Q_i^\alpha/Q_j^{\alpha+1})$, i.e. an extra overall minus sign compared to (4.93). With that sign, (4.138), the formulas for $L_{ij}^{\text{tot}-}$, $\tilde{L}_{ij}^{\text{tot}-}$, $H^-[1] = \operatorname{Tr} L^{\text{tot}-} = q^{-1/2}S^-[1]$, the spectral curve and $F^{\alpha,\beta}$ are correct as written; with the current sign they acquire factors $(-1)^\ell$ resp. $(-1)^{\beta-\alpha}$. Separately: compare with the classical $L$-operators of section 4.2.4 (transposed index structure of $L^{\alpha+}$; classical spin variables use $u^+$, quantum ones $L^-$, $u^-$; $\tilde{L}$ is $Q^{-1}LQ$ classically and $\mu QLQ^{-1}$ here; $H^-[1]$ vs. $(t-1)\operatorname{Tr} L$). Reconcile conventions.] In [AH26c] it was shown that the Poisson brackets of the classical counterparts of these $L$-operators are quadratic in $L^{\alpha\pm}$ and can be written in terms of classical $r$-matrices. Assuming the operator ordering prescribed by equations (4.92) and (4.93), we obtain the following commutation relations

$$R_{ab}^{\alpha+1,\beta+1}(q^{-1})L_b^{\beta+}\bar{R}_{ab}^{\alpha+1,\beta}(q^{-1})^{-1}L_a^{\alpha+} = L_a^{\alpha+}\bar{R}_{ba}^{\beta+1,\alpha}(q^{-1})^{-1}L_b^{\beta+}\underline{R}_{ab}^{\alpha\beta}(q^{-1}), \tag{4.94}$$

$$R_{ab}^{\alpha\beta}(q)L_b^{\beta-}\bar{R}_{ab}^{\beta+1,\alpha}(q)^{-1}L_a^{\alpha-} = L_a^{\alpha-}\bar{R}_{ba}^{\beta,\alpha+1}(q)^{-1}L_b^{\beta-}\underline{R}_{ab}^{\alpha+1,\beta+1}(q). \tag{4.95}$$

These relations provide a quantization of the aforementioned Poisson brackets. Here, $a$ and $b$ denote auxiliary spaces and we introduce the following quantum $R$-matrices:

$$R^{\alpha\beta}(q) := 1+\delta^{\alpha\beta}(1-q^{-1})\times \left[\sum_{i\neq j}\left(\frac{1}{Q_i^\alpha/Q_j^\alpha-1}e_{ii} - \frac{1}{1-Q_j^\alpha/Q_i^\alpha}e_{ij}\right)\otimes(e_{jj}-e_{ji}) + X - X_{21}\right], \tag{4.96}$$

$$\bar{R}^{\alpha\beta}(q) := 1+\delta^{\alpha\beta}\left[(q^{-1/2}-1)+\sum_{i\neq j}\frac{q^{1/2}-q^{-1/2}}{1-qQ_j^\alpha/Q_i^\alpha}(e_{ii}-e_{ij})\otimes e_{jj} + (q^{1/2}-q^{-1/2})X\right], \tag{4.97}$$

$$\underline{R}^{\alpha\beta}(q) := 1+\delta^{\alpha\beta}(1-q^{-1})\sum_{i\neq j}\frac{1}{1-Q_j^\alpha/Q_i^\alpha}(e_{ij}\otimes e_{ji} - e_{ii}\otimes e_{jj}), \tag{4.98}$$

with $X = \sum_{i,j=1}^N e_{ij}\otimes e_{jj}$. These $R$-matrices are identical to the evaluation at zero of the spectral parameter dependent $R$-matrices found in [AKO19] for the spinless case up to a prefactor for $\bar{R}^{\alpha\beta}(q)$. It should be noted that $R^{\alpha\beta}(q)$ satisfies the Yang–Baxter equation

$$R_{ab}^{\alpha\beta}(q)R_{ac}^{\alpha\gamma}(q)R_{bc}^{\beta\gamma}(q) = R_{bc}^{\beta\gamma}(q)R_{ac}^{\alpha\gamma}(q)R_{ab}^{\alpha\beta}(q), \tag{4.99}$$

and $\underline{R}^{\alpha\beta}(q)$ satisfies the dynamical Yang–Baxter equation

$$\underline{R}_{ab}^{\alpha\beta}(q)(P_b^\beta)^{-1}\underline{R}_{ac}^{\alpha\gamma}(q)P_b^\beta\underline{R}_{bc}^{\beta\gamma}(q) = (P_a^\alpha)^{-1}\underline{R}_{bc}^{\beta\gamma}(q)P_a^\alpha\underline{R}_{ac}^{\alpha\gamma}(q)(P_c^\gamma)^{-1}\underline{R}_{ab}^{\alpha\beta}(q)P_c^\gamma, \tag{4.100}$$

where the role of the dynamical parameters is played by the variables $Q_i^\alpha$. Finally, $\bar{R}^{\alpha\beta}(q)$ is the corresponding dynamical twist, i.e.

$$\bar{R}^{\alpha\beta}(q)\underline{R}^{\alpha\beta}(q) = R^{\alpha\beta}(q)\bar{R}_{21}^{\alpha\beta}(q). \tag{4.101}$$

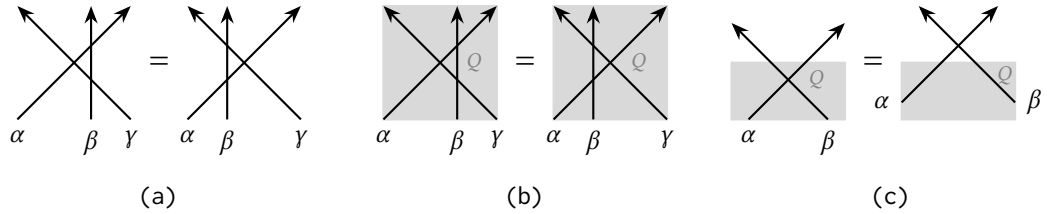



**Figure 4.1:** Diagrammatic representation of the three relations satisfied by the $R$-matrices. (a) The Yang–Baxter equation for the $R$-matrix $R^{\alpha\beta}(q)$, which is depicted as a crossing of two lines labeled by $\alpha$ and $\beta$ on a white background. (b) The dynamical Yang–Baxter equation for the dynamical $R$-matrix $\underline{R}^{\alpha\beta}(q)$, which is depicted as a crossing of two lines labeled by $\alpha$ and $\beta$ on a gray background. The faces with a gray background are labeled by the dynamical parameters $Q$, with their values determined from the reference face (labeled $Q$) by $q$-shifting $Q^\alpha$ when passing over a line labeled $\alpha$. (c) The twist relation for the dynamical twist $\bar{R}^{\alpha\beta}(q)$, which is depicted as two lines labeled by $\alpha$ and $\beta$ crossing from a gray to a white background. Composition is read from bottom to top.

These relations are shown diagrammatically in figure 4.1.

If we define the total $L$-operators to be

$$L^{\text{tot}+} := L^{\ell-1,+} \cdots L^{0,+}, \qquad L^{\text{tot}-} := L^{0-} \cdots L^{\ell-1,-}, \tag{4.102}$$

the relations (4.94) and (4.95) together with equation (4.101) imply

$$R^{00}_{ab}(q^\mp) L^{\text{tot}\pm}_b \bar{R}^{00}_{ab}(q^\mp)^{-1} L^{\text{tot}\pm}_a = L^{\text{tot}\pm}_a \bar{R}^{00}_{ba}(q^\mp)^{-1} L^{\text{tot}\pm}_b \underline{R}^{00}_{ab}(q^\mp), \tag{4.103}$$

which exactly reproduces the $L$-operator algebra of the spinless model found in [AKO19].

### 4.3.3 Commuting Hamiltonians

The commutation relations (4.103) of the total $L$-operator coincide with those obtained in [AKO19] for the $L$-operator of the spinless model. It follows that a family of commuting Hamiltonians is given by the same trace formula as in *loc. cit.*:

$$H^\pm[n] := \mathrm{Tr}_{ab}\, P^{t_b}_{ab} L^{\text{tot}\pm}_a (\bar{R}^{00}_{ba}(q^\mp)^{t_b} R^{00}_{ab}(q^\mp)^{t_b} L^{\text{tot}\pm}_a)^{n-1}, \tag{4.104}$$

where $P = \sum_{i,j=1}^N e_{ij} \otimes e_{ji}$ and the superscript $t_b$ denotes transposition in the auxiliary space $b$. In particular, we have $H^-[1] = \mathrm{Tr}\, L^{\text{tot}-}$, which coincides with the defining Hamiltonian of the trigonometric spin RS model introduced in [AH26c] and generates at the classical level the equations of motion found by Krichever and Zabrodin [KZ95]. We remark that in terms of the 4d $\mathcal{N} = 2$ necklace quiver gauge theory, the Hamiltonian $H^-[1]$ corresponds to the 't Hooft line which has charge $\square$ under all gauge nodes [MOY21].

The Hamiltonians (4.104) can be rewritten in terms of the generalized Macdonald operators

$$\begin{aligned} S^+[n] &:= t^{-n(n-1)/2} \sum_{\substack{I_0,\dots,I_{\ell-1}\subseteq\{1,\dots,N\} \\ |I_\alpha|=n}} \left( \prod_{\alpha=0}^{\ell-1} \frac{\prod_{i\in I_{\alpha-1}, k\notin I_\alpha} (Q_i^{\alpha-1} - (\mu^{\alpha-1})^{-1} Q_k^\alpha)}{\prod_{i\in I_\alpha, k\notin I_\alpha} (Q_i^\alpha - Q_k^\alpha)} \right) \prod_{\alpha=0}^{\ell-1} \prod_{i\in I_\alpha} P_i^\alpha, \\ S^-[n] &:= t^{n(n-1)/2} \sum_{\substack{I_0,\dots,I_{\ell-1}\subseteq\{1,\dots,N\} \\ |I_\alpha|=n}} \left( \prod_{\alpha=0}^{\ell-1} \frac{\prod_{i\in I_{\alpha+1}, k\notin I_\alpha} (Q_i^{\alpha+1} - \mu^\alpha Q_k^\alpha)}{\prod_{i\in I_\alpha, k\notin I_\alpha} (Q_i^\alpha - Q_k^\alpha)} \right) \prod_{\alpha=0}^{\ell-1} \prod_{i\in I_\alpha} (P_i^\alpha)^{-1}. \end{aligned} \tag{4.105}$$

Specifically, we have the identification

$$H^{\pm}[n] = q^{\pm 1/2} \begin{vmatrix} S^{\pm}[1] & 1 & 0 & \cdots & 0 \\ [2]_{q^{\pm 1}} S^{\pm}[2] & S^{\pm}[1] & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ [n]_{q^{\pm 1}} S^{\pm}[n] & S^{\pm}[n-1] & S^{\pm}[n-2] & \cdots & S^{\pm}[1] \end{vmatrix}, \tag{4.106}$$

where we make use of the $q$-numbers $[n]_q = \sum_{k=0}^{n-1} q^k$. This suggests that $S^{\pm}[n]$ are the elementary symmetric counterparts of the power sums $H^{\pm}[n]$. Indeed, we find

$$: \det(q^{\mp \ell/2} L^{\mathrm{tot}\pm} + z) : = \sum_{n=0}^{N} z^{N-n} S^{\pm}[n], \tag{4.107}$$

where $::$ means that the $P_i^{\alpha}$ are normal ordered to the right of the $Q_i^{\alpha}$. Note that introducing a spectral parameter dependence in $L^{\mathrm{tot}\pm}$ would turn equation (4.107) into the quantum spectral curve of the trigonometric spin RS model.

To finish the discussion of the Hamiltonians, let us note that we can build generators of the 1+1d Poincaré Lie algebra from the lowest Hamiltonians $H^{\pm}[1]$ as follows:

$$\mathbf{H} \coloneqq H^{+}[1] + H^{-}[1], \qquad \mathbf{P} \coloneqq H^{+}[1] - H^{-}[1], \qquad \mathbf{B} \coloneqq \sum_{i=1}^{N} \log Q_i^0. \tag{4.108}$$

They obey the expected relations

$$[\mathbf{H}, \mathbf{P}] = 0, \qquad [\mathbf{H}, \mathbf{B}] = \log(q)\mathbf{P}, \qquad [\mathbf{P}, \mathbf{B}] = \log(q)\mathbf{H}, \tag{4.109}$$

and the boost generator $\mathbf{B}$ more generally satisfies

$$[H^{\pm}[n], \mathbf{B}] = \pm n \log(q) H^{\pm}[n], \qquad [S^{\pm}[n], \mathbf{B}] = \pm n \log(q) S^{\pm}[n]. \tag{4.110}$$

This repeats the known pattern of the spinless trigonometric RS model.

### 4.3.4 Quantum loop algebra

In this section, we exhibit a quantum loop algebra structure inside the $K$-theoretic Coulomb branch algebra. As discussed in the introduction, the general theory of [BFN18a, appendix B] and the technology developed in [Tsy23] suggest that the $K$-theoretic Coulomb branch of the necklace quiver should be identified with the $N$-truncated quantum toroidal algebra $U_{q,t}^{(N)}(\ddot{\mathfrak{gl}}_\ell)$. Specializing to zero central charge, the quantum toroidal algebra has two subalgebras isomorphic to the $N$-truncated quantum loop algebra $U_q^{(N)}(\dot{\mathfrak{gl}}_\ell)$, called *horizontal* and *vertical*. Here, we exhibit the horizontal quantum loop subalgebra.

The quantum loop algebra $U_q(\dot{\mathfrak{gl}}_\ell)$ has various useful presentations. Firstly, it may be presented in the Drinfeld–Jimbo generators

$$E^{\alpha}, \qquad (t^{\alpha})^{\pm 1}, \qquad F^{\alpha}, \tag{4.111}$$

which satisfy relations analogous to the relations for $U_q(\mathfrak{gl}_\ell)$ [FT19]:

$$(t^\alpha)^{\pm 1}(t^\alpha)^{\mp 1} = 1, \tag{4.112}$$

$$[t^\alpha, t^\beta] = 0, \tag{4.113}$$

$$[E^\alpha, F^\beta] = \delta^{\alpha\beta}\frac{K^\alpha - (K^\alpha)^{-1}}{q^{1/2} - q^{-1/2}}, \tag{4.114}$$

$$E^\alpha t^\beta = q^{(-\delta^{\alpha\beta}+\delta^{\alpha+1,\beta})/2} t^\beta E^\alpha \tag{4.115}$$

$$F^\alpha t^\beta = q^{(\delta^{\alpha\beta}-\delta^{\alpha+1,\beta})/2} t^\beta F^\alpha, \tag{4.116}$$

$$E^\alpha E^\beta = E^\beta E^\alpha, \qquad \alpha - \beta \neq 0, \pm 1, \tag{4.117}$$

$$F^\alpha F^\beta = F^\beta F^\alpha, \qquad \alpha - \beta \neq 0, \pm 1, \tag{4.118}$$

where $K^\alpha := t^\alpha (t^{\alpha-1})^{-1}$, as well as the $q$-Serre relations [TODO: With $t^\alpha$ as in (4.128), the notebook shows that these relations hold for $(t^\beta)^{-1} = T^+[0]^{\beta\beta}$ instead of $t^\beta$. Minimal fix keeping (4.128): $E^\alpha t^\beta = q^{(\delta^{\alpha\beta}-\delta^{\alpha+1,\beta})/2} t^\beta E^\alpha$, $F^\alpha t^\beta = q^{(-\delta^{\alpha\beta}+\delta^{\alpha+1,\beta})/2} t^\beta F^\alpha$, and $K^\alpha := t^{\alpha+1}(t^\alpha)^{-1}$ (the current $K^\alpha$ is centred at node $\alpha - 1$). The notebook also verifies $E^\alpha K^\beta = q^{-\kappa^{\alpha\beta}/2} K^\beta E^\alpha$. Note that this $K^\alpha$ is inverse to the classical one up to mass factors.]

$$(E^\alpha)^2 E^{\alpha\pm 1} + E^{\alpha\pm 1}(E^\alpha)^2 = (q^{1/2} + q^{-1/2}) E^\alpha E^{\alpha\pm 1} E^\alpha, \tag{4.119}$$

$$(F^\alpha)^2 F^{\alpha\pm 1} + F^{\alpha\pm 1}(F^\alpha)^2 = (q^{1/2} + q^{-1/2}) F^\alpha F^{\alpha\pm 1} F^\alpha, \tag{4.120}$$

in the case $\ell > 2$, and

$$(E^\alpha)^3 E^{\alpha\pm 1} - E^{\alpha\pm 1}(E^\alpha)^3 = (q^{-1} + 1 + q)\Big[(E^\alpha)^2 E^{\alpha\pm 1} E^\alpha - E^\alpha E^{\alpha\pm 1}(E^\alpha)^2\Big], \tag{4.121}$$

$$(F^\alpha)^3 F^{\alpha\pm 1} - F^{\alpha\pm 1}(F^\alpha)^3 = (q^{-1} + 1 + q)\Big[(F^\alpha)^2 F^{\alpha\pm 1} F^\alpha - F^\alpha F^{\alpha\pm 1}(F^\alpha)^2\Big], \tag{4.122}$$

in the case $\ell = 2$. The case $\ell = 1$ does not have Serre relations.

Secondly, the RTT presentation [Mol07] of the quantum loop algebra $U_q(\dot{\mathfrak{gl}}_\ell)$ is given in terms of generator matrices

$$T^\pm(u) = \sum_{\alpha,\beta=1}^{\ell} \sum_{n\geq 0} T^\pm[n]^{\alpha\beta} \otimes u^{\mp n} e^{\alpha\beta} \in U_q(\dot{\mathfrak{gl}}_\ell) \otimes \mathrm{End}(\mathbb{C}^\ell)[\![u^{\mp 1}]\!], \tag{4.123}$$

which satisfy the additional requirements

$$T^\pm[0]^{\alpha\alpha} T^\mp[0]^{\alpha\alpha} = 1, \qquad T^\pm[0]^{\alpha\beta} = 0, \qquad \alpha \gtrless \beta, \tag{4.124}$$

as well as the RTT relations

$$R^{ab}(u/v) T^\pm(u)^a T^\pm(v)^b = T^\pm(v)^b T^\pm(u)^a R^{ab}(u/v), \tag{4.125}$$

$$R^{ab}(u/v) T^-(u)^a T^+(v)^b = T^+(v)^b T^-(u)^a R^{ab}(u/v), \tag{4.126}$$

where we have used the standard unitary trigonometric $R$-matrix for $\mathfrak{gl}_\ell$:

$$\begin{aligned} R(u) = {} & \frac{q^{1/2} - q^{-1/2}u}{q^{-1/2} - q^{1/2}u} \sum_\alpha e^{\alpha\alpha} \otimes e^{\alpha\alpha} + \frac{1-u}{q^{-1/2} - q^{1/2}u} \sum_{\alpha\neq\beta} e^{\alpha\alpha} \otimes e^{\beta\beta} \\ & + \frac{q^{1/2} - q^{-1/2}}{q^{-1/2} - q^{1/2}u} \sum_{\alpha>\beta} e^{\alpha\beta} \otimes e^{\beta\alpha} + \frac{q^{1/2}u - q^{-1/2}u}{q^{-1/2} - q^{1/2}u} \sum_{\alpha<\beta} e^{\alpha\beta} \otimes e^{\beta\alpha}. \end{aligned} \tag{4.127}$$

The isomorphism between the Drinfeld–Jimbo and the RTT presentation is explicitly given in [FT19] for the case of $U_q(\mathfrak{gl}_\ell)$ and has a natural generalization for the case of the quantum loop algebra $U_q(\dot{\mathfrak{gl}}_\ell)$.

We now build a representation of the quantum loop algebra $U_q(\dot{\mathfrak{gl}}_\ell)$ from the abelianized monopole operators $u_i^\alpha$. Indeed, it may be checked that the generators

$$t^\alpha := \prod_{k=1}^{N}((\mu^{\alpha-1})^{-1}Q_k^\alpha/Q_k^{\alpha-1})^{1/2}, \tag{4.128}$$

$$E^\alpha := \frac{(\mu^{\alpha-1})^{-N/2}}{q^{-1/2}-q^{1/2}}\sum_{i=1}^{N}u_i^{\alpha+}, \tag{4.129}$$

$$F^\alpha := \frac{(\mu^{\alpha})^{N/2}}{q^{1/2}-q^{-1/2}}\sum_{i=1}^{N}u_i^{\alpha-}, \tag{4.130}$$

satisfy the relations (4.112)–(4.122). More generally, we can use the $L$-operator algebra to define the combinations

$$E^{\alpha,\beta} := (\mu^{\alpha-1})^{-1}u^{\beta+}L^{\beta-1,+}\cdots L^{\alpha+}e, \qquad F^{\alpha,\beta} := \mu^\beta u^{\alpha-}L^{\alpha-}\cdots L^{\beta-1,-}e, \tag{4.131}$$

where we use the covector $u^{\alpha\pm} := (u_1^{\alpha\pm},\dots,u_N^{\alpha\pm})$ and the vector $e=(1,\dots,1)^t$. [TODO: Prefactors: the notebook uses $\prod_{\rho=\alpha}^{\beta}(\mu^{\rho-1})^{-N/2}$ in $E^{\alpha,\beta}$ and $\prod_{\rho=\alpha}^{\beta}(\mu^{\rho})^{N/2}$ in $F^{\alpha,\beta}$ instead of $(\mu^{\alpha-1})^{-1}$ and $\mu^\beta$; only then are $E^{\alpha,\alpha}$, $F^{\alpha,\alpha}$ proportional to $E^\alpha$, $F^\alpha$ and do the $T^\pm[0]$ formulas below hold.] In terms of the 4d $\mathcal{N}=2$ necklace quiver gauge theory, these may be thought of as monopole operators with fundamental charge under the gauge nodes $\alpha$ to $\beta$, see [BDG15, §6.6.1]. From the integrable spin chain perspective, they may be thought of as the propagation of an excitation along the spin chain from site $\alpha$ to site $\beta$. Using the isomorphism between the Drinfeld–Jimbo and the RTT presentation [FT19], we then find the zeroth and first modes of the generator matrices $T^\pm(u)$ to be of the form

$$T^+[0]^{\alpha\beta}=\begin{cases}(t^\alpha)^{-1}E^{\alpha,\beta-1}, & \alpha<\beta,\\ (t^\alpha)^{-1}, & \alpha=\beta,\\ 0, & \alpha>\beta,\end{cases} \qquad T^-[0]^{\alpha\beta}=\begin{cases}F^{\beta,\alpha-1}t^\beta, & \alpha>\beta,\\ t^\beta, & \alpha=\beta,\\ 0, & \alpha<\beta,\end{cases} \tag{4.132}$$

and

$$T^+[1]^{\alpha\beta}=q^{-(1-\delta^{\alpha\beta})/2}(t^\alpha)^{-1}E^{\alpha,\beta+\ell-1}, \qquad T^-[1]^{\alpha\beta}=q^{(1-\delta^{\alpha\beta})/2}F^{\beta,\alpha+\ell-1}t^\beta. \tag{4.133}$$

One may check that the RTT relations (4.125) of the quantum loop algebra are indeed satisfied to first order in $u$ and $v$. [TODO: The notebook verifies the RTT relations with $T^+(u)=T^+[0]+u^{-1}\cdot$(the expression given here for $T^-[1]$) and $T^-(u)=T^-[0]+u\cdot$(the expression for $T^+[1]$), i.e. with $T^\pm[1]$ swapped, while its quantum determinant check uses the assignment as written here. Either swap $T^\pm[1]$ and recompute $\mathrm{qdet}^\pm[1]$, or keep them and rerun the RTT check.] Now, we can use the quantum Leibniz formula

$$\mathrm{qdet}^\pm(u)=\sum_{n=0}^{\infty}\mathrm{qdet}^\pm[n]u^{\mp n}=\sum_{\sigma\in S_\ell}(-q^{1/2})^{\ell(\sigma)}T^{\sigma(1)1\pm}(u)\cdots T^{\sigma(\ell)\ell\pm}(q^{-\ell+1}u), \tag{4.134}$$

with $S_\ell$ the symmetric group on $\ell$ objects, [TODO: The notebook evaluates instead $\sum_{\sigma\in S_\ell}(-q^{1/2})^{\ell(\sigma)}T^{1\sigma(1)\pm}(u)T^{2\sigma(2)\pm}(qu)\cdots T^{\ell\sigma(\ell)\pm}(q^{\ell-1}u)$ (permutation on the column index,

increasing shifts); the results (4.135) are for that formula.] to extract the first modes of the quantum determinant and find

$$\mathrm{qdet}^{\pm}[0] = t^{\pm N/2}, \qquad \mathrm{qdet}^{\pm}[1] = q^{\mp 1/2}(1 - t^{\mp 1})H^{\pm}[1]. \tag{4.135}$$

It follows that the Bethe subalgebra of the horizontal quantum loop algebra $U_q(\dot{\mathfrak{gl}}_\ell)$ (which includes the quantum determinant) provides a large algebra of commuting Hamiltonians including the defining Hamiltonian $H^-[1]$. By a naive counting, the image of the Bethe subalgebra in the $N$-truncated quotient algebra $U_q^{(N)}(\dot{\mathfrak{gl}}_\ell)$ has $N\ell$ algebraically independent generators, which would provide enough commuting Hamiltonians for Liouville integrability[1], since the $K$-theoretic Coulomb branch algebra is generated by the $2N\ell$ algebraically independent generators $Q_i^\alpha$ and $P_i^\alpha$.

For the first few generalized Macdonald operators (4.105), one can verify that they commute with the Drinfeld–Jimbo generators (4.128)–(4.130). We therefore conjecture that the Hamiltonians $H^{\pm}[n]$ are central in the horizontal quantum loop algebra $U_q(\dot{\mathfrak{gl}}_\ell)$ and hence lie in the horizontal Heisenberg subalgebra defined in [Wen25]. The theory of *loc. cit.* would suggest that eigenstates of the Hamiltonians can be given by wreath Macdonald polynomials.

### 4.3.5 Quantization of physical spin variables

Following the formulas for the classical case elaborated in [AH26c], the spin variables used by Krichever and Zabrodin [KZ95] are defined by the expressions

$$a^\alpha := L^{0-} \cdots L^{\alpha-1,-} e, \qquad c^\alpha := u^{\alpha+1,-} \tilde{L}^{\alpha+1,-} \cdots \tilde{L}^{\ell-1,-} \tag{4.136}$$

for $\alpha = 0, \dots, \ell - 1$, where we use the modified $L$-operator $\tilde{L}^{\alpha-} := \mu^\alpha Q^\alpha L^{\alpha-} (Q^{\alpha+1})^{-1}$, which satisfies the equation[2]

$$q^{1/2} L^{\alpha-} - q^{-1/2} \tilde{L}^{\alpha-} = e u^{\alpha+1,-}. \tag{4.137}$$

[TODO: This equation is correct once the sign of $L^{\alpha-}$ in (4.93) is fixed as in the notebook (see the note there); no change needed here or in (4.138).] The same telescopic argument as in [AH26c] yields a similar identity for the total $L$-operators

$$\sum_{\alpha \in \mathbb{Z}_\ell} q^\alpha a^\alpha c^\alpha = q^{\ell-1/2} L^{\mathrm{tot},-} - q^{-1/2} \tilde{L}^{\mathrm{tot},-}, \tag{4.138}$$

with $\tilde{L}^{\mathrm{tot}-} := \tilde{L}^{0,-} \cdots \tilde{L}^{\ell-1,-} = t Q^0 L^{\mathrm{tot}-} (Q^0)^{-1}$. We can solve equation (4.138) for the matrix coefficients, obtaining

$$L_{ij}^{\mathrm{tot}-} = \frac{q^{1/2-\ell}}{1 - t q^{1-\ell} Q_i^0 / Q_j^0} \sum_{\alpha \in \mathbb{Z}_\ell} q^\alpha a_i^\alpha c_j^\alpha, \qquad \tilde{L}_{ij}^{\mathrm{tot}-} = \frac{q^{1/2}}{t^{-1} q^{\ell-1} Q_j^0 / Q_i^0 - 1} \sum_{\alpha \in \mathbb{Z}_\ell} q^\alpha a_i^\alpha c_j^\alpha. \tag{4.139}$$

We now comment on the commutation relations of the *physical* spin variables $a^\alpha$ and $c^\alpha$, meaning we restrict ourselves to spin states with index $\alpha \neq 0$. We obtain: Moin moin,

[1] We thank J. Lamers and M. Vasilev for this insight.

[2] This can be seen as the trigonometric analog of the quantum moment map equations of the chainsaw quiver variety [FR14, equation (34)].

Sure, happy to do. How about I stay on hold, with a letter ready to submit, and will do so unless Gleb resurfaces in time. (If it were 'at least 2' letters, I can submit a letter in any case, but from your spreadsheet that doesn't seem to be the case.)

This week is fairly full, but I'd have 30min either today or on Friday at your 14h15 or 16h15, or (possibly slightly longer) tomorrow at your 15h.

Best wishes, Jules

$$Q_i^0 a_j^\alpha = a_j^\alpha Q_i^0, \tag{4.140}$$

$$Q_i^0 c_j^\alpha = q^{\delta_{ij}} c_j^\alpha Q_i^0, \tag{4.141}$$

$$a_a^\alpha a_b^\alpha = R_{ab}^{00}(q) a_b^\alpha a_a^\alpha, \tag{4.142}$$

$$a_a^\alpha a_b^\beta = q^{1/2} R_{ab}^{00}(q) a_b^\beta a_a^\alpha, \qquad (\alpha < \beta) \tag{4.143}$$

$$c_a^\alpha \bar{R}_{ba}^{00}(q)^{-1} a_b^\alpha = q^{-1/2} a_b^\alpha c_a^\alpha - q^{-\alpha}(1-q^{-1})\left[\Delta_{ab} \tilde{L}_a^{\mathrm{tot}-} + q^{1/2} \sum_{\rho=0}^{\alpha-1} q^\rho a_b^\rho c_a^\rho\right], \tag{4.144}$$

$$c_a^\alpha \bar{R}_{ba}^{00}(q)^{-1} a_b^\beta = a_b^\beta c_a^\alpha, \qquad (\alpha < \beta) \tag{4.145}$$

$$c_a^\alpha c_b^\alpha = c_b^\alpha c_a^\alpha \underline{R}_{ab}^{00}(q)^{-1}, \tag{4.146}$$

$$c_a^\alpha c_b^\beta = q^{-1/2} c_b^\beta c_a^\alpha \underline{R}_{ab}^{00}(q)^{-1}, \qquad (\alpha < \beta) \tag{4.147}$$

with $\Delta = \sum_{i=1}^N e_i^t \otimes e_i$.

The commutation relations (4.140)–(4.147) show how the spin variables commute given definite spin states $\alpha, \beta$. Physically, however, it makes more intuitive sense to ask how the spin variables of two given particles $i, j$ interchange. This question indeed has a nice answer in the case of the physical spin vectors $c_i$, where we observe a simple exchange relation:

$$c_i^a c_i^b = R^{ab}(-1) c_i^b c_i^a, \qquad c_i^a c_j^b = R^{ab}(Q_j^0/Q_i^0) c_j^b c_i^a, \qquad (i \neq j) \tag{4.148}$$

with $R(u)$ the standard trigonometric $R$-matrix (4.127) for $\mathfrak{gl}_{\ell-1}$. However, a similar formula for the physical spin covectors $a_i$ is not available, repeating the pattern remarked upon in remark 3.14 in the rational degeneration.

We now finish the discussion of the quantization of the physical spin variables by giving their quantum equations of motion (the Heisenberg equations). This can be done simply by utilizing the commutation relations (4.95). For example, we can compute

$$a^\alpha H^-[1] = \mathrm{Tr}_a\, a_b^\alpha L_a^{\mathrm{tot}-} = q^{1/2}\, \mathrm{Tr}_a\, R_{ba}^{00}(q) L_a^{\mathrm{tot}-} \bar{R}_{ba}^{00}(q)^{-1} a_b^\alpha, \tag{4.149}$$

where on the way we have used the identity $\underline{R}_{ba}^{\beta\alpha}(q)^{-1} e_b = q^{1/2} \bar{R}_{ab}^{\alpha\beta}(q)^{-1} e_b$. Writing these identities in components, the Heisenberg equations become

$$\dot{Q}_i^0 = [H^-[1], Q_i^0] = (q^{-1}-1) Q_i^0 L_{ii}^{\mathrm{tot}-}, \tag{4.150}$$

$$\dot{a}_i^\alpha = [H^-[1], a_i^\alpha] = (q^{-1}-1) \sum_{j(\neq i)} L_{ij}^{\mathrm{tot}-} (a_j^\alpha - a_i^\alpha) \frac{1}{1 - Q_j^0/Q_i^0}, \tag{4.151}$$

$$\dot{c}_i^\alpha = [H^-[1], c_i^\alpha] = (q^{-1}-1) \sum_{j(\neq i)} \left( \frac{1}{1 - Q_j^0/Q_i^0} c_i^\alpha L_{ij}^{\mathrm{tot}-} - \frac{1}{1 - Q_i^0/Q_j^0} c_j^\alpha L_{ji}^{\mathrm{tot}-} \right), \tag{4.152}$$

which give the natural operator ordering for the operator form of the classical equations of motion by Krichever and Zabrodin [KZ95].

### 4.3.6 Schur quantization

## 4.4 Elliptic spin Ruijsenaars–Schneider model from 5d $\mathcal{N} = 1$ gauge theory

[MZ25]

## 4.5 Class $S_k$ and the elliptic spin Ruijsenaars–Schneider model

## 4.6 Brane construction

[MOY21]

# 5
# Conclusion

# Appendices

# A
# Elliptic functions

Throughout this thesis, we consider the elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ for some complex number $\tau$ with $\operatorname{Im}\tau > 0$.

**Definition A.1.** (i) The *first Jacobi theta function* of $E_\tau$ is defined by

$$\theta_1(z; e^{i\pi\tau}) := -i\sum_{n\in\mathbb{Z}}(-1)^n e^{i\pi(n+1/2)^2\tau} e^{i(2n+1)z}. \tag{A.1}$$

(ii) The *(normalized) odd Jacobi theta function* of $E_\tau$ is defined by

$$\theta(z) := \frac{\theta_1(\pi z, e^{i\pi\tau})}{\pi\theta_1'(0, e^{i\pi\tau})}. \tag{A.2}$$

(iii) We denote the logarithmic derivative of $\theta(z)$ by

$$E_1(z) := \frac{\theta'(z)}{\theta(z)}. \tag{A.3}$$

At $z = 0$, $E_1(z)$ has a pole of residue one. Taking the regular part, we set $E_1(0) := 0$.

(iv) The *Weierstraß $\wp$-function* is

$$\wp(z) := E_1(z)^2 - \frac{\theta''(z)}{\theta(z)} + \frac{\theta'''(0)}{3}. \tag{A.4}$$

It has a double pole at $z = 0$ with coefficient one.

(v) The *infinite $p$-Pochhammer symbol* is defined by

$$(x; p)_\infty := \prod_{n=0}^{\infty}(1 - p^n x). \tag{A.5}$$

**Lemma A.2.** *(i) The odd Jacobi theta function $\theta(z)$ satisfies the quasi-periodicity*

$$\theta(z+1) = -\theta(z), \quad \theta(z+\tau) = -e^{-i\pi\tau-2\pi iz}\theta(z), \tag{A.6}$$

*and is normalized such that $\theta'(0) = 1$.*

*(ii) Letting* $x := e^{2\pi \mathrm{i} z}$ *and* $p = e^{2\pi \mathrm{i}\tau}$*, we have*

$$\theta(z) = \frac{x^{1/2} - x^{-1/2}}{2\pi\mathrm{i}} \prod_{n=1}^{\infty} \frac{(1-p^n x)(1-p^n x^{-1})}{(1-p^n)^2} = -\frac{x^{-1/2}}{2\pi\mathrm{i}(p;p)_\infty^2}(x;p)_\infty (p x^{-1};p)_\infty, \tag{A.7}$$

*(iii) In the limit* $\tau \to \mathrm{i}\infty$, i.e. $p \to 0$*, we have* $\theta(z) \to \frac{1}{\pi}\sin(\pi z)$ *and* $E_1(z) \to \pi\cot(\pi z)$*.*

*(iv) The logarithmic derivative* $E_1(z)$ *satisfies the quasi-periodicity*

$$E_1(z+1) = E_1(z), \qquad E_1(z+\tau) = E_1(z) - 2\pi\mathrm{i}. \tag{A.8}$$

*(v) The Weierstraß* $\wp$*-function is doubly periodic:*

$$\wp(z+1) = \wp(z), \qquad \wp(z+\tau) = \wp(z). \tag{A.9}$$

**Definition A.3.** The *Kronecker elliptic function* is defined by

$$\Phi(z,u) := \frac{\theta(z+u)}{\theta(z)\theta(u)}. \tag{A.10}$$

At $z = 0$ or $u = 0$, we define the Kronecker elliptic function by its regular part:

$$\Phi(z,0) := E_1(z), \qquad \Phi(0,u) := E_1(u), \qquad \Phi(0,0) := 0. \tag{A.11}$$

**Lemma A.4.** *The Kronecker elliptic function satisfies the following properties:*

*(i)* $\Phi(z,u) = \Phi(u,z)$*.*

*(ii)* $\Phi(z,u)$ *is quasi-periodic:*

$$\Phi(z+1,u) = \Phi(z,u), \qquad \Phi(z+\tau,u) = e^{-2\pi\mathrm{i}u}\Phi(z,u). \tag{A.12}$$

*(iii)* $\Phi(z,u) = \frac{1}{z} + E_1(u) + O(z)$*.*

*(iv)* $\Phi(z,u)$ *satisfies the* Fay identity*:*

$$\Phi(z,x)\Phi(w,y) = \Phi(z,x-y)\Phi(z+w,y) + \Phi(w,y-x)\Phi(z+w,x), \tag{A.13}$$

*which, combined with the symmetry* $\Phi(z,u) = \Phi(u,z)$*, becomes in the limit of coinciding second arguments* $(x \to y)$

$$\Phi(z,x)\Phi(z,y) = \Phi(z,x+y)(\Phi(z,0) + \Phi(x,0) + \Phi(y,0) - \Phi(z+x+y,0)). \tag{A.14}$$

*(v) In the limit* $\tau \to \mathrm{i}\infty$*, we have*

$$\Phi(z,u) \to \pi\cot(\pi z) + \pi\cot(\pi u). \tag{A.15}$$

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